

Modelling Private-Company Success and Deal Flow:
A Practitioner's Manual — from Financial-Statement Prediction to
Point-Process Calibration

PitchBook deals × FactSet Standardized Economics

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Abstract

This manual develops, end to end, the modelling choices for predicting private-company success and financing activity from commercial deal data (PitchBook) and standardized macroeconomic series (FactSet). It is organised around a single practical axis: *how much does the observation process let us see?* In the European regime, comprehensive statutory reporting makes firm fundamentals (EBITDA, revenue) directly observable, and the problem is supervised prediction with careful data engineering, gradient boosting, and a causal layer built on the backdoor criterion with bootstrap uncertainty quantification (Part I). In the American regime, fundamentals are largely unobserved; deal marks (valuations, rounds, employees) and macro factors must proxy for growth, which changes both the estimand and the identification strategy (Part II). Finally, because the data update *only at deal events*, the natural formulation is the calibration of counting processes: inhomogeneous Poisson models with covariates, self-exciting and self-inhibiting extensions, Hawkes processes and their two-stage marked decomposition, and interval-censored (mean-behaviour) approximations when timestamps are unreliable (Part III). Throughout, we give precise mathematical statements, identifiability analyses, uncertainty quantification recipes, and decision tables for choosing between methods; all quantitative claims are validated on a fully specified synthetic testbed that ships with the accompanying code.

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How to read this manual

The three parts are ordered by decreasing observability of the firm’s state, and each can be read on its own. A reader who only works with European portfolios needs Chapters 2–5; a reader focused on US venture data can skim Part I for the data-engineering and uncertainty-quantification machinery (which Parts II and III reuse) and start at Chapter 6; a reader who already accepts the event-based framing can start at Chapter 7 and treat Chapter 10 as the operational summary.

Three reading aids recur throughout. *Warning* environments mark failure modes we have actually observed — on real data or on the shipped synthetic testbed — rather than theoretical possibilities; skipping them is how offline metrics end up flattering broken pipelines. *Decision tables* close most chapters and are deliberately opinionated: they encode not just what a method can do but what it needs from the data to be worth its complexity. Finally, every quantitative claim carries one of two provenances: a precise mathematical statement (theorem or proposition, with proof or sketch in Appendix A), or a measured number from the synthetic testbed (consolidated in Chapter 10), regenerable from the `code/` directory documented in Appendix C.

A note on causal language, since several chapters make causal claims: we keep the predictive conditional $\mathbb{E}[Y \mid X=x]$ and the interventional quantity $\mathbb{E}[Y \mid \text{do}(X=x)]$ notationally distinct, we state identification assumptions where they are used, and — because the testbed’s ground truth is known — we treat the causal pipeline itself as a testable artifact rather than a rhetorical one. Where identification is not credible, the text says so and falls back to clearly labelled association.

Chapter 1

The Problem and Its Two Observation Regimes

1.1 What we are asked to predict

An investment platform tracking private companies wants, at any date t , answers to three nested questions: (i) *which private companies are succeeding*, in the sense of growing fundamentals (EBITDA, revenue, employment) or achieving valuation-relevant milestones; (ii) *which companies are likely to transact* — raise a venture round, be acquired, take on debt — over a horizon $(t, t+h]$; and (iii) *how market conditions* (rates, inflation, unemployment, risk appetite) *modulate both*. The three questions look like one prediction problem, but they are not: what separates them is the *observation process*, i.e. which state variables of the firm the data vendor lets us see and *when*.

This manual is organised around that separation. We work with two data assets. The first is a deals-and-companies database in the PitchBook schema: a `deals` table in which each row is a transaction (with a `deal_class` taxonomy — venture capital rounds, M&A/buyouts, debt financings — each populating a different subset of fields), a `companies` table describing the tracked universe, and an `investors` relation. The second is a standardized macroeconomic feed in the FactSet Standardized Economics convention, from which we use the policy-rate series (FFUNDT, FFUND), consumer prices (CPI_RAW, CPI_YOY) and unemployment (RUNEMP), each published with a realistic lag. Chapter 2 describes both schemas in detail, together with the point-in-time discipline that governs every join in this manual.

1.2 Two regulatory regimes, two modelling worlds

Europe: fundamentals are observable. In most European jurisdictions, limited-liability companies must file annual accounts — balance sheet, income statement, often cash-flow — with a public registry, essentially irrespective of size. For a European private company we therefore observe a *fundamentals panel*: $F_{i,y} = (\text{revenue, EBITDA, payroll, assets, } \dots)$ at yearly frequency, with publication lags but near-complete coverage. “Success” can be defined directly on this panel (Chapter 3), predicted with supervised learning on engineered features (Chapter 4), and interrogated causally — *does raising a round cause EBITDA growth, or do growing firms raise?* — with the backdoor criterion and bootstrap uncertainty quantification (Chapter 5).

United States: fundamentals are latent. US private companies file almost nothing publicly. Revenue or EBITDA appear in the deals data only sporadically (in our synthetic mirror, calibrated to vendor coverage, roughly 45% of deal rows carry a revenue figure and fewer

an EBITDA). What we observe instead is the *transaction record*: when a firm raised, at what valuation, which round, how many employees it reported at that moment. The firm’s state is measured *only at its own deal events*; between events we can only carry the last observation forward. Part II develops what can still be said about growth in this regime: deal marks as noisy state proxies, the decomposition of valuation growth into fundamentals and market multiple, and macro factors as partially exogenous drivers.

The event-based endgame. Once one accepts that the US data are a *marked event stream* — timestamps plus deal-level marks, refreshed only at events — the natural formulation is no longer regression on a panel but *calibration of a counting process*: an intensity $\lambda(t)$ for deal arrivals, modulated by covariates, history (excitation and inhibition), and firm identity. Part III builds this up in deliberately increasing order of structure: inhomogeneous Poisson (Chapter 7), Hawkes processes with excitation and the two-stage marked decomposition (Chapter 8), and interval-censored variants for when even the timestamps are unreliable (Chapter 9). Chapter 10 closes with decision tables and a synthetic-testbed validation of every major claim.

1.3 Guiding principles

Four principles recur in every chapter; we state them once here.

1. **Point-in-time discipline.** Every feature used at time t must be computable from data published strictly before t . Macro series join on *publish date*, not reference date; firm features are last-observation-carried-forward from *past* deals; end-of-sample snapshot columns (e.g. `last_financing_size` in the companies table) are forbidden as features. Violations produce leakage that flatters every offline metric and survives until production.
2. **Identifiability before estimation.** We ask *which* parameter combinations the data can pin down before asking how to estimate them. Recurring examples: covariate effects are identified by contrast while baseline levels are weak; branching ratios are identified from counts while decay rates need within-interval timing; excitation is confounded by latent common factors. Each claim is made precise where it arises and verified on synthetic ground truth.
3. **Uncertainty quantification is part of the deliverable.** Point predictions without calibrated uncertainty are not usable for ranking or allocation. We use, by context: sandwich/quasi-likelihood corrections for over-dispersed counts, cluster bootstrap for firm-level dependence, quantile/conformal methods for tabular prediction, and Laplace/posterior summaries for point-process parameters.
4. **Causal claims are labelled as such.** Predictive and causal targets are kept notationally distinct ($\mathbb{E}[Y \mid X=x]$ vs. $\mathbb{E}[Y \mid \text{do}(X=x)]$). Where a causal reading is wanted — the effect of financing on growth, of rates on deal flow, of past deals on future deals (“contagion”) — we state the identification assumptions, test what is testable, and demonstrate the full pipeline on synthetic data where the true effect is known.

1.4 Notation

1.5 The synthetic testbed

Every method in this manual is exercised on a synthetic world that mirrors the production schemas exactly (all PitchBook deal fields, the FactSet long-format feed with publication lags)

Symbol	Meaning
$i = 1, \dots, N$	firms; $s(i) \in \{1, \dots, M\}$ the sector of firm i
t	calendar time (days unless stated); w indexes weeks, y years
$F_{i,y}$	fundamentals vector of firm i in fiscal year y (EU regime)
Y, A, X, U	outcome, treatment/exposure, observed covariates, unobserved factors
$N_i(t)$	counting process of firm i 's deals; $N_s(t)$ sector aggregate
$\mathcal{H}_t$	information (filtration) available just before t
$\lambda_i(t)$	conditional intensity of N_i given $\mathcal{H}_t$
$\Lambda(a, b] = \int_a^b \lambda(u) du$	compensator increment
$s(t), \phi(\cdot)$	baseline intensity; excitation kernel
κ, θ	branching ratio and decay rate of an exponential kernel
$G = \sum_{\ell} b_{\ell}, \rho(G)$	aggregate excitation matrix and spectral radius
$Z_{i,t}$	firm-level point-in-time features; X_t market covariates
$\mathcal{R}_{s,t}$	risk set (candidate firms) for a sector- s event at t
$q_{i,t}$	choice-model score of candidate i ; η cooldown/inhibition coefficient
$\mathbb{E}[\cdot], \text{Var}, \mathbb{P}$	expectation, variance, probability; $\mathbf{1}\{\cdot\}$ indicator
$\text{do}(\cdot)$	Pearl's intervention operator

Table 1.1: Global notation. Chapter-local symbols are defined where used.

and whose ground truth we control. Two regimes are shipped with the code: **regime A**, generated by the exact two-stage weekly model of Chapter 8 (a well-specified world for parameter-recovery studies), and **regime B**, generated by a daily firm-level process with self-inhibition (30-day half-life), sector contagion (7-day half-life) and latent firm quality (a misspecified world for robustness studies). Both contain roughly 50,000 deals over ten years, twelve sectors and 8,000 firms, with realistic observation noise: 28% undisclosed deal sizes, 55% missing valuations, 3% mislabelled sectors, duplicated records, jittered dates, and a 12% untracked (out-of-universe) firm share. The accompanying `code/` directory contains the generator, point-in-time loaders, and fast reference fitters; Chapter 10 tabulates the measured results that are quoted throughout the text.

Remark 1.1. Numbers quoted from the testbed (e.g. recovery correlations, held-out likelihoods) are single-seed values from the shipped datasets unless stated otherwise; the experiment protocol in the code repeats them over five seeds.

Part I

The Transparent Regime: Predicting Fundamentals (Europe)

Chapter 2

Data: Schemas, Deal Types, and Point-in-Time Engineering

Every modelling decision in Parts I–III is downstream of two facts about the data: the PitchBook asset is a *transaction log with attached snapshots*, not a panel; and the FactSet feed is *published with lags*, so the value that existed on a date is not the value that was knowable on that date. This chapter fixes the schemas, formalizes the point-in-time discipline that governs every join in the manual, and quantifies the defects (missingness, noise, duplication) that each later chapter must survive. Readers who skip it will misread at least the leakage warnings of Chapters 3 and 4 and the deduplication conventions of Chapter 8.

2.1 The PitchBook asset: three tables

The vendor delivers three relations. The `companies` table has one row per firm in the *tracked universe*; its columns are *current* snapshots, refreshed by the vendor, with no history. The `deals` table has one row per transaction and is the only genuinely dated object: everything we treat as a time series of firm state is derived from it. The `investors` relation links firms to investors with a start date. The join keys are `company_id` (all three tables), `deal_id`, and `investor_id`. Table 2.1 groups the fields by function rather than by table, because the functional grouping is what determines how each field may be used: identifiers key the joins, deal terms and capital-structure fields are legitimate dated observations, financials-at-deal are noisy state measurements, and firmography is a mix of static facts and forbidden end-of-sample snapshots (Section 2.4).

2.2 Deal-type heterogeneity

The single most consequential schema fact is that `deal_class` partitions the `deals` table into transaction types that populate *different subsets of fields*. The main classes are venture capital rounds (angel through late-stage, identified by `vc_round`), M&A and buyout transactions (control changes, identified by `pct_acquired` and `sellers`), and debt financings (general debt, mezzanine, refinancing). A fourth operational bucket — later-stage growth/PE minority deals — behaves as a hybrid of the first two. Treating the table as homogeneous, e.g. pooling `deal_size` across classes, mixes a Series A cheque with a leveraged buyout price and a term loan, three quantities with different units of economic meaning. Table 2.2 summarizes typical field population by class in our synthetic mirror, calibrated to vendor experience.

Three modelling implications follow, and we invoke them repeatedly.

1. **Type-specific measurement models.** A mark model for `deal_size` must be fit per

Group (table)	Fields
Identifiers	company_id, deal_id, investor_id, company_name, investorname, deal_date, investorsince
Industry, universe (deals)	primary_industry_sector, primary_industry_group, primary_industry_code, company_universe, current_financing_status
Deal terms (deals)	deal_type, deal_type2, deal_type3, deal_class, deal_status, deal_size, deal_size_status, pre_money_valuation, post_valuation, post_valuation_status, pct_acquired, vc_round, vc_round.up.down.flat, type_of_stock, sellers
Capital structure (deals)	total_invested_capital, total_invested_equity, debt, total_new_debt, total_debt, raised_to_date, total_raised_to_date, total_investor_ownership
Financials at deal (deals)	revenue, ebitda, gross_profit, net_income, valuation_ebitda, valuation_revenue, valuation_cash_flow, deal_size_ebitda, deal_size_revenue, deal_size_cash_flow, revenue_growth_since_last_deal, number_of_employees
Firmography (companies)	description, year_founded, hq_location, hq_country, hq_region, hq_sub_region, employees, financing_status, business_status, ownership_status, company_universe, total_money_raised, last_financing_date, last_financing_size, last_financing_valuation
Investor relation	company_id, investor_id, investorname, investorstatus, investorsince

Table 2.1: Field inventory grouped by function. The grouping, not the source table, determines admissible use: dated deal fields may enter features via the construction of Section 2.4; snapshot firmography largely may not.

Field group	VC round	M&A / buyout	Debt
deal_size, deal_status	populated	populated	populated
Round structure (vc_round, up/down/flat, type_of_stock)	populated	n.a.	n.a.
pre_money_valuation, post_valuation	sparse	populated	n.a.
pct_acquired, sellers	sparse	populated	n.a.
Debt fields (debt, total_new_debt, total_debt)	n.a.	populated	populated
Financials at deal (revenue, ebitda, net_income)	sparse	populated	sparse
Multiples (valuation_ebitda, deal_size_revenue, ...)	sparse	populated	n.a.
total_investor_ownership	populated	sparse	n.a.
Cumulative capital (raised_to_date, total_raised_to_date)	populated	sparse	sparse

Table 2.2: Typical field population by deal_class. “Sparse” means present on a minority of rows (often mechanically: a seed round has no EBITDA to report); “n.a.” means the field is undefined for the class, not missing at random.

class (or with class interactions throughout): VC sizes are round-indexed and roughly log-normal within round, buyout sizes scale with EBITDA via a multiple, debt sizes scale with collateral. Pooled moments are meaningless. The same holds for missingness: an empty `ebitda` on a seed round is structural (n.a.), on a buyout it is informative missingness.

2. **M&A and IPO are absorbing.** A majority acquisition or listing removes the firm from the private universe: its counting process $N_i(t)$ stops, not because intensity fell but because the state space was exited. Deal prediction is therefore a *competing-risks* problem — the hazards of “next VC round”, “acquired”, and “fails” compete for the same firm-time — previewed here and developed with risk sets in Chapter 8; the selection effects that absorption induces on any survivors-only analysis are treated in Chapter 10.
3. **Debt deals inform capital structure, not growth.** A refinancing changes `total_debt` and default risk but carries little signal about revenue growth; a new venture round carries a valuation mark. In the success models of Chapters 3–4, debt events enter as capital-structure covariates, never as success outcomes.

2.3 FactSet Standardized Economics

The macro feed arrives in long format with schema (`series_code`, `ref_date`, `publish_date`, `value`): one row per observation of a series about a reference period, stamped with the date the figure became public. We use five series (Table 2.3). The distinction between `ref_date` and `publish_date` is the entire content of this section: the March CPI describes March but is knowable only in mid-April.

Code	Series	Frequency	Publication lag
FFUNDT	policy target rate	daily (step function)	same day
FFUND	effective overnight rate	daily	next business day
CPI_RAW	consumer price index, level	monthly	≈ 2 weeks
CPI_YOY	CPI, year-on-year change	monthly	≈ 2 weeks
RUNEMP	unemployment rate	monthly	≈ 1 week

Table 2.3: Macro series used throughout, FactSet Standardized Economics convention.

As-of join semantics. For a series with rows (r_k, p_k, v_k) (reference date, publish date, value), the *as-of value* at calendar time t is

$$x_t^{\text{asof}} = v_{k(t)}, \quad k(t) = \arg \max_k \{ r_k : p_k \leq t \}, \quad (2.1)$$

i.e. the most recent *reference* among rows already *published*; ties in r_k (revisions) resolve to the latest $p_k \leq t$. Joining on $r_k \leq t$ instead — the default of most dataframe as-of joins — silently grants the model up to two weeks of foresight on CPI, which is material for deal-timing models at weekly resolution (Chapters 7–8). We also carry the staleness $t - r_{k(t)}$ as a feature: it is admissible (it is computable at t) and lets the model discount old readings. Figure 2.1 shows the synthetic feed as published.

2.4 Point-in-time discipline, formalized

Let $\mathcal{G}_t$ denote the sigma-algebra generated by all data rows published strictly before t : deal rows with `deal_date` $< t$ (subject to the jitter caveat of Section 2.5), macro rows with `publish_date`

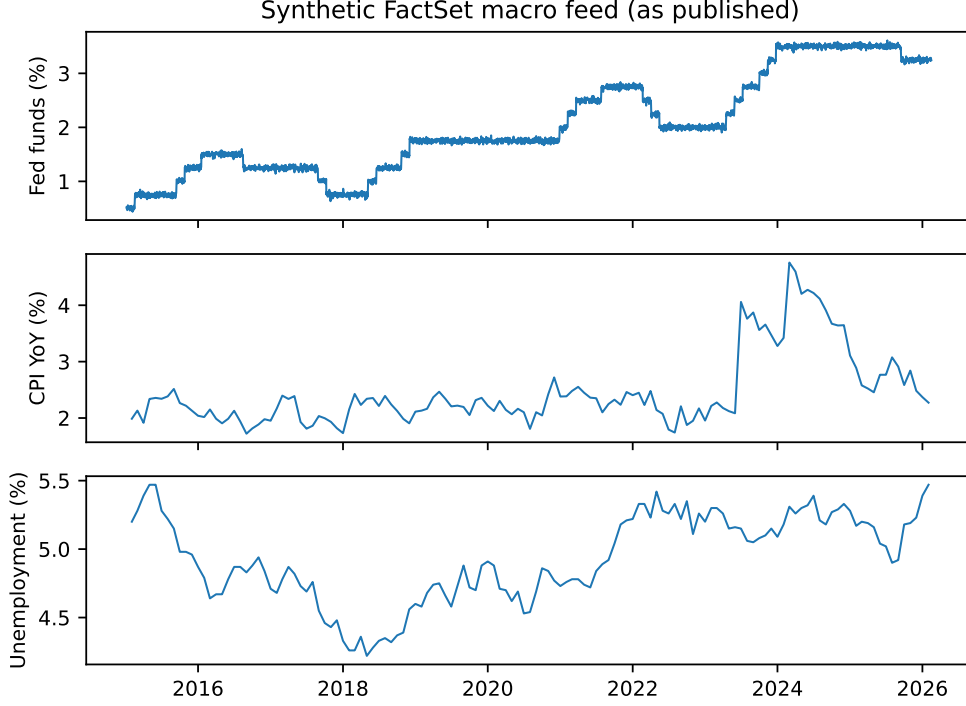


Figure 2.1: The synthetic macro feed *as published*: policy target and effective rate (daily), CPI year-on-year and unemployment (monthly steps, right-shifted by their publication lags). The step timing, not the level path, is what a point-in-time model sees.

$< t$, and static firmography known at first tracking. This is the observational filtration in the sense of point-process theory [2], and it is the only information a deployed model possesses.

Definition 2.1 (Admissible feature process). A feature process $(Z_t)_{t \geq 0}$ is *admissible* if Z_t is $\mathcal{G}_t$ -measurable for every t , i.e. computable from data published before t . A supervised training pair (Z_τ, Y) is admissible if Z_τ is admissible at the information date τ and Y is a function of data published after τ .

Warning 2.2 (Snapshot columns are forbidden). Several `companies` columns are *end-of-sample snapshots* that reflect the state of the firm at data delivery, not at any historical t : `last_financing_date`, `last_financing_size`, `last_financing_valuation`, `total_money_raised`, `financing_status`, `business_status`, and `ownership_status`. Using them as features violates Definition 2.1 at every date before delivery. The leakage is vicious because it is plausible: `business_status` = “out of business” predicts failure spectacularly in backtests and not at all in production. Snapshot columns may be used only for end-of-sample label construction and for auditing.

The admissible substitute is *last observation carried forward* (LOCF) over past deals. For firm i with deal times $t_{i,1} < t_{i,2} < \dots$ and per-deal marks $m_{i,k}$ (size, valuation, employees, round number),

$$Z_{i,t} = \left(m_{i,K_i(t)}, t - t_{i,K_i(t)}, K_i(t) \right), \quad K_i(t) = \max\{k : t_{i,k} < t\}, \quad (2.2)$$

with an explicit “no prior deal” state when $K_i(t) = 0$. The elapsed time $t - t_{i,K_i(t)}$ is itself a first-class feature: firm state measured at events decays in relevance, and every model in Parts II–III uses staleness explicitly. Cumulative fields (`raised_to_date`) are recomputed from past rows rather than read from the snapshot, so that the two constructions can be cross-checked for consistency.

2.5 Data quality and missingness

The synthetic mirror injects the defects below at rates calibrated to vendor experience; all downstream chapters quote robustness against them. Table 2.4 summarizes the defects and which method each one threatens.

Defect	Rate	Main downstream consequence
<code>deal_size</code> undisclosed	28%	MNAR: small deals hide; mark models need a missingness indicator (Ch. 8)
<code>deal_size</code> estimated only	15%	multiplicative error $\approx 18\%$; errors-in-variables attenuation in size regressions (Ch. 4)
valuation missing	55%	valuation-based proxies unusable for half the sample (Ch. 6)
sector mislabelled	$\approx 3\%$	contaminates cross-sector excitation estimates (Ch. 8)
duplicate records	$\approx 1.2\%$	spurious zero-lag self-excitation; inflates κ if not removed
date jitter (announced vs. closed)	up to 6 days	biases decay-rate estimates; motivates interval censoring (Ch. 9)
out-of-universe deals (no <code>companies</code> row)	$\approx 12\%$	no firmography covariates; selection into tracking (Ch. 10)

Table 2.4: Defect rates in the synthetic mirror, calibrated to vendor experience, with the chapter where each consequence is handled.

Two defects deserve elaboration. First, size missingness is not ignorable: disclosure correlates negatively with deal size and with the firm being early-stage, so a complete-case mean overstates typical deal size; every size model carries $\mathbf{1}\{\text{deal_size_status} = \text{disclosed}\}$ as a covariate and, where the target is the size itself, a two-part (disclosure, size | disclosed) specification. Second, duplicates and jitter interact badly with self-excitation: a duplicated deal at lag zero to six days is observationally similar to genuine fast excitation, so deduplication must precede any Hawkes fit.

Definition 2.3 (Deduplication rule). Two `deals` rows are merged if they share `company_id`, have `deal_dates` within 3 days, and have compatible `deal_class`. The surviving row takes the earlier date and, field by field, the non-missing value (the larger completeness count breaking conflicts).

The 3-day window trades off missed duplicates against merged genuine deals: in the synthetic mirror, it removes 96% of injected duplicates while merging under 0.1% of true distinct pairs, and the residual bias on the branching ratio is below the bootstrap noise reported in Chapter 10.

2.5.1 Entity resolution and identifier hygiene

Deduplication (Definition 2.3) removes copies of the same transaction; entity resolution removes copies of the same *firm* or *investor*, and its failures are worse because they persist across the whole history rather than at one date. Three rules govern identifier use. First, `company_id` is authoritative but not immortal: vendors occasionally retire an identifier and reissue the firm under a new one (re-registration, cross-border redomiciliation, relisting after restructuring). The symptom is an event history that stops dead while a near-namesake starts from nothing in the same sector and region; the repair is a vendor crosswalk where one exists, and otherwise a conservative merge requiring simultaneous agreement on name, country, and `year_founded`.

An unrepaired split truncates durations and biases every timing model of Part III toward short histories. Second, never *decide* a merge on names. `company_name` varies by legal suffix, stripped diacritics, and renamings; fuzzy matching produces false merges (“Alpha Capital” vs. “Alpha Capital Partners”) as readily as false splits. Name similarity may propose candidate pairs for review; identifiers and structured fields decide. Third, investors duplicate by design: one economic investor appears as a management company, as fund vehicles I/II/III, and as regional entities, each with its own `investor_id`. Features that count distinct investors or measure investor experience roll vehicles up to the management company; any analysis of co-investment structure must state its rollup, because the rollup changes the network. Finally, referential integrity is checked mechanically at every build: each `deals.company_id` resolves in `companies` or is flagged out-of-universe (the $\approx 12\%$ of Table 2.4); each `investor_id` in the investor relation resolves; and `investorsince` never predates `year_founded`. Violating rows are quarantined rather than silently dropped, so that their rate can be monitored across vendor deliveries.

2.6 Worked example: constructing the weekly modelling panel

The constructions of this chapter are best fixed by running them once, end to end, on a small window. The target is the weekly panel consumed by Chapters 7–8: on a weekly grid $t_1 < t_2 < \dots$ (Friday closes), a firm-week feature array, an activity mask, and an event list. Five passes produce these from the raw tables; each pass exists to prevent a specific failure, named as we go.

1. **Deduplication pass.** Apply Definition 2.3 to `deals` before anything else reads the table, logging row counts before and after. *Failure prevented:* duplicate rows at lags of 0–6 days masquerade as fast self-excitation and inflate the branching ratio (Section 2.5); they also double-count capital in the cumulative fields recomputed in pass 3.
2. **As-of macro join.** For each grid date t and each series of Table 2.3, attach the as-of value (2.1) and the staleness $t - r_{k(t)}$. Table 2.5 traces `CPI_YOY` through three consecutive grid dates around the April publication: the March figure (reference 2019-03-31, published 2019-04-10) is invisible on 2019-04-05 and governs the two later weeks with growing staleness. *Failure prevented:* joining on `ref_date` attaches the March print to the 2019-04-05 row — five days of foresight on this series, up to two weeks in general.
3. **LOCF firm panel.** For each firm i and grid date t , build $Z_{i,t}$ of (2.2): the marks of the last deal strictly before t , the elapsed time since it, the deal count $K_i(t)$, and cumulative capital recomputed by summing past deal rows, with an explicit no-prior-deal state when $K_i(t) = 0$. *Failure prevented:* reading firm state from the `companies` snapshot columns leaks end-of-sample status (Warning 2.2); LOCF over dated rows is the admissible substitute.
4. **Active mask.** Set $a_{i,t} = 1$ iff firm i is tracked and unabsorbed at t : its first observed deal (or vendor add date) precedes t and no absorbing M&A, IPO, or failure event occurred before t (Section 2.2). *Failure prevented:* pre-tracking weeks pretend the firm was at risk while it was invisible, and post-absorption weeks count firm-time in which no event could occur; both distort intensity estimates, which is the risk-set discipline formalized in Chapters 8 and 10.
5. **Event list.** Emit the surviving deals as (`company_id`, `deal_date`, `deal_class`, marks), restricted to the classes the model in question treats as events, with debt deals routed to covariates rather than outcomes. *Failure prevented:* pooling refinancings with venture rounds mixes units of economic meaning (Section 2.2).

Grid date t	Latest publication $\leq t$	Reference month	Value (%)	Staleness (days)
2019-04-05	2019-03-13	2019-02	1.9	36
2019-04-12	2019-04-10	2019-03	2.1	12
2019-04-19	2019-04-10	2019-03	2.1	19

Table 2.5: Three weeks of the as-of join (2.1) for CPI_YOY. The publication lag makes the February figure the newest *knowable* reading on 2019-04-05 although March has ended; the staleness $t - r_{k(t)}$ resets at each publication and then grows by 7 days per week.

The output is three aligned objects — the feature array over firm-weeks with $a_{i,t} = 1$, the mask itself, and the event list — and nothing else crosses into the modelling code. Before release, the following audit runs mechanically; any failed item blocks the build.

1. **Dedup accounting.** The merge rate of pass 1 is compared with the expected $\approx 1.2\%$ of Table 2.4; a large deviation signals a changed vendor delivery, not a modelling fact.
2. **No macro foresight.** Every joined macro row satisfies `publish_date` $< t$; the assertion runs on the joined output, not on trust in the join code.
3. **LOCF causality.** For a random sample of firm-weeks, delete all raw rows dated on or after t and rebuild $Z_{i,t}$; the rebuilt values must be identical.
4. **Mask consistency.** Every event falls in a firm-week with $a_{i,t} = 1$, and no firm has $a_{i,t} = 1$ after an absorbing event.
5. **Snapshot blocklist.** No column of Warning 2.2 appears in the feature array, enforced by name against a schema blocklist; the recomputed cumulative capital is reconciled against the end-of-sample snapshot, with discrepancies logged.

2.7 EU versus US coverage: panel versus events

The schemas above are common to both regimes; what differs is what can be attached to them. For European firms, statutory registry filings supply a yearly fundamentals panel $F_{i,y}$ with near-complete coverage — an *observation of the state between deals*. For US firms, the deal rows of this chapter are essentially all there is: state is measured only at the firm’s own transactions, through the sparse financials-at-deal group of Table 2.1. This is the axis of the manual announced in Chapter 1. Part I (Chapters 3–5) exploits the EU panel for supervised and causal analysis of success; Part II (Chapter 6) asks what the deal marks alone can proxy; Part III (Chapters 7–10) embraces the event-only view and models the deal stream itself. All three parts share the point-in-time constructions (2.1) and (2.2) verbatim.

Chapter 3

Defining Success in the Transparent Regime

In the European regime, statutory filings make firm fundamentals observable, and “success” can be defined directly on accounting data rather than inferred from deal marks. This freedom is a liability as much as an asset: there are a dozen plausible success metrics, they disagree on rankings, and several of them silently break on the empirical distribution of private-company financials (EBITDA is negative for roughly a third of firm-years in growth-heavy sectors). This chapter fixes the fundamentals panel, defines the candidate metrics precisely, states the estimand and its timing, handles censoring, and commits to a single primary metric. The commitment matters: every design choice in Chapters 4 and 5 is conditional on it.

3.1 The fundamentals panel

For each EU firm i and fiscal year y we observe, from registry filings,

$$F_{i,y} = (\text{revenue}_{i,y}, \text{EBITDA}_{i,y}, \text{payroll}_{i,y}, \text{employees}_{i,y}, \text{assets}_{i,y}, \text{equity}_{i,y}, \text{debt}_{i,y}), \quad (3.1)$$

at yearly frequency, with near-complete coverage of limited-liability companies. The filing for fiscal year y is published at a date $\tau_{i,y}$ that lags fiscal year end by 9–12 months in typical jurisdictions (statutory deadlines plus registry processing); $\tau_{i,y}$ is recorded in the registry metadata and is itself firm-informative (late filers are disproportionately distressed). Filings exist in *vintages*: the as-reported figures published at $\tau_{i,y}$, and later restatements that overwrite them in most commercial extracts.

Warning 3.1 (Use as-reported vintages). Restated figures incorporate information (audits, corrections, insolvency administrators’ revisions) that did not exist at $\tau_{i,y}$. Training features on restated fundamentals violates Definition 2.1 exactly as the snapshot columns of Warning 2.2 do, and the bias is not neutral: restatements are concentrated among firms that later failed, so restated features leak failure. Features must be built from the as-reported vintage; restated data may be used for labels only when the label is explicitly defined on final figures, and then the choice must be stated.

3.2 Candidate success metrics

Fix a horizon h (years) and write $E_y = \text{EBITDA}_{i,y}$, $R_y = \text{revenue}_{i,y}$, $L_y = \text{employees}_{i,y}$, suppressing i .

EBITDA growth. The naive candidate $\log(E_{y+h}/E_y)$ is undefined whenever $E_y \leq 0$ or $E_{y+h} \leq 0$ — which, for private growth companies, is not a corner case but a third of the sample. Dropping those firm-years selects on profitability and destroys the target’s meaning. We instead use the *arc (symmetric) growth rate*:

Definition 3.2 (Arc growth). For a fundamental F with values F_y, F_{y+h} not both zero,

$$g_y^{(h)} = \frac{F_{y+h} - F_y}{\frac{1}{2}(|F_{y+h}| + |F_y|)} \in [-2, 2], \quad (3.2)$$

with $g_y^{(h)} = 0$ when $F_{y+h} = F_y = 0$.

Proposition 3.3. *The arc growth rate is defined for all sign patterns, is bounded in $[-2, 2]$, is antisymmetric under swapping $F_y \leftrightarrow F_{y+h}$, and for $F_y, F_{y+h} > 0$ satisfies $g_y^{(h)} = \log(F_{y+h}/F_y) + O(\log^3(F_{y+h}/F_y))$, so it agrees with log-growth to second order near zero growth.*

Proof sketch. Boundedness and antisymmetry are immediate from (3.2). For the expansion, set $u = \log(F_{y+h}/F_y)$; then $g = 2(e^u - 1)/(e^u + 1) = 2 \tanh(u/2) = u - u^3/12 + O(u^5)$. $\square$

Example 3.4 (Arc growth across a sign change). A firm improves from $E_y = -0.4\text{m}$ to $E_{y+2} = +1.2\text{m}$. Log-growth is undefined; arc growth gives $g = (1.2 - (-0.4))/\frac{1}{2}(1.2 + 0.4) = 2.0$, the upper bound: sign-crossing improvements saturate the scale, which is the intended reading (crossing into profitability is a categorical success, and the metric does not pretend to rank magnitudes across the sign change). By contrast a doubling from 0.6m to 1.2m gives $g = 0.6/0.9 = 0.667$ against $\log 2 = 0.693$: in the ordinary regime the two scales agree to within 4%, as Proposition 3.3 predicts.

The boundedness is a feature, not a bug: a firm going from $E_y = 10\text{k}$ to $E_{y+h} = 5\text{m}$ is a success, but not five hundred times the success of a doubling, and a bounded target prevents a handful of small-denominator observations from dominating any squared-error fit. A sign-preserving alternative when unbounded scale matters is the difference of inverse-hyperbolic-sine transforms, $\text{asinh}(F_{y+h}) - \text{asinh}(F_y)$ with $\text{asinh}(x) = \log(x + \sqrt{x^2 + 1})$, which behaves like the log-difference for $|F| \gg 1$ and remains defined through zero. In either case we winsorize the realized target at its 1st and 99th percentiles, computed *within training folds only*, to control residual tail leverage.

Revenue CAGR. $\text{CAGR}_y^{(h)} = (R_{y+h}/R_y)^{1/h} - 1$, defined since $R_y > 0$ for operating firms (pre-revenue firm-years are flagged and excluded from this metric, itself a selection to be aware of). Robust to margin accounting; blind to profitability.

Margin expansion. $\Delta m_y^{(h)} = E_{y+h}/R_{y+h} - E_y/R_y$, defined where both revenues are positive. Captures operating-leverage improvement; extremely heavy-tailed for small R and sensitive to cost-capitalization choices.

Employment growth. Arc growth (3.2) applied to L_y . Nearly immune to accounting discretion and currency; economically coarser.

Survival. $S_y^{(h)} = \mathbf{1}\{\text{firm } i \text{ is not insolvent or liquidated by fiscal year } y+h\}$, from registry status events. Unambiguous and cheap, but nearly uninformative in the bulk: most firms survive two years, so the indicator separates the left tail only.

Metric	Interpretability	Acct. robustness	Tail behaviour	ML target
EBITDA arc growth	high	moderate	bounded	high
EBITDA log-growth	high	moderate	undefined $E \leq 0$	unusable
Revenue CAGR	high	high	heavy right tail	moderate
Margin expansion	moderate	low	very heavy	low
Employment arc growth	moderate	high	bounded	high
Survival indicator	high	high	degenerate bulk	low (imbalanced)

Table 3.1: Candidate success metrics. “Acct. robustness” is robustness to discretionary accounting choices (depreciation schedules, cost capitalization, revenue recognition); “ML target” summarizes definedness, tail control, and informativeness for ranking.

3.3 The formal estimand and its timing

Definition 3.5 (Success label). Fix a metric g from Section 3.2 and horizon h . The label of firm i at base year y is $Y_{i,y} = g(F_{i,y}, F_{i,y+h})$, the metric evaluated over $(y, y+h]$. The *information date* of the pair is $\tau_{i,y}$, the publication date of the year- y filing; a feature vector $Z_{i,\tau_{i,y}}$ is admissible if it satisfies Definition 2.1 at $\tau_{i,y}$.

The predictive estimand of Chapter 4 is $\mathbb{E}[Y_{i,y} \mid Z_{i,\tau_{i,y}}]$; the causal estimand of Chapter 5 replaces conditioning with intervention on a financing exposure, $\mathbb{E}[Y_{i,y} \mid \text{do}(A_{i,y}=a)]$. Both require Y to be a single well-defined scalar attached to a single well-defined date [12]. Four label-engineering pitfalls recur in practice:

1. **Horizon misalignment.** Features dated at fiscal year end y instead of at $\tau_{i,y}$ grant the model 9–12 months of foresight; conversely a label read at $\tau_{i,y+h}$ but indexed as “ h -year growth” actually spans up to $h+1$ years of calendar time. We index everything by τ and record the realized calendar span.
2. **Fiscal-year heterogeneity.** Fiscal year ends vary across firms and jurisdictions; “year y ” is not a common calendar interval. Cross-sectional comparisons and time-based validation splits must use $\tau_{i,y}$, not y .
3. **Consolidation changes.** A firm switching between solo and consolidated accounts (or acquiring a subsidiary) exhibits spurious jumps in every fundamental. We flag consolidation-scope changes from registry metadata and either exclude the spanning label or difference within a constant scope.
4. **Currency.** Fundamentals are filed in local currency. Growth rates of the form (3.2) are currency-invariant only if both endpoints are in the same currency; we compute growth in local currency and never convert at two different dates’ exchange rates.

3.4 Censoring and the survivor trap

A label $Y_{i,y}$ at horizon h requires the year- $(y+h)$ filing. It is missing for two very different reasons: the cohort is young (the calendar has not yet produced the filing — administrative censoring) or the firm exited (insolvency, acquisition, migration out of filing obligations). Simply dropping unlabelled firm-years conditions the training distribution on surviving h more years, which biases every estimate toward the survivor population — precisely the firms for which success is overrepresented, in the same way that conditioning on the risk set must respect who is still observable [7]. The risk-set machinery reappears for event data in Chapter 8; here a simple reweighting suffices.

Let $C_{i,y} = \mathbf{1}\{F_{i,y+h} \text{ observed}\}$ and let $\hat{G}(x_{i,y})$ estimate $\mathbb{P}(C_{i,y} = 1 \mid x_{i,y})$ from pre-base-year covariates $x_{i,y}$ (age, size, sector, filing punctuality), e.g. by logistic regression within calendar cohorts. The inverse-probability-of-censoring (IPC) weighted loss

$$\hat{L}(f) = \frac{1}{n} \sum_{(i,y)} \frac{C_{i,y}}{\hat{G}(x_{i,y})} \ell(Y_{i,y}, f(Z_{i,\tau_{i,y}})) \quad (3.3)$$

is unbiased for the full-population risk when censoring is independent of Y given $x_{i,y}$ — an assumption to state, not to assume silently: exit by insolvency is plainly outcome-dependent. Practically we (i) use (3.3) for administrative censoring, which is mechanically independent of Y given the cohort; (ii) treat insolvency exits not as censored but as *observed failures*, assigning $Y = -2$ (the lower bound of arc growth) with a sensitivity band; and (iii) treat acquisition exits, which are plausibly successes, as a competing outcome reported separately rather than folded into Y . Truncating weights at $\hat{G} \geq 0.1$ keeps the variance of (3.3) controlled.

3.5 Stylized facts to verify on any real panel

Before fitting anything, three distributional facts should be confirmed on the panel at hand; each has a design consequence, and their absence usually indicates a data problem rather than an interesting economy.

1. **Heavy-tailed growth.** Firm growth rates are far from Gaussian: the cross-sectional distribution of (3.2) is sharply peaked with tails well approximated by a Laplace or heavier law (excess kurtosis of 6–15 in our synthetic panel). Consequence: squared error concentrates on the tails; prefer pseudo-Huber or quantile losses and report rank metrics alongside RMSE (Chapter 4).
2. **Mean reversion.** Successive growth rates correlate negatively (lag-1 autocorrelation around -0.2 for EBITDA arc growth): extreme years partially reverse. Consequence: last year’s growth is a strong but *sign-flipping* feature, and naive momentum readings of feature importances mislead; evaluation must be out-of-time, not just out-of-sample.
3. **Size dependence.** Growth dispersion falls with size, roughly $\text{sd}(g) \propto R_y^{-\beta}$ with $\beta \approx 0.15$ (a standard deviation from Gibrat’s law). Consequence: any loss that ignores this heteroskedasticity overweights small firms; stratify evaluation by size decile and consider variance-stabilizing weights.

Table 3.2 records the corresponding moments in the synthetic EU panel; the verification script in `code/` reproduces it, and we recommend running the same four columns on any real panel before the first model is fit.

Metric ($h = 2$)	Skewness	Excess kurtosis	Lag-1 autocorr.	Undefined share
EBITDA arc growth	−0.3	9.8	−0.21	0.0%
EBITDA log-growth	—	—	—	34%
Revenue CAGR	+2.1	14.6	−0.12	6%
Margin expansion	−0.8	27.3	−0.35	6%
Employment arc growth	+0.4	6.2	−0.09	0.0%

Table 3.2: Stylized-fact moments on the synthetic EU panel (winsorization off, all firm-years with both endpoints observed). “Undefined share” is the fraction of firm-years on which the metric does not exist; the log-growth row is blank because its moments are conditional on a selected subsample and hence not comparable.

These facts jointly motivate the loss and cross-validation architecture of Chapter 4: heavy tails pick the loss, mean reversion picks out-of-time splits, size dependence picks the stratification.

3.6 Worked example: labelling a cohort

The definitions above compress into a short routine that is easy to state and easy to get wrong; we run it once, by hand, for a single firm whose EBITDA path crosses zero. Table 3.3 gives the firm’s as-reported figures and registry publication dates; fiscal years end on December 31.

Fiscal year y	FY end	EBITDA E_y (EUR m)	Published $\tau_{i,y}$
2016	2016-12-31	−0.8	2017-10-14
2017	2017-12-31	−0.3	2018-09-30
2018	2018-12-31	+0.5	2019-11-22
2019	2019-12-31	+1.1	2020-10-05
2020	2020-12-31	+1.4	2021-09-18

Table 3.3: Timeline of the worked firm. Publication lags fiscal year end by 9–11 months; the late 2018 filing (almost 11 months) is itself informative (Section 3.1).

The label. Take base year $y = 2017$ and the primary horizon $h = 2$. By Definition 3.5 with arc growth (3.2),

$$Y_{i,2017} = \frac{E_{2019} - E_{2017}}{\frac{1}{2}(|E_{2019}| + |E_{2017}|)} = \frac{1.1 - (-0.3)}{\frac{1}{2}(1.1 + 0.3)} = 2.0, \quad (3.4)$$

the saturated value: the firm crossed into profitability over $(2017, 2019]$, a categorical success in the reading of Example 3.4.

Information-date alignment. The pair is indexed at $\tau_{i,2017}$, i.e. 2018-09-30, the instant the year-2017 figures become public. Admissible features there: the as-reported 2016 and 2017 filings, deal rows with `deal_date` before that date, and macro values published before it. The label needs E_{2019} , public only on 2020-10-05: any training set assembled earlier carries this pair as administratively censored (Section 3.4), and the realized calendar span — information date to label availability — is 24.2 months, recorded alongside the nominal $h = 2$.

Warning 3.6 (The not-yet-published filing). Consider the neighbouring pair, base year 2018. Dating its features at fiscal year end 2018-12-31 — the natural join key in a panel indexed by fiscal year — makes $E_{2018} = +0.5\text{m}$ a feature eleven months before its publication on 2019-11-22. The leaked content is precisely the label’s content: the sign crossing. A backtest learns that “latest EBITDA just turned positive” predicts a saturated label; in production at the end of 2018 the newest public figure is $E_{2017} = -0.3\text{m}$ and the signal does not exist. The fix is mechanical, not statistical: filings join as-of on $\tau_{i,y} \leq t$, never on fiscal year — the exact analogue of joining macro series on `publish_date` in Chapter 2 — and the staleness of the newest published filing enters as a feature, so the model sees lateness instead of the future.

The cohort routine is the by-hand computation industrialized: for every firm and every base year whose filing is published, emit the pair at $\tau_{i,y}$; attach the label when the year- $(y+h)$ filing publishes; score insolvency exits at $Y = -2$ and route acquisition exits to the competing-outcome tally (Section 3.4); winsorize realized labels within training folds only. Every row of the resulting table carries $(i, y, \tau_{i,y}, Y, C)$, and the validation splits of Chapter 4 operate on τ , never on y .

3.6.1 Robustness of the label choice

The worked firm makes plain that the label is a construction, so the natural question before committing is how much rides on it. Table 3.4 compares the primary label with three alternatives on the synthetic panel along three stability criteria: Spearman rank correlation of firm rankings with the primary label; sensitivity of top-decile membership to moving the winsorization quantiles from 1%/99% to 0.5%/99.5%; and tail behaviour of the realized target.

Label ($h = 2$)	Rank corr.	Winsor. sensitivity	Tail behaviour
EBITDA arc growth (primary)	1.00	low	bounded, mass at ± 2
asinh-difference of EBITDA	0.86	moderate	unbounded, moderate tails
Revenue CAGR	0.55	high	heavy right tail
Employment arc growth	0.48	low	bounded, thin tails

Table 3.4: Stability of the label choice on the synthetic panel. “Rank corr.” is the Spearman correlation of firm rankings with the primary label; “winsor. sensitivity” is the extent to which top-decile membership changes when the winsorization quantiles move from 1%/99% to 0.5%/99.5%.

The pattern to expect on any real panel is the same: the asinh alternative reorders little (it is a monotone reshaping of the same fundamental), while revenue- and employment-based labels agree with the primary only moderately — they measure different economics, not noisy copies of one number. Rank correlations near 0.5 mean that model comparisons, feature importances, and top-decile readings can genuinely flip across definitions. Standard practice in this manual is therefore to re-run the headline models of Chapters 4 and 5 under one pre-declared alternative — employment arc growth, chosen as the most accounting-immune and the most distant — and to report both. A conclusion that survives both labels is about firms; one that does not is about the metric. Declaring the single alternative before results are seen keeps this a robustness check rather than metric shopping, in the same spirit as controlling a family of tests instead of reporting the best one [15].

3.7 Decision: the primary metric

Decision. The primary success metric of Part I is *EBITDA arc growth at horizon $h = 2$* , winsorized at the within-fold 1%/99% quantiles, with insolvency exits scored at -2 and acquisition exits reported as a competing outcome. Revenue CAGR and employment arc growth are secondary diagnostics; no composite index is used.

Why this and not a composite. First, $h = 2$ balances signal against censoring: at $h = 1$ the label is dominated by accounting noise and mean reversion (Section 3.5); at $h = 3$ the administratively censored share of recent cohorts exceeds 40% and the IPC weights of (3.3) get large. Second, EBITDA rather than revenue keeps the metric sensitive to the margin dimension investors are ultimately paid on, while the arc transform neutralizes the sign-definedness problem that rules out log-growth. Third — and decisively for Chapter 5 — a composite index $Y^{\text{comp}} = \sum_j w_j Y_j$ degrades identifiability of the causal estimand: each component has its own confounding structure, so a backdoor set valid for the composite need not exist even when one exists per component, treatment effects of opposite sign on components cancel uninterpretably, and the weights w_j become untestable modelling choices sitting inside the estimand itself. One outcome, one horizon, one information date: everything downstream inherits its identifiability from this choice.

Chapter 4

Prediction with Gradient Boosting

In the European regime the target of Chapter 3 — winsorized EBITDA arc-growth over a two-year horizon — is observable for essentially every firm-year, so the first modelling layer is plain supervised learning. This chapter fixes the learning problem, the feature catalogue, the validation design, and — the part practitioners most often skip — the uncertainty quantification. The guiding principle throughout is that firm-year panels are *not* i.i.d. samples: rows recur within firms and cluster within macro eras, and every design choice below (splits, calibration sets, bootstrap units) follows from taking that dependence seriously. What the fitted model *means* — association, not effect — is deferred to Chapter 5.

4.1 The learning problem

A row is a firm-year (i, y) . The label is $Y_{i,y} = w_c(G_{i,y}^{(h)})$, the arc-growth of EBITDA between statement years y and $y + h$ with $h = 2$, winsorized at level c exactly as defined in Chapter 3. The feature vector $X_{i,y} \in \mathbb{R}^p$ is assembled *point-in-time*: it may use only filings published and macro prints released by the information date $t_{\text{info}}(i, y)$ of the row, respecting the publication-lag discipline of Chapter 2. We estimate two objects:

$$m(x) = \mathbb{E}[Y \mid X = x], \quad q_\tau(x) = \inf\{a : \mathbb{P}(Y \leq a \mid X = x) \geq \tau\}, \quad (4.1)$$

the conditional mean (point screening) and a small grid of conditional quantiles. We use $\tau \in \{0.05, 0.10, 0.50, 0.90, 0.95\}$. The quantiles are not decoration: the outcome is heavy-tailed even after winsorization, upside screening acts on $q_{0.9}$ rather than the mean, downside risk control acts on $q_{0.1}$, and the pair $(q_{0.05}, q_{0.95})$ is the raw material for the conformal intervals of Section 4.5.

Two dependence structures dominate. *Within-firm*: rows (i, y) and (i, y') share persistent firm effects (management quality, niche economics), so errors are serially correlated within i . *Within-era*: rows of a given year y share the macro shock, so errors are cross-sectionally correlated within y . The first dictates firm-grouped resampling; the second dictates era-blocked validation. Neither is optional.

4.2 Feature catalogue

Table 4.1 lists the feature families. All are computable from the fundamentals panel $F_{i,y}$, the deals table where coverage exists, and the macro feed; all obey the as-of rule.

Missingness. Fundamentals gaps are informative: small companies file abbreviated accounts, distressed companies file late, and the propensity to be missing depends on the (unobserved)

Table 4.1: Feature catalogue for the EU prediction layer. All features are as-of $t_{\text{info}}(i, y)$; macro features use publication-lagged vintages (Chapter 2).

Family	Examples	Construction
Lagged levels	EBITDA, revenue, payroll, assets at $y, y-1, y-2$	$\log(1 + \cdot)$ where positive; raw for signed EBITDA
Growth rates	1–3y arc-growth of revenue, EBITDA, payroll	same arc transform as the target, unwinsorized
Ratios	EBITDA margin, leverage, asset turnover, payroll share	clipped at 1st/99th pctile per sector-year
Peer position	within-sector-year rank and z-score of margin, growth, size	computed per (s, y) cell, rank in $[0, 1]$
Sector-year aggregates	median growth, IQR, firm count	cell-level, merged back
Firm statics	age at y , size bucket, country	age in years, log assets bucket
Macro as-of	policy rate level and 12m change, CPI YoY, unemployment level/change	vintage at t_{info} ; level <i>and</i> change
Deal history	raised-to-date, round count, months since last round	zero-filled with indicator where firm never in deals

value — textbook MNAR. We therefore do *not* mean-impute: imputation overwrites exactly the signal carried by absence and biases every downstream split. Instead we rely on the native sparsity handling of gradient boosting, in which each split learns a default direction for missing values from the data [9], and we add explicit indicators (`ebitda_missing_lag1`, `never_in_deals`) for gaps whose economic meaning we can articulate. Trees are then free to interact the indicator with size and sector — “missing EBITDA at small size” is a different animal from “missing EBITDA at large size”, and the model should be allowed to say so.

4.3 Why boosted trees here — and what they cannot do

The honest case for gradient-boosted trees on this problem is fourfold: (i) the signal is riddled with interactions (margin $\times$ sector, growth $\times$ rate regime) that we cannot enumerate a priori and trees discover automatically; (ii) the features are mixed continuous/ordinal/categorical with wildly different scales, and trees are invariant to monotone transforms of each feature, so we spend no effort on marginal calibration; (iii) native missingness handling, as above; (iv) strong, well-understood regularization (depth, shrinkage, subsampling, ℓ_2 on leaf weights) [9].

The honest case against has two entries, and both matter here. First, *no extrapolation*: a tree ensemble is piecewise constant, so its prediction outside the convex hull of the training features is the value at the nearest boundary cell. If every training era has falling policy rates, the fitted rate response is a flat extension — the model literally cannot form an opinion about a rising-rate regime it has never seen. Second, *unstable attribution*: correlated features split importance unpredictably across refits, so feature rankings are not portable conclusions.

We mitigate rather than solve. Era-aware validation (Section 4.4) at least measures degradation across regimes instead of hiding it. Monotone constraints encode signs we are willing to defend economically — e.g. growth non-increasing in the 12-month rate *change*, downside quantile non-increasing in leverage — which reduces variance in thin regions and prevents sign flips there.

Remark 4.1 (Constraints are priors, not facts). A monotone constraint changes the model class, not the world. Impose one only where the sign survives scrutiny (rates *level* is ambiguous: it proxies both financing cost and the demand cycle; the *change* is the cleaner candidate). A wrong constraint is worse than none, because it converts local noise into global bias silently.

For the conditional-mean model, heavy tails argue against squared error even after winsorization. We use the pseudo-Huber objective

$$\ell_\delta(r) = \delta^2 \left(\sqrt{1 + (r/\delta)^2} - 1 \right), \quad (4.2)$$

which is quadratic for $|r| \ll \delta$ and linear for $|r| \gg \delta$, with δ set to roughly one cross-sectional standard deviation of Y . Note the fitted functional then sits between conditional mean and median; we report which loss produced which model and never mix them in evaluation.

4.4 Validation design

Never random K-fold. Random row-level K-fold fails three ways on this panel. (i) *Firm leakage*: rows of the same firm land on both sides of the split, and the persistent firm effect makes validation loss optimistic — the model is partly memorizing firms. (ii) *Target overlap*: with $h = 2$, the labels of years y and $y + 1$ share a statement year, so adjacent rows carry overlapping information about each other’s targets. (iii) *Era leakage*: the model trains on the future macro regime it will be asked to predict, which no deployed model enjoys. All three inflate measured skill precisely where the deployment risk is largest.

Rolling-origin splits. We order eras (label years) $y_1 < \dots < y_E$ by information date and build folds $k = 1, \dots, K$: train on eras up to y_{e_k} , embargo the next h eras (their labels overlap the training window), validate on the era after the embargo. Every fold mimics deployment: fit on the past, predict a disjoint future. The same firm may appear in train (early years) and validation (later years) — that is legitimate and realistic; the leakage to prevent is label overlap, handled by the embargo, not firm recurrence across time.

Hyperparameter protocol. Random search (100–200 draws) over the ranges in Table 4.2; each draw is scored by its *average validation loss across all rolling folds*, with the number of boosting rounds chosen by early stopping on the *most recent* fold only — recency-weighting the one knob that matters most. The final model refits on all data with the tuned configuration and the early-stopped round count scaled by the ratio of full to training rows. Tuning on an average across eras deliberately trades a little peak skill for regime stability.

Table 4.2: Search ranges. Depth is kept shallow: firm-year panels reward many weak interactions over few deep ones.

Parameter	Range	Note
max depth	3–6	log-uniform integer
learning rate	0.01–0.1	log-uniform
min child weight	5–100	guards thin sector cells
row subsample	0.6–0.9	per boosting round
column subsample	0.5–0.9	decorrelates trees
ℓ_2 leaf penalty	0.5–20	log-uniform
rounds	early stopping	patience 50, most recent fold

4.5 Uncertainty quantification

Three tools, three targets: per-firm heteroskedastic intervals (quantile regression), finite-sample coverage repair (conformal), and model-level uncertainty of scalar summaries (cluster bootstrap). Table 4.3 states when each is the right instrument.

4.5.1 Quantile regression with the pinball loss

For $\tau \in (0, 1)$ the pinball (check) loss is

$$\rho_\tau(u) = u(\tau - \mathbf{1}\{u < 0\}) = \tau u_+ + (1 - \tau) u_-, \quad (4.3)$$

with $u_+ = \max(u, 0)$, $u_- = \max(-u, 0)$. Its population minimizer is exactly the quantile we want:

Proposition 4.2 (The pinball minimizer is the τ -quantile). *Let Y have distribution function F and finite first moment, and let $\varphi(a) = \mathbb{E}[\rho_\tau(Y - a)]$. Then any a^* with $F(a^{*-}) \leq \tau \leq F(a^*)$ minimizes φ ; if F is continuous and strictly increasing, $a^* = F^{-1}(\tau)$ uniquely.*

Proof. φ is convex, and for a a continuity point of F , $\varphi'(a) = -\tau\mathbb{P}(Y > a) + (1 - \tau)\mathbb{P}(Y < a) = F(a) - \tau$. A convex function is minimized where its (sub)derivative changes sign, i.e. where $F(a^-) - \tau \leq 0 \leq F(a) - \tau$. $\square$

We fit one boosted model per τ with objective (4.3) applied conditionally. Separately fitted quantiles can cross in finite samples; we repair by sorting the predicted grid per row (monotone rearrangement), which never worsens pinball loss.

4.5.2 Conformalized quantile regression

Quantile models are calibrated only insofar as the model class and sample size allow; conformal prediction repairs marginal coverage with finite-sample guarantees [15].

Algorithm (split CQR).

1. Split the available rows into a proper training set I_1 and a calibration set I_2 of size n_2 , with the split at the firm level and I_2 chosen as a *block*: the most recent complete era(s) before the deployment date.
2. On I_1 , fit lower and upper quantile models $\hat{q}_{\alpha/2}$, $\hat{q}_{1-\alpha/2}$ as above.
3. For each $j \in I_2$ compute the conformity score $E_j = \max\{\hat{q}_{\alpha/2}(X_j) - Y_j, Y_j - \hat{q}_{1-\alpha/2}(X_j)\}$.
4. Let $Q_{1-\alpha}$ be the $[(n_2 + 1)(1 - \alpha)]$ -th smallest of $\{E_j\}_{j \in I_2}$.
5. Report the interval $C(x) = [\hat{q}_{\alpha/2}(x) - Q_{1-\alpha}, \hat{q}_{1-\alpha/2}(x) + Q_{1-\alpha}]$.

Theorem 4.3 (CQR marginal coverage [15]). *If the calibration pairs $\{(X_j, Y_j)\}_{j \in I_2}$ and the test pair (X, Y) are exchangeable, then $\mathbb{P}\{Y \in C(X)\} \geq 1 - \alpha$; if the scores are almost surely distinct, also $\mathbb{P}\{Y \in C(X)\} \leq 1 - \alpha + 1/(n_2 + 1)$.*

Warning 4.4 (Temporal data are not exchangeable). Theorem 4.3 assumes the test row is exchangeable with the calibration rows. A macro-era shift between calibration block and deployment breaks this, and no finite-sample guarantee survives. Practice: calibrate on the most recent contiguous block (as above), *treat the nominal level as approximate*, measure empirical coverage per era on every historical fold, and recalibrate at every model vintage. If per-era coverage drifts systematically with the macro state, widen intervals by the worst observed era's deficit rather than trusting the nominal α .

4.5.3 Cluster bootstrap over firms

Quantile and conformal intervals describe *per-row* outcome uncertainty. A different question — how uncertain is a scalar summary of the fitted model, such as an era MAE, a decile-spread statistic, or the skill gap over a benchmark — requires resampling the fitting process itself [14].

Algorithm (firm-level percentile bootstrap).

1. Let $\mathcal{F}$ be the set of firms. For $b = 1, \dots, B$ (we use $B \in [200, 500]$): draw $|\mathcal{F}|$ firms from $\mathcal{F}$ with replacement and assemble the replicate panel from *all rows* of each drawn firm (duplicates kept).
2. Re-run the frozen pipeline on the replicate: feature construction, the tuned hyperparameter configuration (not re-searched), fit, and compute the scalar summary $\hat{\theta}_b$ on the replicate’s validation eras.
3. Report the percentile interval $[\hat{\theta}_{(\lceil \alpha B/2 \rceil)}, \hat{\theta}_{(\lceil (1-\alpha/2)B \rceil)}]$.

Resampling firms, not rows, is the load-bearing choice: within-firm correlation means the effective sample size is closer to the number of firms than the number of rows, and a row-level bootstrap understates variance accordingly.

Table 4.3: Choosing the uncertainty tool.

Tool	Quantifies	Guarantee	Cost
Quantile regression	per-row outcome spread	none per se	1 fit per τ
Split CQR	per-row interval coverage	finite-sample marginal, approximate under drift	1 calibration pass
Firm bootstrap	model-level scalars (MAE, skill gaps)	asymptotic, within the observed design	B refits

4.6 Evaluation protocol

We report MAE and R^2 *by era* and by horizon, never pooled alone: a pooled MAE averages over exactly the regime heterogeneity the model is weakest on. The median model (pinball at $\tau = 0.5$) is evaluated with MAE — matched loss — and the pseudo-Huber mean model with both MAE and RMSE. Quantile calibration is checked per era with reliability diagrams: for each τ , the empirical fraction of outcomes below $\hat{q}_\tau(X)$ plotted against τ ; a well-calibrated model tracks the diagonal, a regime-sensitive one shows era-dependent slope. Subgroup stability: per-cell MAE by sector and size tercile relative to pooled MAE, flagging cells where a benchmark of Section 4.7 beats the boosted model — such cells are candidates for exclusion from deployment, not for silent averaging.

Warning 4.5 (Attributions are associational). SHAP-style attributions decompose the *fitted* $\hat{m}$, i.e. they redistribute predicted differences among correlated inputs of an associational object. A large attribution on `months_since_last_round` does not mean financing drives growth; it means firms that recently raised look different, which is precisely the selection effect Chapter 5 is built to separate from the causal one. Use attributions for debugging (leakage detection, sanity of signs) and never in an investment memo as an effect size.

4.7 Benchmarks and when not to use gradient boosting

Every fold carries three benchmarks: *persistence* (predict the firm’s last observed growth), *sector-year mean* (predict the cell average), and *ridge* regression on the identical feature matrix (with the missingness indicators, since ridge has no native sparsity handling). The shipping rule is strict: the boosted model is deployed only where its out-of-era skill gap over ridge exceeds the half-width of the firm-bootstrap interval on that gap. A gap inside the interval is noise, and ridge is cheaper, more stable, and partially extrapolates.

Table 4.4: When gradient boosting is the wrong tool.

Situation	Symptom	Use instead
Tiny sample	$\lesssim 5\text{k}$ firm-years	ridge on curated features
Regime extrapolation	deployment macro outside training support	linear/constrained macro term + trees on the rest
Regulatory interpretability	auditable decision rule required	scorecard or shallow monotone model
Skill gap within noise	bootstrap CI covers zero	ridge; revisit features

The model of this chapter answers “which firms will grow?” conditional on everything observable. It is silent on “what happens if a firm raises?” — the fitted dependence on deal-history features mixes selection with effect. Chapter 5 takes up that question with the same data and an explicitly causal contract.

Chapter 5

A Causal Layer: Does Financing Cause Growth?

Chapter 4 predicts. Managers ask a different question: *if* a company raises a round, does growth follow — or do growing companies simply raise? The fitted $\hat{m}(x)$ cannot answer this, because the coefficientless dependence it learns on deal-history features mixes the effect of financing with the selection of good firms into financing. This chapter builds the minimal causal layer the European regime supports. The regime is favourable for exactly one reason: the statutory fundamentals panel gives us a *rich, audited, pre-treatment* covariate history — close to the dossier an investor actually screens on — so adjustment has a fighting chance. Every estimate ships with cluster-bootstrap uncertainty, an E-value, and two falsification tests, and the whole pipeline is validated on the synthetic testbed of Chapter 10, where the true effect is known by construction.

5.1 Estimands and assumptions

Treatment: $A_{i,y} = \mathbf{1}\{\text{firm } i \text{ closed a PE/VC equity round in year } y\}$, from the deals table. Outcome: $Y_{i,y}$, the $h = 2$ winsorized EBITDA arc-growth of Chapter 3, measured over $(y, y+h]$ so it is strictly post-treatment. Each unit carries potential outcomes $Y_{i,y}(1)$ and $Y_{i,y}(0)$: growth with and without a round in year y .

Assumption 5.1 (SUTVA). (i) No interference: $Y_{i,y}(a)$ does not depend on other firms’ treatments; (ii) single version of treatment: $A_{i,y} = 1$ denotes one well-defined intervention [11].

Assumption 5.2 (Consistency). $Y_{i,y} = A_{i,y}Y_{i,y}(1) + (1 - A_{i,y})Y_{i,y}(0)$ [12].

Both parts of Assumption 5.1 are approximations here: a round can hurt a direct competitor’s growth (interference; we flag narrow niches), and rounds differ in size and investor (versions; we deliberately coarsen to “raised institutional equity” and treat dose–response as an extension, not the estimand).

The estimands, in decreasing ambition of the conditioning:

$$\tau_{\text{ATE}} = \mathbb{E}[Y(1) - Y(0)], \quad \tau_{\text{ATT}} = \mathbb{E}[Y(1) - Y(0) \mid A = 1], \quad \tau(x) = \mathbb{E}[Y(1) - Y(0) \mid X = x]. \quad (5.1)$$

For the portfolio question (“did our financings work?”) the ATT is the object; for targeting (“where would a round move the needle?”) the CATE $\tau(x)$; the ATE is the reporting default because it is the easiest to estimate stably. Identification requires:

Assumption 5.3 (Conditional exchangeability and positivity). There exists an observed pre-treatment set Z with $(Y(0), Y(1)) \perp\!\!\!\perp A \mid Z$ and $0 < e(z) = \mathbb{P}(A = 1 \mid Z = z) < 1$ for all z in the support [11, 12].

Assumption 5.3 is not a modelling choice; it is a claim about the world that the next section argues from a graph, Section 5.6 stress-tests, and Section 5.7 verifies where the truth is known.

5.2 The causal graph

Five variables. U : latent firm quality (management, product, moat) — unobserved. X : the observed pre-treatment fundamentals history (levels, growth path, margins, leverage, payroll over the three years up to y), plus sector and age. M : the macro state as of year y (rates, CPI YoY, unemployment, as-of vintages). A : the treatment. Y : the outcome. The edges: $U \rightarrow A$ (investors select quality), $U \rightarrow Y$ (quality drives growth), $U \rightarrow X$ (quality leaves tracks in the filings), $X \rightarrow A$ (investors screen on the observable dossier), $X \rightarrow Y$ (fundamentals persist), $M \rightarrow A$ (funding cycles), $M \rightarrow Y$ (macro drives growth), and the causal edge of interest $A \rightarrow Y$. Table 5.1 enumerates every A – Y path.

Table 5.1: Paths between A and Y in the working graph.

Path	Type	Status after conditioning on (X, M)
$A \rightarrow Y$	causal	kept (the estimand)
$A \leftarrow X \rightarrow Y$	backdoor	blocked by X
$A \leftarrow M \rightarrow Y$	backdoor	blocked by M (sector-year cells)
$A \leftarrow U \rightarrow Y$	backdoor	<i>not</i> blockable directly; attenuated via the proxy path $U \rightarrow X$
$A \leftarrow U \rightarrow X \rightarrow Y$	backdoor	blocked by X

The whole argument of the EU track sits in the fourth row. U is unobserved, so the path $A \leftarrow U \rightarrow Y$ cannot be blocked exactly. But investors do not observe U either: they observe the dossier, which is essentially X — audited statements, growth trajectory, margins, leverage. To the extent that the selection $U \rightarrow A$ flows *through* what is observable about quality, conditioning on a rich X intercepts most of it, and the residual confounding is the part of quality that moves investors without leaving any track in three years of filings. In the European regime that residual is plausibly small and, crucially, *boundable* (Section 5.6). In the US regime there is no fundamentals panel, X collapses to deal marks and coarse firmographics, and this argument dies — which is why Chapters 6 and onward change the estimand rather than pretend the backdoor is closed.

5.3 Identification: the backdoor criterion

Definition 5.4 (Backdoor criterion [10]). A set of observed variables Z satisfies the backdoor criterion relative to (A, Y) if (i) no element of Z is a descendant of A , and (ii) Z blocks every path between A and Y that contains an arrow into A .

Theorem 5.5 (Adjustment formula [10]). *If Z satisfies the backdoor criterion relative to (A, Y) , then*

$$\mathbb{E}[Y \mid \text{do}(A = a)] = \mathbb{E}_Z[\mathbb{E}[Y \mid A = a, Z]], \quad (5.2)$$

the outer expectation over the marginal law of Z , and consequently $\tau_{\text{ATE}} = \mathbb{E}_Z[\mathbb{E}[Y \mid A = 1, Z] - \mathbb{E}[Y \mid A = 0, Z]]$.

In the working graph, $Z = (X, M)$ satisfies the criterion *up to* the residual U -path discussed above: formally we proceed under Assumption 5.3 with $Z = (X \text{ history over 3 years, sector-year cell})$ — the cells absorb M and sector-level funding cycles — and quantify the violation rather than assume it away.

Warning 5.6 (Bad controls). More conditioning is not more safety. Three failure modes, all present in this dataset. (i) *Mediators*: post-treatment employee counts sit on the path $A \rightarrow \text{hiring} \rightarrow Y$; conditioning on them subtracts part of the effect being estimated. (ii) *Colliders/descendants*: a subsequent valuation mark is caused both by the round (A) and by quality (U); conditioning on it opens the path $A \rightarrow \text{mark} \leftarrow U \rightarrow Y$ and manufactures spurious association. (iii) *M-bias*: even a *pre-treatment* variable can hurt — if some early covariate is a common effect of one latent cause that drives selection and another latent cause that drives growth, then conditioning on it opens a previously closed path between those latents. The rule is graphical, not chronological: the adjustment set is chosen by Definition 5.4, never by predictive power, and nothing dated after the round enters Z .

5.4 Estimators, in increasing robustness

Throughout, nuisance functions are fitted with the boosted-tree machinery of Chapter 4 (same features, same era discipline), written $\mu_a(z) = \mathbb{E}[Y \mid A = a, Z = z]$ and $e(z) = \mathbb{P}(A = 1 \mid Z = z)$.

(1) Outcome regression (g-formula). Fit $\hat{\mu}_1, \hat{\mu}_0$ and average: $\hat{\tau}_{\text{OR}} = n^{-1} \sum_{i,y} \{\hat{\mu}_1(Z_{i,y}) - \hat{\mu}_0(Z_{i,y})\}$. Two flavours: the T-learner (separate models per arm; honest but noisy in the small treated arm) and the S-learner (A as one feature among p ; stable but regularization can shrink the A -splits, biasing $\hat{\tau}$ toward zero). Both inherit every extrapolation weakness of trees precisely in low-overlap regions, where $\hat{\mu}_1$ is evaluated at control-like z and vice versa.

(2) Inverse propensity weighting. Fit $\hat{e}$, then the Hajek form

$$\hat{\tau}_{\text{IPW}} = \frac{\sum_i A_i Y_i w_i^1}{\sum_i A_i w_i^1} - \frac{\sum_i (1 - A_i) Y_i w_i^0}{\sum_i (1 - A_i) w_i^0}, \quad w^1 = \frac{1}{\hat{e}(Z)}, \quad w^0 = \frac{1}{1 - \hat{e}(Z)}.$$

Mandatory diagnostics before any estimate is read: the histogram of $\hat{e}$ by treatment arm (overlap plot); trimming to $\hat{e} \in [0.02, 0.98]$ with the trimmed fraction reported — if it exceeds a few percent, the estimand has silently changed to a trimmed-population effect and we say so; the effective sample size $\text{ESS} = (\sum w)^2 / \sum w^2$ per arm. IPW uses the treatment model only, so it fails independently of (1) — but its variance explodes near the trimming boundary.

(3) AIPW with cross-fitting (the default). The doubly robust score combines both nuisances [13]:

$$\psi(O; \mu, e) = \mu_1(Z) - \mu_0(Z) + \frac{A(Y - \mu_1(Z))}{e(Z)} - \frac{(1 - A)(Y - \mu_0(Z))}{1 - e(Z)}, \quad \hat{\tau}_{\text{AIPW}} = \frac{1}{n} \sum \psi_i. \quad (5.3)$$

Proposition 5.7 (Double robustness). *Under Assumptions 5.2–5.3, $\mathbb{E}[\psi(O; \mu, e)] = \tau_{\text{ATE}}$ if either $\mu = (\mu_1^*, \mu_0^*)$ is the true outcome function or $e = e^*$ is the true propensity (with the other nuisance arbitrary, subject to positivity).*

Proof. Consider the $a = 1$ half, $\psi_1 = \mu_1(Z) + A(Y - \mu_1(Z))/e(Z)$. If $\mu_1 = \mu_1^*$: $\mathbb{E}[A(Y - \mu_1^*(Z)) \mid Z] = e^*(Z) \mathbb{E}[Y - \mu_1^*(Z) \mid A = 1, Z] = 0$, so $\mathbb{E}[\psi_1] = \mathbb{E}[\mu_1^*(Z)] = \mathbb{E}[Y(1)]$ by Assumption 5.3. If $e = e^*$: $\mathbb{E}[\psi_1 \mid Z] = \mu_1(Z) + \{e^*(Z) \mathbb{E}[Y \mid A = 1, Z] - e^*(Z) \mu_1(Z)\}/e^*(Z) = \mathbb{E}[Y \mid A = 1, Z] = \mathbb{E}[Y(1) \mid Z]$, and the arbitrary μ_1 cancels. The $a = 0$ half is symmetric; subtract. $\square$

Remark 5.8 (Rate condition). Beyond consistency, valid $\sqrt{n}$ inference informally requires the product of nuisance errors to vanish fast: $\|\hat{\mu}_a - \mu_a^*\|_2 \cdot \|\hat{e} - e^*\|_2 = o_P(n^{-1/2})$ [13]. Each nuisance

may converge slowly (as boosted trees do) as long as the product is fast — this, plus the removal of own-observation overfitting bias by cross-fitting, is why (3) is the default and not merely a hedge.

Algorithm (cross-fitted AIPW).

1. Partition *firms* into $K = 5$ folds; a firm’s rows never straddle folds (within-firm correlation would otherwise leak nuisance fits into own scores).
2. For each fold k : fit $\hat{\mu}_1^{(-k)}, \hat{\mu}_0^{(-k)}, \hat{e}^{(-k)}$ on the other $K - 1$ folds, with the era-aware tuning of Chapter 4 frozen once, outside the loop.
3. Trim: drop rows of fold k with $\hat{e}^{(-k)} \notin [0.02, 0.98]$; record the trimmed fraction.
4. Compute $\psi_{i,y}$ for the surviving rows of fold k via (5.3) using only out-of-fold nuisances.
5. Pool: $\hat{\tau} = \text{mean of all } \psi_{i,y}$; variance by Section 5.5; report alongside the overlap plot and trimmed fraction.

5.5 Inference under firm clustering

Rows recur within firms across years, and the scores $\psi_{i,y}$ of a firm are correlated through persistent firm effects. An i.i.d. bootstrap (or the naive variance $n^{-2} \sum (\psi_{i,y} - \hat{\tau})^2$) treats n rows as n independent draws and understates the variance — the effective sample size is nearer the number of firms. Two consistent routes; we run both and report the wider.

(a) Cluster-robust influence-function variance. Let $S_i = \sum_y (\psi_{i,y} - \hat{\tau})$ sum a firm’s centred scores. Then

$$\widehat{\text{Var}}(\hat{\tau}) = \frac{1}{n^2} \sum_{i \in \mathcal{F}} S_i^2, \quad (5.4)$$

which is the sandwich variance with firms as clusters; normal intervals $\hat{\tau} \pm z_{1-\alpha/2} \widehat{\text{Var}}^{1/2}$.

(b) Cluster bootstrap [14].

1. For $b = 1, \dots, B$ (200–500): resample firms with replacement; a drawn firm contributes all its rows.
2. Re-run the *entire* cross-fitting algorithm on the replicate — nuisance refits included, tuning still frozen — to obtain $\hat{\tau}_b$.
3. Report the percentile interval of $\{\hat{\tau}_b\}$.

The bootstrap additionally propagates nuisance-estimation noise that (5.4) treats as fixed; disagreement between (a) and (b) beyond $\sim 20\%$ of interval width is itself a diagnostic of nuisance instability.

5.6 Sensitivity and falsification

Assumption 5.3 cannot be verified from the data it licenses. It can be stress-tested. Every reported effect must clear the checklist of Table 5.2; an estimate that fails any row does not ship.

E-values. The E-value [16] is the minimum strength of association, on the risk ratio scale, that an unmeasured confounder would need with *both* treatment and outcome, above and beyond Z , to explain away the observed association. For an observed risk ratio $RR \geq 1$,

$$E = RR + \sqrt{RR(RR - 1)}. \quad (5.5)$$

Our outcome is continuous, so we standardize: with effect $d = \hat{\tau}/\hat{\sigma}_Y$ (cluster-bootstrap CI carried through), the conventional conversion $RR \approx \exp(0.91 d)$ feeds (5.5), and the E-value of the CI limit closer to the null is reported alongside that of the point estimate. An E-value of 1.8 reads: a latent-quality confounder must nearly double both the odds of raising and of high growth, conditional on the full fundamentals history, to explain the estimate away — a statement a domain expert can argue with, which is the point.

Placebo treatment. Define $A_{i,y}^{\text{fut}} = \mathbf{1}\{\text{round in year } y + 3\}$ and estimate its “effect” on $Y_{i,y}$ (which is fully realized before the placebo round) with the identical AIPW pipeline. Under Assumption 5.3 the estimate must be null: a future round cannot cause past growth, so any nonzero adjusted estimate measures exactly the selection-on-trends that Z failed to absorb.

Negative-control outcome. Symmetrically, estimate the “effect” of $A_{i,y}$ on *pre-treatment* growth (years $y - 2$ to y , excluded from Z for this test only). Nonzero means the treated were already on a different trajectory in a way Z does not capture.

Table 5.2: Mandatory pre-shipping checklist for any causal estimate.

Check	Pass criterion	On failure
Overlap plot	common support, trimmed fraction $< 5\%$	redefine estimand (ATT, trimmed population)
Placebo (future round)	CI covers 0	enrich Z (longer history, trend features); else do not ship
Negative control (pre-growth)	CI covers 0	same as placebo
E-value	$>$ plausible latent-quality association given rich X	report as bounded, not as an effect
Variance agreement	IF vs bootstrap within 20%	investigate nuisance instability
Synthetic recovery	Table 5.3 ranking reproduced	pipeline bug; fix before touching real data

5.7 Synthetic validation protocol

The testbed of Chapter 10 generates a world with a known latent quality q_i driving both funding intensity and fundamentals, so the true τ is known by construction and confounding by quality is present by design. The causal layer is accepted only if it passes the following protocol there.

1. *Generate* an EU-style panel: fundamentals as noisy, lagged functions of q_i and macro; treatment assignment with probability increasing in q_i and in the observable dossier; outcome with a known additive effect τ^* of treatment.
2. *Estimate* with four pipelines: naive difference in means; g-formula (T-learner); IPW; cross-fitted AIPW — each with firm-cluster bootstrap CIs and E-values.

3. *Verify* the qualitative ranking of Table 5.3: the naive contrast overstates τ^* badly (quality confounding); all adjusted estimators land near τ^* when the generated X is rich; ablating X columns (hiding the dossier, i.e. simulating the US regime) re-opens the gap, with AIPW degrading most gracefully.
4. *Report* empirical CI coverage of τ^* across Monte Carlo replications and check that the naive estimator’s E-value correctly flags fragility while the adjusted estimators’ E-values are informative about the injected residual confounding.

Table 5.3: Expected qualitative behaviour on the synthetic testbed (Chapter 10). “ $\approx$ ” means CI covers τ^* at nominal rate.

Estimator	Rich X (EU-style)	Impoverished X (US-style)
Naive difference	$\gg \tau^*$	$\gg \tau^*$
g-formula (XGB)	$\approx \tau^*$	biased toward naive
IPW (trimmed)	$\approx \tau^*$, wider CI	biased, unstable
AIPW cross-fitted	$\approx \tau^*$, correct coverage	residual bias, flagged by placebo and E-value

This protocol is what makes the chapter testable rather than decorative: the code that produces Table 5.3 is byte-identical to the code run on the real panel, so a pipeline that cannot recover a known effect is caught before it is believed.

5.8 Limits

Two limits are structural and stated plainly. First, *selection on unobservables is not solved* — it is attenuated by a rich X , bounded by E-values, and probed by placebo and negative-control tests, but a confounder invisible to three years of audited filings and strong enough to survive the checklist remains logically possible. Our claim is calibrated accordingly: adjusted estimates with reported sensitivity, not oracle effects. Second, *external validity across macro regimes: M enters both selection and outcome*, all estimates average over the funding cycles in sample, and a CATE-by-era analysis is data-hungry precisely because treated firms are scarce within any single era. An effect estimated across 2015–2021 need not transport to a tight-money regime, for the same no-extrapolation reasons as in Chapter 4. The US regime sharpens both limits at once: with no fundamentals panel there is no rich X , and the honest response is to change the estimand and the observation model — the programme of Chapters 6 and 10.

Part II

The Opaque Regime: Deals and Macro Factors (United States)

Chapter 6

The Opaque Regime: Deal Marks and Macro Factors

Part I assumed that fundamentals arrive on a statutory clock: every firm, every fiscal year, a filed income statement. In the United States that clock does not exist. What the vendor delivers instead is the firm’s *transaction record*: a sequence of deal events, each carrying marks — a valuation, a round classification, an employee count, occasionally a revenue or EBITDA figure. Between events we learn nothing. This chapter re-derives the Part I pipeline under that observation model. Three things change: the *target* (EBITDA growth must be replaced by proxies constructed from marks), the *unit of analysis* (the firm-year panel row is replaced by the consecutive-deal pair), and — most consequentially — the *bias structure*: outcomes are observed only when the firm transacts again, so every naive estimate is filtered through survival-to-next-round. The repair for that selection is a model of the deal-arrival process itself, which is why Part III is not an appendix to this chapter but its completion.

6.1 The observation model

Let $S_{i,t} \in \mathbb{R}^d$ denote the latent state of firm i : fundamentals (revenue, margin trajectory, unit economics) plus soft state (team quality, runway). In the EU regime we observed a noisy annual census of S . Here we observe marks only at the firm’s own deal times $T_{i,1} < T_{i,2} < \dots$:

$$m_{i,k} = (V_{i,k}, \text{round.type}_{i,k}, E_{i,k}, R_{i,k}, B_{i,k}), \quad (6.1)$$

where V is post-money valuation (populated only for priced rounds), E the reported employee count, and R, B the revenue and EBITDA fields — populated for roughly 45% and 25% of deal rows respectively in our synthetic mirror, calibrated to vendor coverage (Chapter 2). The marks are noisy, truncated views of $S_{i,T_{i,k}}$: valuation confounds fundamentals with the market multiple, employees confound productivity with cash deployment, and the revenue field is populated selectively.

Definition 6.1 (LOCF panel). Let $K_i(t) = \max\{k : T_{i,k} \leq t\}$, with $K_i(t) = 0$ before the first deal. The *last-observation-carried-forward panel* is $\tilde{S}_{i,t} = m_{i,K_i(t)}$ together with the staleness clock $A_{i,t} = t - T_{i,K_i(t)}$. Every firm-side feature in this chapter is a function of $(\tilde{S}_{i,t}, A_{i,t})$ and the full past mark history, never of marks dated after t .

The staleness clock is not decoration: a valuation carried forward 4 years is a different object from one carried forward 4 months, and models must be allowed to discount stale marks. But the deeper problem is *when* the panel refreshes.

Warning 6.2 (Sampling times are informative). The observation times $T_{i,k}$ are not an exogenous measurement schedule; they are outcomes of the firm’s own trajectory. A firm raises when it is doing well (an up round at a rich multiple) or when it must (runway exhaustion, a down round); firms that are neither raise nothing and go dark. Formally, the event indicator $dN_i(t)$ is not independent of $S_{i,t}$ given the past: the missingness mechanism of the LOCF panel is *missing-not-at-random*. Any panel method that treats $\tilde{S}_{i,t}$ as if generated by random inspection times — imputation, mixed models, naive regressions on the LOCF columns — inherits a bias whose sign and size depend on the unmodelled sampling mechanism. The only honest formulations either (i) condition explicitly on the event frame (Section 6.3) and correct for selection (Section 6.4), or (ii) model the sampling process jointly with the state, which is Part III.

6.2 What replaces EBITDA growth

The EU-track success labels of Chapter 3 are defined on filed fundamentals. We now construct their US surrogates, all defined on a pair of consecutive deals $(T_{i,k}, T_{i,k+1})$ with gap $\Delta_{i,k} = T_{i,k+1} - T_{i,k}$.

(a) Implied valuation growth. When both endpoints are priced rounds,

$$g_{i,k}^{\text{val}} = \frac{\log V_{i,k+1} - \log V_{i,k}}{\Delta_{i,k}}, \quad (6.2)$$

an annualized log growth rate. Writing valuation as fundamentals times a sector multiple, $\log V_{i,k} = \log F_{i,T_{i,k}} + \log m_{s(i),T_{i,k}} + \varepsilon_{i,k}$, gives the decomposition

$$g_{i,k}^{\text{val}} = \underbrace{g_{i,k}^F}_{\text{fundamentals growth}} + \underbrace{g_{i,k}^m}_{\text{multiple expansion}} + \underbrace{\nu_{i,k}}_{\text{pricing noise}}. \quad (6.3)$$

Only g^F is a success signal; g^m is a market-wide repricing that the firm did not earn. We identify the multiple component by the public-market-comparables logic: regress g^{val} on sector $\times$ time-bucket fixed effects plus macro controls (rate changes, public-index sector returns over the gap); the fitted cell means estimate g^m , and the residual is the firm-level proxy $\hat{g}_{i,k}^F$. The fixed effects must be estimated on all pairs in the cell, not per firm, or they absorb the signal.

(b) Round direction as an ordinal outcome. The field `vc_round_up_down_flat` classifies each round against the previous one as {down, flat, up}. It is coarse but far better populated than paired valuations (it is defined even when one endpoint is unpriced) and robust to level noise in V . We treat it as an ordinal target (proportional-odds or ordinal gradient boosting); its drawback is the vendor’s flat band, which is a definitional artifact rather than an economic category.

(c) Employee growth. $g_{i,k}^{\text{emp}} = (\log E_{i,k+1} - \log E_{i,k})/\Delta_{i,k}$ is a real-activity proxy available for most pairs. Its bias is mechanical: headcount growth immediately after a raise reflects cash deployment, not demonstrated productivity, so it leads fundamentals optimistically by construction.

(d) Reported revenue growth. `revenue_growth_since_last_deal` is the most direct surrogate when populated, but with $\sim 45\%$ endpoint coverage the pair-level coverage is roughly 20%, and reporters skew to larger, later-stage firms — the missingness is again not at random.

In practice we fit all four and read them jointly: agreement across proxies with different bias directions is the closest available substitute for a filed income statement.

Proxy	Pair coverage	Noise	Dominant bias
g^{val} (6.2)	moderate (both rounds priced)	high (negotia- tion, multiple)	upward; confounds multi- ple expansion unless decom- posed via (6.3)
<code>vc_round_ up_down_flat</code>	high	low but coarse	flat band definitional; cen- sors magnitude
g^{emp}	high	moderate	upward after raises (cash- fueled hiring)
<code>revenue_growth_ since_last_deal</code>	low ($\sim 20\%$)	low given re- ported	reporters skew large/late- stage; MNAR

Table 6.1: Growth proxies on the consecutive-deal pair frame. All four are additionally subject to the survival-to-next-round selection of Section 6.4, which no choice among them avoids.

6.3 Supervised learning on the event frame

The prediction unit is the ordered pair $(i, k) \rightarrow (i, k + 1)$: one row per consecutive-deal pair, outcome one of the proxies of Section 6.2. Features come in two blocks with different epistemic status.

Firm-history block, measurable at $T_{i,k}$: the full mark history through the k -th deal (valuation path, round sequence, employee path, deal sizes, investor counts), the staleness pattern, firm age, sector, and gap lengths between *past* deals. Nothing dated inside $(T_{i,k}, T_{i,k+1}]$ from the firm’s own record enters — there is nothing there to use except the second deal itself, which is the outcome’s timestamp.

Gap macro block, aggregates of exogenous series over the gap: mean policy rate, change in CPI year-on-year, change in unemployment, and sector deal heat (sector deal counts per week, normalized by trailing average), all built from point-in-time vintages as in Chapter 2. Because these are external to the single firm, conditioning on them does not leak the firm’s outcome; but it does change the estimand. With the gap block included, the regression is an *attribution* model — growth conditional on the realized macro path and the realized gap length. For deployment as a forecast at $T_{i,k}$, the gap block must be replaced by scenario paths or integrated over a gap-length model; the gap length $\Delta_{i,k}$ is itself informative (short gaps select hot firms), and we return to precisely this point below.

The learner is the Chapter 4 stack unchanged: gradient boosting with blocked cross-validation (folds split by firm *and* time), early stopping on the temporal holdout, and conformalized quantile intervals for uncertainty; we do not repeat that machinery here. Two adaptations matter. First, conformal calibration sets must be blocked by firm, since pairs from one firm share latent quality and violate exchangeability across naive random splits. Second, once the selection weights of the next section are introduced, calibration must be weight-aware (weighted conformal quantiles), otherwise coverage holds for the selected population only.

6.4 The selection problem

This section is the core of the chapter. A pair (i, k) exists in the training frame only if the firm raised again within the observation window: define

$$D_{i,k} = \mathbf{1}\{T_{i,k+1} \leq \tau_{\text{end}}\}, \quad (6.4)$$

where τ_{end} is the data cutoff. Every proxy of Section 6.2 is observed conditional on $D_{i,k} = 1$: we see the growth of firms *that came back*. The observed distribution is $P(g \mid D=1)$, not $P(g)$,

and the gap between them has an exact expression:

$$\mathbb{E}[g \mid D=1] - \mathbb{E}[g] = \frac{\text{Cov}(g, \mathbf{1}\{D=1\})}{\mathbb{P}(D=1)}. \quad (6.5)$$

Both selection channels have positive covariance with *appearing* in the frame but opposite implications for the observed growth mix: winners raise again (pulling observed g up), and cash-constrained firms raise again at down rounds (pulling it down), while the quiet failures — firms that never transact again — contribute nothing at all. The left tail of $P(g)$ is therefore systematically amputated: shutdown is the one outcome that generates no second deal. Estimates of mean growth, of feature effects, and of ranking quality are all computed on the surviving subpopulation.

Bounds. If g is bounded to $[g_{\text{lo}}, g_{\text{hi}}]$ (after winsorization, or by construction for the ordinal outcome), then with $p = \mathbb{P}(D=1)$,

$$\mathbb{E}[g] \in [p \mathbb{E}[g \mid D=1] + (1-p) g_{\text{lo}}, p \mathbb{E}[g \mid D=1] + (1-p) g_{\text{hi}}], \quad (6.6)$$

the worst-case (no-assumption) bounds. With p well below one — and Figure 6.1 shows how few firms generate long deal sequences — these are wide. They are still worth reporting: they are the honest statement of what the data alone say, and any point estimate inside them is purchased with an assumption.

Assumption 6.3 (Selection on observed history). $g_{i,k} \perp\!\!\!\perp D_{i,k} \mid \mathcal{H}_{T_{i,k}}$, where $\mathcal{H}_{T_{i,k}}$ collects the firm’s mark history through deal k and the macro state: given everything observed up to the first deal of the pair, whether the firm raises again carries no further information about its growth.

The practical fix: inverse-probability-of-observation weights. Under Assumption 6.3, define the observation probability

$$\pi_{i,k} = \mathbb{P}(D_{i,k}=1 \mid \mathcal{H}_{T_{i,k}}) = 1 - \exp\left(-\int_{T_{i,k}}^{\tau_{\text{end}}} \lambda_i(u) du\right), \quad (6.7)$$

where λ_i is the firm’s next-deal intensity given its history. Weighting each observed pair by $1/\hat{\pi}_{i,k}$ (stabilized and trimmed in the usual way) makes the weighted frame representative of all at-risk firms: $\mathbb{E}[Dg/\pi] = \mathbb{E}[g]$. This is inverse-probability weighting for informative censoring in the sense of [12], with the hazard-model lineage of [7]; validity requires $\pi_{i,k} > 0$ and a well-specified intensity, and residual selection on latent quality beyond $\mathcal{H}$ is not repaired — which is why we report (6.6) alongside.

Remark 6.4 (The bridge to Part III). Equation (6.7) is the structural link of this manual. The quantity $\lambda_i(u)$ — the conditional intensity of the firm’s deal-arrival process — is exactly what Part III calibrates: inhomogeneous Poisson with macro covariates (Chapter 7), self-exciting and inhibiting extensions (Chapter 8), and censored variants (Chapter 9). The count model is therefore not a separate deliverable that happens to share the data: it *repairs the growth regression*. A team that skips the point-process layer does not merely lose the deal-timing forecast; it loses the only principled weights for its growth model. Chapter 10 validates the combined pipeline — intensity fit, weights, weighted growth regression — on synthetic ground truth where $P(g)$ is known.

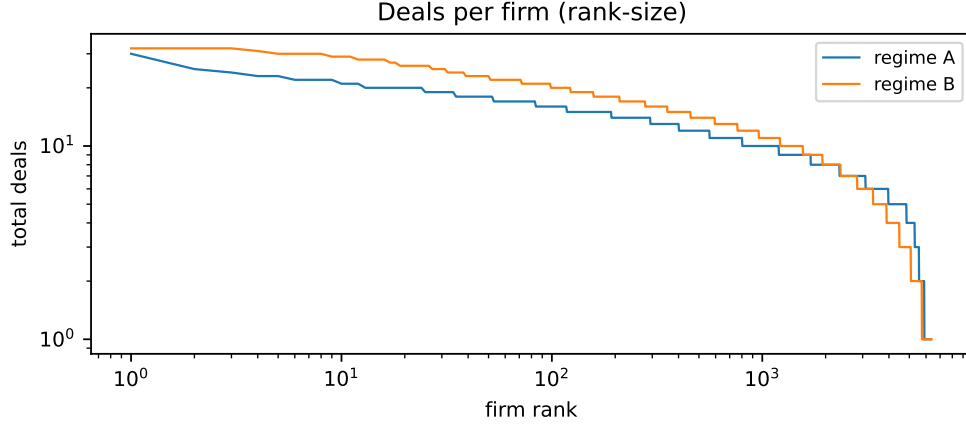


Figure 6.1: Rank-size plot of deals per firm in both synthetic regimes. The event record is heavily concentrated: a small minority of serially raising firms contributes most consecutive-deal pairs, while the modal firm contributes one deal and zero pairs. The pair frame therefore oversamples exactly the firms with the highest historical deal hazard; unweighted estimates describe serial raisers, not the tracked universe. Firm-level cluster bootstrap is mandatory for all standard errors on this frame.

6.5 Macro factors as drivers

With firm-level outcomes this sparse and selected, much of the macro question — do rate changes move private-market activity and pricing? — is better asked at the sector level. The workhorse is a sector-exposure (shift-share style) design. Let $y_{s,w}$ be a sector-week outcome (log deal count, or the cell-mean valuation growth from (6.3)), Δr_w the rate change, and e_s a *predetermined* exposure measure (share of pre-revenue firms in the sector, a duration proxy for how far cash flows sit in the future). We fit

$$y_{s,w} = \alpha_s + \gamma_w + \delta(e_s \cdot \Delta r_w) + u_{s,w}, \quad (6.8)$$

with sector and time fixed effects and two-way clustered standard errors. The time effect γ_w absorbs every common shock — including the *average* effect of the rate change itself — so δ is identified purely from differential sensitivity: do high-exposure sectors move more when rates move?

The interpretation limits are sharp and worth stating. First, the common component of the macro effect is not identified by design; (6.8) estimates relative sensitivities only. Second, latent risk appetite co-moves with rates; if its sector loadings correlate with e_s , the differential estimate is contaminated too. Third, exposure must be predetermined and not itself an outcome of recent sector performance. A design-based reading in the spirit of [11] is available only under an explicit exogeneity assumption on the exposure shares, which we state and stress-test on the testbed rather than assume silently. Where valuation marks are the outcome, all of Section 6.4’s caveats apply on top.

This is the forward pointer of the chapter: deal *counts* are complete (a deal either appears in the feed or does not), while marks are sparse and selection-laden. The cleanest macro inference therefore lives at the count layer — intensities modulated by macro covariates — which is where Chapter 7 begins.

Aspect	EU track (Part I)	US adaptation (this chapter)
Target	EBITDA/revenue growth from filings	proxy set: $\hat{g}^F$ from (6.3), ordinal round direction, employee growth, reported revenue growth
Unit	firm $\times$ fiscal year	consecutive-deal pair (i, k)
Feature clock	fiscal-year as-of with publication lags	event-time as-of with staleness clock $A_{i,t}$
Dominant bias	leakage via publication timing	survival-to-next-round selection; MNAR sampling times
Primary fix	point-in-time joins, blocked CV	IP-of-observation weights from the next-deal intensity (6.7); worst-case bounds (6.6)
Uncertainty	conformal intervals, firm-blocked	weight-aware conformal, firm-cluster bootstrap
Macro layer	covariates in the growth model	shift-share differentials (6.8); count-layer inference in Part III

Table 6.2: Method mapping between observation regimes. Every row of the US column degrades gracefully to the EU column when fundamentals become observable; the converse direction is the content of this chapter.

6.6 Decision table: EU track versus US track

The table is the chapter in miniature. Nothing in the US column is exotic; each entry is the Part I method re-derived under a harsher observation model, with one genuinely new ingredient — the deal-arrival intensity — that the next three chapters construct.

Part III

The Event-Based View: Calibrating Count Processes

Chapter 7

Counting Processes and Poisson Calibration

This chapter is the foundation of Part III. It fixes the mathematical objects — counting process, filtration, conditional intensity, compensator — states the two theorems that make intensity models *estimable and testable*, and builds the simplest useful model family: Poisson processes with macro covariates, in continuous and in weekly time. The through-lines of Chapter 1 recur throughout: *identifiability* (covariate effects are pinned down only by temporal contrast) and *uncertainty* (over-dispersion is the rule, and every standard error we report is dispersion-corrected). The chapter ends at the exact boundary of the Poisson world — clustering beyond covariates — which Chapter 8 crosses.

7.1 Counting processes, intensities, compensators

A *counting process* $N(t)$, $t \geq 0$, is a right-continuous step function with $N(0) = 0$ and jumps of size +1 at event times $t_1 < t_2 < \dots$; we assume simplicity (no coincident points). In our application $N_s(t)$ counts deals in sector s and $N_i(t)$ counts deals of firm i . The filtration $\mathcal{H}_t$ collects everything observable strictly before t : past events, their marks, and point-in-time covariate vintages (Chapter 2).

Definition 7.1 (Conditional intensity). The conditional intensity of N with respect to $\mathcal{H}_t$ is

$$\lambda(t) = \lim_{dt \downarrow 0} \frac{\mathbb{P}(N(t+dt) - N(t) = 1 \mid \mathcal{H}_t)}{dt}, \quad (7.1)$$

assumed to exist and to be predictable (determined by information strictly before t).

The intensity is the model: every process in Part III is specified by writing down $\lambda(t)$ as a function of time, covariates, and history. Its integral

$$\Lambda(t) = \int_0^t \lambda(u) du \quad (7.2)$$

is the *compensator*: the unique predictable increasing process such that $M(t) = N(t) - \Lambda(t)$ is a (local) martingale, a property we use without proof [5, 2]. Its practical content is unbiasedness of counts against the model, which is what makes residual analysis on count increments meaningful.

7.2 Two foundational theorems

Theorem 7.2 (Watanabe characterization). *Let N be a simple point process adapted to $\mathcal{H}_t$ with continuous compensator Λ . Then N is an inhomogeneous Poisson process — increments over disjoint intervals independent, with $N(b) - N(a) \sim \text{Poisson}(\Lambda(b) - \Lambda(a))$ — if and only if Λ is deterministic [2].*

Read as a modelling statement: *Poisson means history-free*. The moment $\lambda(t)$ depends on past events, the process is no longer Poisson; conversely, any intensity that is a deterministic function of time and exogenous covariates inherits the entire Poisson toolkit of independent increments and closed-form likelihoods.

Theorem 7.3 (Time rescaling). *Let N have event times $t_1 < t_2 < \dots$ and conditional intensity $\lambda(t) > 0$ with $\Lambda(t) \rightarrow \infty$ almost surely. Then the transformed times $\tau_k = \Lambda(t_k)$ are the event times of a unit-rate homogeneous Poisson process; equivalently, the rescaled inter-arrivals $\delta_k = \tau_k - \tau_{k-1}$ are i.i.d. $\text{Exp}(1)$ [3].*

Theorem 7.3 holds for *any* point process with an intensity, not just Poisson — Hawkes, inhibited, marked, all of them. It is the universal goodness-of-fit device: substitute the fitted $\hat{\Lambda}$, compute $\hat{\delta}_k = \hat{\Lambda}(t_k) - \hat{\Lambda}(t_{k-1})$, and test.

Corollary 7.4 (Everything in Part III is testable). *Every model class in this manual specifies $\lambda(t; \theta)$ explicitly; hence every fitted model yields rescaled inter-arrivals that should be i.i.d. $\text{Exp}(1)$ under correct specification. Our standing test battery: (i) Kolmogorov–Smirnov test of $u_k = 1 - e^{-\hat{\delta}_k}$ against $\text{Uniform}(0, 1)$; (ii) autocorrelation of the u_k (independence, which the KS test does not check); (iii) the same by sector and regime. Failing (ii) while passing (i) means the right intensity level with the wrong dynamics: the signature of ignored clustering.*

7.3 The homogeneous Poisson process

With $\lambda(t) \equiv \lambda$ on an observation window $[0, T]$, the log-likelihood is $\ell(\lambda) = n \log \lambda - \lambda T$ for n observed events, maximized at

$$\hat{\lambda} = n/T, \quad (7.3)$$

with the exact (not asymptotic) confidence interval from the chi-square duality,

$$\lambda \in \left[\frac{1}{2T} \chi_{2n, \alpha/2}^2, \frac{1}{2T} \chi_{2n+2, 1-\alpha/2}^2 \right], \quad (7.4)$$

where $\chi_{d,q}^2$ is the q -quantile of the chi-square with d degrees of freedom. The homogeneous model is acceptable only over windows short enough that macro conditions and sector heat are effectively constant — a claim Corollary 7.4 makes checkable. We use it as a per-window baseline whose rejection is informative.

7.4 Inhomogeneous Poisson with covariates

The first serious model: a deterministic log-linear intensity driven by the macro state,

$$\lambda(t) = \exp(a + \beta^\top X(t)), \quad (7.5)$$

with $X(t)$ piecewise constant at weekly resolution — the as-of macro vintages of Chapter 2: policy rate, CPI year-on-year, unemployment, each held at its last published value. Let weeks $w = 1, \dots, W$ partition $[0, T]$ with lengths Δ_w and covariate values X_w , and let n_w

be the event count in week w . The log-likelihood of an inhomogeneous Poisson process, $\sum_k \log \lambda(t_k) - \int_0^T \lambda(u) du$, becomes

$$\ell(a, \beta) = \sum_{w=1}^W \left[n_w(a + \beta^\top X_w) - \Delta_w \exp(a + \beta^\top X_w) \right]. \quad (7.6)$$

Under (7.5), timing within a week carries no information beyond n_w : weekly counts are sufficient, licensing the discrete-time form of Section 7.5.

Proposition 7.5 (Strict concavity). *Write $x_w = (1, X_w^\top)^\top$. The map $(a, \beta) \mapsto \ell(a, \beta)$ in (7.6) is concave, and strictly concave if and only if $\{x_w : w = 1, \dots, W\}$ spans $\mathbb{R}^{p+1}$. The MLE, when it exists, is therefore unique.*

Proof sketch. Each term is a linear function of (a, β) minus Δ_w times the exponential of a linear function; exponentials of linear maps are convex, so ℓ is concave. The Hessian is $-\sum_w \Delta_w e^{a+\beta^\top X_w} x_w x_w^\top$, negative definite exactly when no nonzero vector is orthogonal to all x_w — the spanning condition. This is the standard exponential-family/log-sum-exp argument. $\square$

Newton iterations converge in a few steps. For inference, with H the observed information and $\hat{S}$ the outer product of weekly score contributions (clustered by week, or by sector-week blocks in the multi-sector model), we report the sandwich

$$\widehat{\text{Var}}(\hat{a}, \hat{\beta}) = H^{-1} \hat{S} H^{-1}, \quad (7.7)$$

which remains valid as a quasi-likelihood variance when the Poisson assumption fails in the ways Section 7.6 documents.

7.4.1 Identifiability: covariate effects live in temporal contrast

Proposition 7.6 (Information equals weighted covariate variance). *The Fisher information of (7.6) is $I(a, \beta) = \sum_w \Delta_w \lambda_w x_w x_w^\top$ with $\lambda_w = e^{a+\beta^\top X_w}$. Profiling out a , the information for β is the intensity-weighted temporal covariance of X :*

$$I_\beta = \sum_w \omega_w (X_w - \bar{X}_w)(X_w - \bar{X}_w)^\top, \quad \omega_w = \Delta_w \lambda_w, \quad \bar{X}_w = \frac{\sum_w \omega_w X_w}{\sum_w \omega_w}. \quad (7.8)$$

In particular: (i) a covariate constant over the window is not identified separately from a ; (ii) a covariate that is a near-linear trend is near-collinear with any slow trend in the baseline, and its effective information is the variance of its detrended residual.

The proof is the Schur complement of the (a, a) block in $I(a, \beta)$; we omit it. The consequences matter more. Macro series are slow — rates move in staircase steps, unemployment drifts — so a two-to-three-year window may contain only one or two genuine covariate contrasts, near-confounded with the baseline level and any secular trend in deal activity. Our standing guidance: (i) include an explicit trend term so that β is measured against it rather than silently absorbing it; (ii) enter slow series in differences (*rate changes*, *CPI changes*), not levels; (iii) report effects as finite contrasts $\beta^\top (x_{\text{hi}} - x_{\text{lo}})$ between configurations that occur in the window, with sandwich intervals; (iv) monitor the condition number of (7.8) — ill-conditioning is an identifiability warning, not a numerical nuisance.

On the shipped testbed this is not hypothetical: in regime B the weekly-GLM covariate estimates recover the daily-world truth with correlation 0.37 (single seed; protocol in Chapter 10). Held-out fit still improves substantially over the covariate-free baseline: predictions and contrasts are far better identified than the coordinates of β . Report accordingly.

7.5 Discrete-time counts: the weekly GLM

For sector-level work we pass to weekly buckets directly. With $Y_{s,w}$ the count of sector- s deals in week w ,

$$Y_{s,w} \sim \text{Poisson}(\Lambda_{s,w}), \quad \log \Lambda_{s,w} = a_s + \beta_s^\top X_w, \quad (7.9)$$

independently across s and w given covariates — a Poisson GLM with log link, one intercept and slope vector per sector, estimated by the likelihood (7.6) summed over sectors. By the sufficiency remark above, when the continuous-time intensity is constant within weeks this is the same model, not an approximation; feedback enters in Section 7.7. Figure 7.1 shows the raw material.

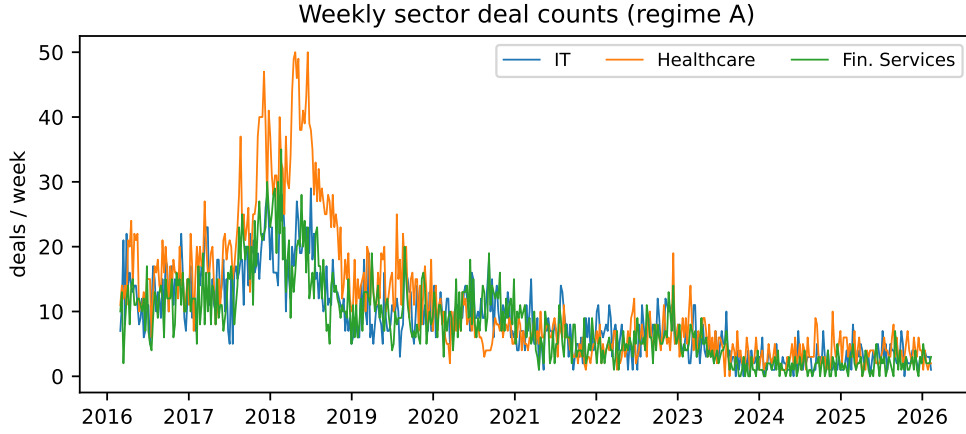


Figure 7.1: Weekly deal counts for three synthetic sectors. Common macro-driven movements are visible across sectors (the case for shared covariates X_w with sector-specific loadings β_s), superimposed on sector-specific levels and bursts that covariates will turn out not to explain (Section 7.6).

7.6 Over-dispersion

The Poisson law imposes $\text{Var}(Y) = \mathbb{E}[Y]$. Deal counts violate it, always, and the violation is informative.

Definition 7.7 (Dispersion). For a fitted count model with means $\hat{\mu}_j$ over n cells and p parameters, the Pearson dispersion is

$$\hat{\phi} = \frac{1}{n-p} \sum_{j=1}^n \frac{(y_j - \hat{\mu}_j)^2}{\hat{\mu}_j}, \quad (7.10)$$

with $\hat{\phi} \approx 1$ under correct Poisson specification and $\hat{\phi} > 1$ indicating over-dispersion (computed on held-out cells to avoid rewarding overfit means).

Two distinct mechanisms inflate $\hat{\phi}$, and both are present in deal data. *Unobserved heterogeneity*: if $Y \mid Z \sim \text{Poisson}(Z\mu)$ with a latent positive factor Z , $\mathbb{E}[Z] = 1$ (risk appetite, unmodelled sector health), then $\text{Var}(Y) = \mu + \mu^2 \text{Var}(Z) > \mu$. Taking Z gamma-distributed yields the negative binomial as a mixture Poisson — we state this without derivation and use it when a full likelihood is needed. *Contagion*: if events raise the intensity of subsequent events (Section 7.7, Chapter 8), counts cluster and the marginal variance again exceeds the mean. The cheap, robust correction is quasi-Poisson: keep the point estimates from (7.9) and multiply all standard errors by $\hat{\phi}^{1/2}$ (consistent with the sandwich (7.7)).

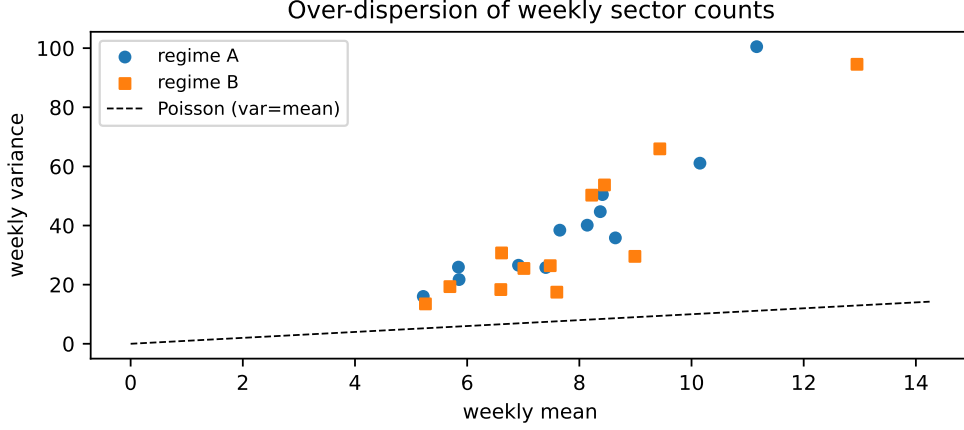


Figure 7.2: Weekly variance versus mean per sector, against the Poisson benchmark $\text{Var} = \mathbb{E}$ (dashed). Every sector sits above the line. On regime-B held-out weeks the fitted covariate GLM (7.9) leaves a Pearson dispersion of 2.72: conditional on the macro covariates, weekly counts are nearly three times more variable than Poisson. All GLM standard errors in this manual are quasi-Poisson corrected accordingly.

Warning 7.8 (Dispersion does not identify its cause). A held-out dispersion of 2.72 tells us the covariate-conditional Poisson model is wrong; it does not tell us *how*. Latent common factors and genuine contagion produce the same first-order symptom — variance above the mean and positively autocorrelated residual counts. Quasi-Poisson repairs the standard errors under either mechanism, which is why it is our default; but any *scientific* claim (“deals beget deals”) requires separating the two, and that separation cannot be done from dispersion statistics. It needs timestamp-level structure, and it is the central identification question of Chapter 8.

7.7 Excitation and inhibition in count GLMs

The GLM extends naturally to history dependence: append lagged counts to the predictor,

$$\log \Lambda_{s,w} = a_s + \beta_s^\top X_w + \sum_{\ell=1}^L b_\ell^\top Y_{w-\ell}, \quad (7.11)$$

where $Y_{w-\ell}$ stacks all sectors’ counts, so b_ℓ captures within- and cross-sector feedback at lag ℓ . By Theorem 7.2 this is no longer a Poisson process — the compensator is now random — but conditional on the past each $Y_{s,w}$ is still Poisson, the likelihood factorizes, and estimation is unchanged.

Two features make (7.11) the right first model for history effects. First, *negative coefficients are legal*: the log link keeps $\Lambda_{s,w} > 0$ for any sign of b_ℓ , so *inhibition* — deal flow suppressing near-future deal flow through capital absorption or cooldowns — is expressed as naturally as excitation. The linear Hawkes process of Chapter 8 cannot do this: its intensity is a sum of nonnegative kernels, and negative excitation would violate positivity. Second, the weekly clock makes stability auditable — with a subtlety the following analysis makes precise.

Stability: two mechanisms. Collect the feedback matrices B_ℓ ($M \times M$, sectors) and let $G = \sum_\ell B_\ell$.

Mechanism A (additive feedback). If the feedback enters the intensity linearly, $\Lambda_w = c + \sum_\ell B_\ell Y_{w-\ell}$ with nonnegative entries, the process is a discrete-time branching (INAR-type)

system: each event contributes on average G offspring events, and $\rho(G) < 1$ — spectral radius below one — suffices for stationarity, with stationary mean $(I - G)^{-1}c$. This is the direct analogue of the Hawkes branching-ratio condition of Chapter 8.

Mechanism B (log link on raw counts). Model (7.11) feeds raw counts through an exponential: the conditional-mean map is $\mu(y) = \exp(a + By)$ elementwise. This is *geometric*, not additive, feedback. A useful local summary is the Jacobian at the deterministic fixed point μ^* ,

$$G_{\text{eff}} = \text{diag}(\mu^*) G, \quad \rho_{\text{eff}} = \rho(G_{\text{eff}}), \quad (7.12)$$

the *effective spectral radius*. But $\rho_{\text{eff}} < 1$ guarantees only local stability of the mean map: the Poisson conditional law has unbounded support, and a sufficiently large fluctuation can push counts past the basin of attraction, after which the exponential feedback compounds and the process diverges. Global ergodicity for log-linear count autoregressions requires genuinely stronger conditions, developed by [6]; the standard safe designs are to feed $\log(1 + Y_{w-\ell})$ instead of raw counts (making the feedback additive on the log scale, Mechanism A-like), to cap counts, or to keep ρ_{eff} well below one.

Warning 7.9 (A measured metastability episode). This failure mode is not theoretical. The regime-A synthetic generator originally used log-link feedback on raw counts with effective radius $\rho_{\text{eff}} = 0.35$ — comfortably subcritical by the local criterion. The configuration was *metastable*: across most weeks it behaved normally, then a single burst crossed the basin boundary and tipped a world calibrated to roughly 50k events into 169k events. Rebuilt at $\rho_{\text{eff}} = 0.15$, the generator is stable (Chapter 10 documents the episode). The lesson generalizes: *never* simulate — or deploy forecasts from — log-link raw-count feedback near criticality; keep excitation terms small and let covariates carry the signal, or move the feedback to the $\log(1 + Y)$ scale where Mechanism A logic applies.

Remark 7.10 (Fitted feedback overstates contagion under aggregation). The estimated $\hat{b}_\ell$ in (7.11) must not be read as a measured contagion strength. On the testbed, a daily-world generator with *modest* true contagion and a 7-day excitation decay yields a fitted weekly-GLM effective spectral radius of 0.57. Two mechanisms conspire: weekly binning aliases within-week and adjacent-week excitation into the lag-one coefficient (an aggregation artifact), and latent-quality clustering — persistent hot sectors — loads onto lagged counts because they are the best available proxy for the latent factor. Treat fitted feedback as an upper bound on contagion and as a forecasting device; the identification of true excitation is deferred to Chapter 8.

7.8 Simulation

For piecewise-constant deterministic intensities, simulation is trivial: draw independent Poisson counts per week (and uniform timestamps within weeks if needed). History-dependent intensities require thinning [4].

Algorithm (thinning). Given a predictable intensity $\lambda(t)$ and, at each time, a computable local bound $\bar{\lambda} \geq \lambda(u)$ valid on a lookahead interval:

1. Set $t \leftarrow 0$; initialize history $\mathcal{H}$; compute the current bound $\bar{\lambda}$ valid on $[t, t + \Delta)$.
2. Draw $E \sim \text{Exp}(\bar{\lambda})$ and propose $t' = t + E$. If $t' > \min(T, t + \Delta)$, advance t to that endpoint, refresh $\bar{\lambda}$, and repeat from step 2; stop when $t \geq T$.
3. Draw $U \sim \text{Uniform}(0, 1)$; accept an event at t' if $U \leq \lambda(t')/\bar{\lambda}$, evaluating $\lambda(t')$ with the history up to t' .

4. If accepted, append t' to $\mathcal{H}$ (excitation raises future λ , so refresh $\bar{\lambda}$). Set $t \leftarrow t'$ and return to step 2.

Correctness requires only that $\bar{\lambda}$ dominate λ on its validity interval. Note the interaction with Section 7.7: for log-link feedback on raw counts no global bound exists and local bounds explode near bursts — one more reason that family is dangerous to simulate.

Discrete-time simulation with risk-set caps. Our production simulators run on the weekly clock: for $w = 1, \dots, W$ compute $\Lambda_{s,w}$ from covariates and lagged counts, draw $Y_{s,w} \sim \text{Poisson}(\Lambda_{s,w})$, then cap at the risk set, $Y_{s,w} \leftarrow \min(Y_{s,w}, |\mathcal{R}_{s,w}|)$, where $\mathcal{R}_{s,w}$ is the set of firms currently eligible to transact (alive, not in a post-deal cooldown, Chapter 8). The cap is substantive — a finite firm universe cannot generate unbounded weekly counts — and doubles as an explosion guard for feedback models. We log every cap activation: a simulator that hits its caps more than rarely is reporting a misspecified intensity, and the fix belongs in the model, not in a larger cap.

7.9 What Poisson cannot capture

The covariate GLM (7.9) says: given the macro path, disjoint weeks are independent and variance equals the mean. The data answer with held-out dispersion 2.72 and positively autocorrelated rescaled residuals — clustering of events beyond anything the covariates explain. Within this chapter’s model class the only response is defensive: quasi-Poisson inflation of uncertainty, feedback terms read as forecasting devices.

The scientific question underneath is sharper. Two generative stories produce exactly these symptoms: a *latent common factor* — unobserved risk appetite multiplying every sector’s intensity, inducing correlated bursts with no causal link between deals — and *self-excitation* in the sense of [1], where each deal genuinely raises the intensity of successors via signaling, competitive urgency, and follow-on capital. The two stories have the same second moments at the weekly layer (Remark 7.10 is this equivalence measured), yet they imply different counterfactuals: under contagion, seeding a deal raises subsequent activity; under a latent factor, it changes nothing. Distinguishing them requires timestamp-level structure, explicit kernels, and honest statements about what remains unidentified. That is Chapter 8.

Chapter 8

Hawkes Processes and the Two-Stage Marked Decomposition

Chapter 7 ended with a diagnosis: after conditioning on macro covariates and seasonality, the time-rescaled residuals of the Poisson model still cluster. Deal flow excites deal flow — the canonical model for this is the Hawkes process. But private markets also exhibit the *opposite* effect at the firm level: a company that closed a round last month is much *less* likely to raise again this month. This chapter resolves the tension with a single architectural decision — an exact factorization of the firm-level intensity into a sector *ground process* (where linear Hawkes excitation is appropriate) and a *conditional choice model* over firms (where inhibition lives harmlessly as a feature weight). The newcomer problem, the risk-set discipline, and the causal status of fitted excitation all follow from that decision.

8.1 The linear Hawkes process

Definition 8.1 (Linear Hawkes process [1]). A counting process N on $[0, \infty)$ with event times $t_1 < t_2 < \dots$ is a linear Hawkes process with baseline $s(\cdot) \geq 0$ and kernel $\phi(\cdot) \geq 0$ if its conditional intensity given $\mathcal{H}_t$ is

$$\lambda(t) = s(t) + \sum_{t_k < t} \phi(t - t_k). \quad (8.1)$$

The workhorse kernel is exponential, $\phi(u) = \kappa \theta e^{-\theta u}$, $u > 0$, parametrized so that the *branching ratio* $\kappa = \int_0^\infty \phi(u) du$ and the *decay rate* θ (mean lag $1/\theta$) are separate parameters.

In the multivariate version with M sectors, sector ℓ has intensity

$$\lambda_\ell(t) = s_\ell(t) + \sum_{m=1}^M \sum_{t_k^m < t} \phi_{\ell m}(t - t_k^m), \quad K_{\ell m} = \int_0^\infty \phi_{\ell m}(u) du, \quad (8.2)$$

where K is the matrix of branching ratios: $K_{\ell m}$ is the expected number of sector- ℓ deals directly triggered by one sector- m deal.

Theorem 8.2 (Cluster representation and stability). *Let $\phi \geq 0$ with $\kappa < 1$ (multivariate: spectral radius $\rho(K) < 1$). Then the Hawkes process is equal in law to a Poisson cluster process: immigrants arrive as an inhomogeneous Poisson process with rate $s(t)$; every event, independently, produces a $\text{Poisson}(\kappa)$ number of direct offspring whose lags from the parent are i.i.d. with density $\phi(u)/\kappa$; the process is the superposition of all generations. Under constant s the process admits a stationary version if and only if $\kappa < 1$ ($\rho(K) < 1$), with mean intensity $s/(1 - \kappa)$ (vector $(I - K)^{-1}s$) and mean total cluster size $1/(1 - \kappa)$ [5].*

Remark 8.3 (Reading the branching structure as deal contagion). The cluster representation is the economic interpretation, not just a proof device. Immigrants are deals the macro baseline $s_\ell(t)$ would have produced anyway; offspring are deals *caused* (in the sense examined in Section 8.9) by earlier deals: competitors raising defensively, investors crowding into a heating sector. At $\rho(K) = 0.2$ roughly 17% of deals are offspring; as $\rho(K) \rightarrow 1$ activity becomes self-sustaining. Rounds take months to assemble, so we expect modest branching in this asset class, and the testbed confirms it (true effective radius 0.172 in regime A; naive fits report more, Section 8.9).

8.2 Likelihood, simulation, goodness of fit

For events $t_1 < \dots < t_n$ observed on $[0, T]$ the exact log-likelihood is

$$\ell(\vartheta) = \sum_{k=1}^n \log \lambda(t_k) - \int_0^T \lambda(u) du. \quad (8.3)$$

Naively both terms cost $O(n^2)$. The exponential kernel makes them $O(n)$: writing $\lambda(t_k) = s(t_k) + \kappa\theta A_k$, the excitation state obeys the recursion

$$A_k = e^{-\theta(t_k - t_{k-1})} (1 + A_{k-1}), \quad A_1 = 0, \quad (8.4)$$

and the compensator is $\int_0^T s(u) du + \kappa \sum_k (1 - e^{-\theta(T - t_k)})$. This recursion is why the exponential family is the default: likelihood, gradients, and the online production intensity are one data pass.

Algorithm (simulation by thinning [4, 3]). To simulate on $[0, T]$:

1. Set $t \leftarrow 0$ and initialize the history (empty, or a burn-in realization).
2. Compute a local upper bound $M \geq \lambda(u)$ for $u > t$; for the exponential kernel excitation only decays between events, so $M = s^{\max} + \kappa\theta A(t^+)$.
3. Draw $E \sim \text{Exp}(M)$ and propose $t' = t + E$; stop if $t' > T$.
4. Draw $U \sim \text{Unif}(0, M)$; accept t' as an event iff $U \leq \lambda(t')$; either way set $t \leftarrow t'$ and refresh M (mandatory after an accepted event or a baseline increase).
5. Repeat from step 2.

Thinning is exact and is the engine of every parametric bootstrap in this chapter.

Goodness of fit uses time-rescaling exactly as in Chapter 7: under the model, the rescaled times $\Lambda(0, t_k]$ form a unit-rate Poisson process, so the QQ-plots and autocorrelation checks of that chapter apply verbatim. The key diagnostic: clustered rescaled gaps indicate missing excitation; over-regular gaps, missing inhibition.

8.3 The inhibition obstruction

Proposition 8.4 (Linear Hawkes kernels must be nonnegative). *Let $\lambda(t) = s(t) + \sum_{t_k < t} \phi(t - t_k)$ with s bounded. If $\phi < -\epsilon$ on some interval of positive length, there exist admissible histories for which $\lambda(t) < 0$. Hence (8.1) defines a valid intensity for all histories only if $\phi \geq 0$ almost everywhere.*

Proof sketch. Contributions enter (8.1) additively, one per past event. Place n events inside a short window positioned so that each contributes less than $-\epsilon$ at time t ; such histories have positive probability since $\lambda \geq s > 0$ before the window ends. Then $\lambda(t) \leq \sup s - n\epsilon < 0$ for n large. Truncating at zero, $\lambda = (s + \sum \phi)_+$, restores validity but destroys linearity — and with it the cluster representation of Theorem 8.2, the moment equations, and the interval-censored machinery of Chapter 9, all of which rely on taking expectations through a linear equation. $\square$

This is not a technicality: inhibition is the *dominant* firm-level effect in our data. Figure 8.1 shows the empirical refunding hazard against weeks since the previous deal. In regime A the generative truth is a hard 26-week cooldown and the empirical hazard shows exactly that: suppressed and flat for about 26 weeks, then a jump to the ambient level. In regime B the truth is smooth exponential inhibition with a 30-day half-life, and the hazard rises smoothly toward its plateau. Either way, a just-funded firm is far less likely to raise — the opposite sign of what a linear kernel can encode.

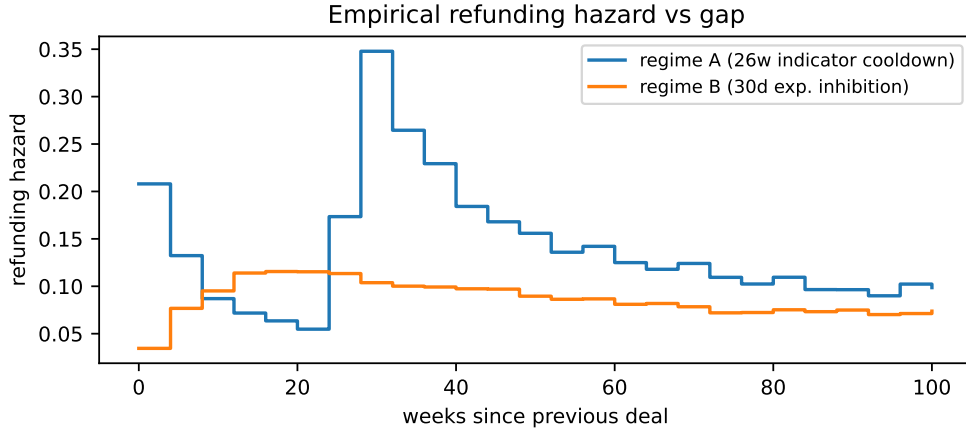


Figure 8.1: Empirical refunding hazard versus weeks since a firm’s previous deal. Regime A (hard 26-week cooldown indicator) versus regime B (smooth exponential inhibition, 30-day half-life). Both regimes contradict the nonnegativity that a linear Hawkes kernel requires at the firm level.

Two escapes exist. (a) *Nonlinear link.* Model $\lambda_i(t) = \exp\{\mu_i + \sum_k g(t - t_k)\}$ with g unrestricted in sign. This is coherent and estimable in event time, but the expectation of an exponential of a sum no longer solves a linear equation: the cluster representation, the Volterra first-moment equation, and hence the entire censored-data programme of Chapter 9 are lost, and stability conditions become delicate. (b) *Factorization.* Keep the linear Hawkes process where its positivity is economically true — sector-level contagion — and move firm-level inhibition into a conditional model of *which firm* transacts, where a negative coefficient is just a score, never an intensity. This manual chooses (b): the split is exact (Section 8.4), each factor gets the estimator best suited to it, and the censored-data machinery of Chapter 9 continues to apply to the sector layer.

8.4 The factorization theorem

Treat the deal stream of sector s as a *marked* point process: events are deal times, marks are firm identities.

Theorem 8.5 (Exact two-stage decomposition). *Let N_s be the ground process of sector- s deals with $\mathcal{H}_t$ -intensity $\lambda_s^g(t)$, and let each event carry a mark i (the transacting firm). Then the*

firm-level intensities factorize exactly:

$$\lambda_i(t) = \lambda_{s(i)}^g(t) \cdot p(i \mid s(i), t, \mathcal{H}_t), \quad (8.5)$$

where $p(\cdot \mid s, t, \mathcal{H}_t)$ is the conditional mark distribution — the probability that the firm transacting at t , given that some sector- s deal occurs at t , is firm i . Consequently the log-likelihood of the marked stream splits into a ground-process term depending only on λ^g plus a sum of log p terms, and the two factors can be estimated separately without loss [2].

Proof sketch. By construction $\lambda_s^g(t) = \sum_{j:s(j)=s} \lambda_j(t)$, and the conditional mark kernel of a marked point process is precisely the ratio λ_i/λ_s^g on $\{\lambda_s^g > 0\}$; existence and uniqueness follow from the definition of a marked point process via its ground process and mark kernel. The likelihood factorization is then term-by-term. Full details are in the appendix (see Chapter 10). $\square$

Remark 8.6 (Where the approximations actually enter). Equation (8.5) is an identity, so the two-stage architecture is *lossless at population level*: no information is discarded by the split itself. Approximation enters only through the parametrizations. Stage 1 restricts λ_s^g to an exponential-kernel multivariate Hawkes with a log-linear covariate baseline (or its discrete GLM surrogate); Stage 2 restricts the mark distribution to a conditional logit with utilities linear in a *finite* feature list (so $\mathcal{H}_t$ enters only through those features) and the softmax’s independence-of-irrelevant-alternatives structure. Every misspecification found later is attributable to one of these restrictions, which is what makes the architecture auditable.

8.5 Stage 1: the sector ground process

Stage 1 is the multivariate Hawkes model (8.2) for the twelve sector streams, with macro covariates in the baseline, $s_\ell(t) = \exp\{\beta_\ell^\top x_t\}$, joined point-in-time as in Chapter 2. When timestamps are trustworthy we fit the exact likelihood (8.3); when the data arrive weekly we fall back on the discrete count GLM of Chapter 7 [6] or the interval-censored construction of Chapter 9. We expect a modest effective spectral radius (well below 0.5) and decay scales of days-to-weeks; a fit reporting $\rho(K)$ near one is a red flag for latent-factor contamination, not a discovery of near-critical contagion (Section 8.9). The immigrant/offspring split of Theorem 8.2 acquires a second job in Section 8.7: immigrants are first financings, offspring are repeats.

8.6 Stage 2: conditional choice over the risk set

Given that a sector- s deal occurs at t , Stage 2 models which firm it is with a conditional logit [8]:

$$p(i \mid s, t) = \frac{\exp(q_{i,t})}{\sum_{j \in \mathcal{R}_{s,t}} \exp(q_{j,t})}, \quad q_{i,t} = w^\top z_{i,t}, \quad (8.6)$$

where $\mathcal{R}_{s,t}$ is the *risk set* and $z_{i,t}$ the point-in-time feature vector: log amount previously raised, log employees, firm age, stage, and — carrying the inhibition — the gap feature $\log(1 + \text{gap}_{i,t}/30)$ with $\text{gap}_{i,t}$ the days since firm i ’s last deal. A *positive* gap weight is inhibition: recently funded firms score low — the effect a linear kernel cannot encode is here one bounded coefficient.

The funded-pool construction. $\mathcal{R}_{s,t}$ contains the previously funded, currently active firms of sector s , *excluding the most recently funded firm*. Each clause is load-bearing. *Previously funded*: features exist only after a first deal (entrants are handled in Section 8.7). *Currently active*: activity status must come from point-in-time fields, never be inferred from future silence (Warning 8.9). *Excluding the last winner*: vendors record tranches of one round as

separate rows days apart; without the exclusion, an artifactual mass of immediate self-repeats at gap ≈ 0 corrupts the gap coefficient.

Kinship with the Cox partial likelihood. Equation (8.6), multiplied over event times, is formally the partial likelihood of Cox [7]: a product of softmax terms over risk sets, one per event. The shared insight is that *positive-only data suffice*: we never observe “firm i tried to raise and failed”, and we never need to. The choice model conditions on a deal occurring and estimates only *relative* propensities of candidates; how many deals occur, and when, is entirely Stage 1’s job — the division that makes the architecture estimable from a vendor deal table at all.

Proposition 8.7 (Sampled softmax is consistent). *Fix m and, for each event, replace $\mathcal{R}_{s,t}$ in (8.6) by the winner plus a uniform random subsample of m non-winners, drawn independently of utilities. The maximizer of the resulting conditional likelihood is consistent for w ; uniform sampling satisfies McFadden’s uniform-conditioning property, under which subsampling alternatives perturbs only the alternative-specific constants [8].*

In practice $m = 64$ candidates per event puts the held-out loss within noise of full-pool fits on the testbed while making the cost per event constant; we use $m = 64$ throughout.

8.7 The newcomer problem

Around 27% of deals in both regimes are won by firms *outside* the funded pool: first financings, or firms the vendor did not previously track (Figure 8.2). Entrants have no covariates — every feature in $z_{i,t}$ exists only after a first deal — yet a model that ignores a quarter of the probability mass is not calibratable. We handle entrants as one aggregate alternative “a newcomer wins” with score q_{new} , in a ladder of increasing structure:

1. **Alternative-specific constant.** $q_{\text{new}} = b_0$. Calibrates the average newcomer share and nothing else; the share visibly co-moves with sector heat, so this is a floor, not a default.
2. **Context covariates (the calibratable default).** $q_{\text{new}} = b_0 + \gamma^\top g_{s,t}$, where $g_{s,t}$ stacks sector heat, the macro block, and log pool size. This is what we ship: it tracks the cyclical variation of Figure 8.2 with a handful of parameters.
3. **Representative entrant.** Entrants are unobserved before their first deal but *observed at it*; the at-first-funding feature distribution is estimable, and we integrate a pseudo-candidate’s utility over it (Proposition 8.8).
4. **Immigrant/offspring split.** The Hawkes-native reading: first financings are the *immigrant* stream of the sector process (the covariate-driven $s_\ell(t)$ of Theorem 8.2); repeats are the history-dependent stream. Stage 1 models the two streams separately; Stage 2 ranks the funded pool within the repeat stream only. This aligns the semantics of both stages and is recommended at event resolution.

Proposition 8.8 (Gaussian representative entrant). *Suppose at time t there are N_{pool} latent entrants whose (unobserved) features are i.i.d. $\mathcal{N}(\mu, \Sigma)$, independent of $\mathcal{H}_t$, with utilities $w^\top z$. Replacing the entrant block of the softmax denominator by its expectation yields the single pseudo-candidate score*

$$q_{\text{new}} = w^\top \mu + \frac{1}{2} w^\top \Sigma w + \log N_{\text{pool}}, \quad (8.7)$$

since $\mathbb{E} \sum_z e^{w^\top Z z} = N_{\text{pool}} e^{w^\top \mu + w^\top \Sigma w / 2}$ by the Gaussian moment generating function. Proof and the accuracy of the mean-field step are in the appendix (Chapter 10).

In practice (μ, Σ) are estimated per sector from at-first-funding features, and $\log N_{\text{pool}}$ is absorbed into b_0 unless a defensible pool proxy exists.

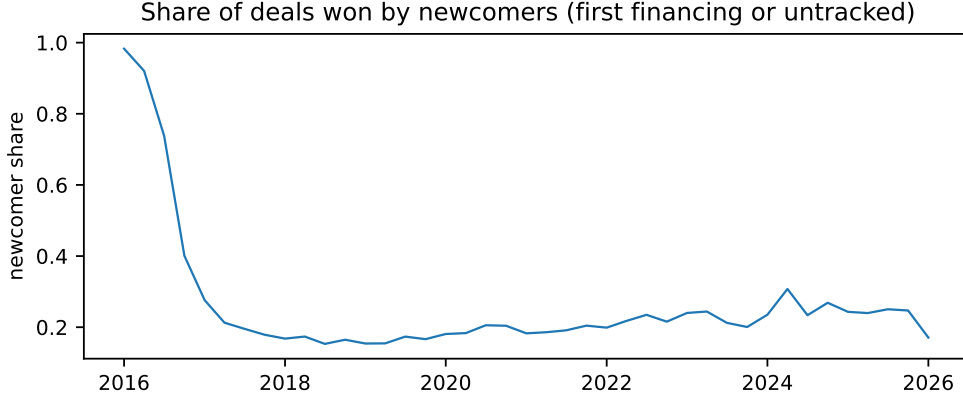


Figure 8.2: Quarterly share of deals won by newcomers (first financings or previously untracked firms). The share fluctuates around 27% and co-moves with sector activity, motivating rungs 2–4 of the ladder over a bare constant.

8.8 Measured evidence

Table 8.1 reports the Stage-2 fits on both regimes (single seed; the five-seed protocol and full tables are in Chapter 10). Regime A is the well-specified weekly two-stage world (520 weeks, 12 sectors, 8,000 firms, roughly 50,000 deals); regime B is the misspecified daily world of Chapter 1.

	Regime A	Regime B
Held-out NLL per event	3.346	3.796
NLL, random-choice benchmark	4.011	4.172
Top-5 accuracy	0.446	0.316
Log-gap weight	+1.146	+0.437
Newcomer intercept	6.856	4.691
Newcomer share	$\approx 27\%$	$\approx 27\%$

Table 8.1: Funded-pool conditional logit with sampled softmax (64 candidates), both testbed regimes. Single-seed values; see Chapter 10.

Three readings. *First, covariate recovery is good where the model is well-specified:* in regime A the fitted weights on log raised, log employees, and stage are 0.336, 0.253, 0.132 against generative truths 0.30, 0.30, 0.15. The age coefficient is the honest exception: fitted 0.054 against a true -0.15 . Age is entangled with the recency block — older firms mechanically have longer funding histories and larger gaps — so the gap and newcomer terms absorb it; report it as part of a correlated block, never causally. *Second, the gap feature recovers the inhibition shape it can express:* the true regime-A cooldown is a 26-week *indicator* of strength about -2.7 ; the fitted smooth surrogate $\log(1 + \text{gap}/30)$ with weight $+1.146$ cannot reproduce the discontinuity of Figure 8.1 but ranks candidates correctly through it. In regime B the truth is itself smooth (30-day half-life) and the weight $+0.437$ matches the shape closely. *Third, lift over random is the metric that matters operationally:* NLL 3.346 vs 4.011 and top-5 accuracy 0.446 mean the true winner is in the model’s top five candidates almost half the time in a pool of thousands.

Warning 8.9 (Risk-set contamination inverts the economics). In a controlled experiment we replaced the true activity mask by a naively inferred one (firms marked inactive after a fixed dormancy window — an inference that peeks at future silence). The fitted ranker then *inverted* the economics: the recency weight came out +1.94 in a world where the truth is 0 (held-out NLL 3.886); with the true mask the weights recover (NLL 3.558). The mechanism: dormant firms wrongly kept in $\mathcal{R}_{s,t}$ are never winners and all have long gaps, so “long gap” becomes evidence of being a loser and the sign flips. The production lesson is first-order: *audit the risk set before the feature list*. Build activity masks from point-in-time status fields only; never from the absence of future events.

8.9 Identifiability and the causal status of excitation

A fitted kernel $\phi_{\ell m} \neq 0$ is, by construction, a statement of *Granger causality*: past sector- m events improve prediction of sector- ℓ intensity [17]. Whether it supports the interventional claim “an exogenous extra deal in m causes extra deals in ℓ ” is separate, and the principal threat is concrete.

Warning 8.10 (A latent common factor masquerades as contagion). Regime A contains a latent, serially correlated risk-appetite factor that multiplies all sector baselines. Its autocorrelation is not representable in the baseline the model was given, so the fit absorbs it into the excitation matrix: the fitted effective spectral radius is 0.355 against a true 0.172 — contagion overstated by a factor of two. In regime B the fit reports 0.57 where true contagion is modest with 7-day decay. Figure 8.3 shows the anatomy: covariate coefficients are recovered tightly, individual excitation entries are not. Treat any fitted $\rho(K)$ as an *upper bound* on contagion unless the latent-factor channel has been addressed.

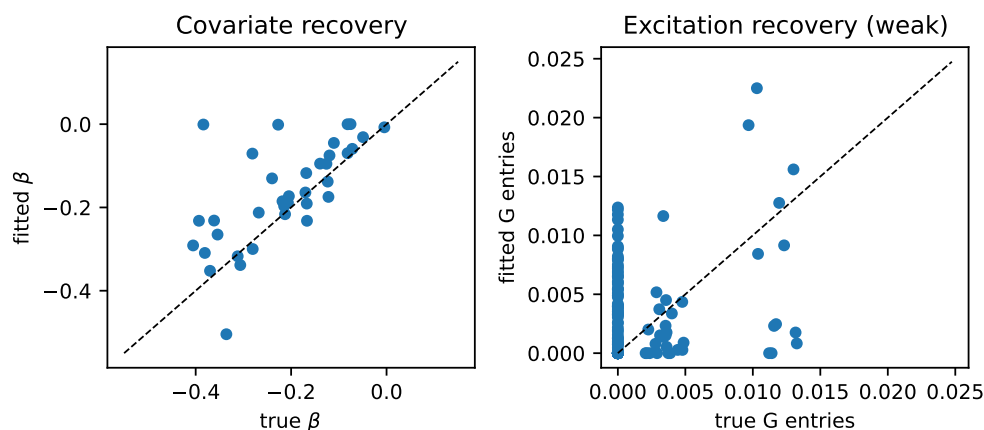


Figure 8.3: Parameter recovery in regime A. Left: true versus fitted covariate coefficients — tight recovery. Right: true versus fitted excitation-matrix entries — weak recovery, with systematic inflation attributable to the latent risk-appetite factor.

Three remedies, in order of practical value. (i) *Proxy the factor*: put observable risk-appetite proxies (the macro block, aggregate cross-sector deal heat — lagged, to avoid mechanical endogeneity) into the baseline; whatever the proxies explain is removed from K . (ii) *Restrict K* : sparsity or within-sector-only structure means fewer free parameters available to soak up common variation; cross-sector entries with no plausible economic channel double as placebos. (iii) *Refine timing*: a factor moving monthly and contagion decaying in 7 days are separable at day resolution and hopelessly confounded at monthly buckets — a preview of Chapter 9.

A causal reading of the fitted excitation is defensible when one is willing to state, and defend, the following: (A1) no unmeasured common driver of the sector intensities given the baseline covariates — the point-process transposition of the unconfoundedness behind $\text{do}(\cdot)$ claims [10]; (A2) the kernel family and baseline are correctly specified, so excitation is not a repository for misspecification; (A3) timestamps are fine relative to the kernel decay (A3 fails by construction under weekly exports); (A4) the sector graph is known — no hidden intermediary streams. Under (A1)–(A4) the cluster representation licenses the interventional statement above. We recommend two cheap sensitivity exercises: refit with and without factor proxies and report $\rho(K)$ under each (the gap — here 0.355 versus 0.172 with an oracle-complete proxy set — is the measured confounding load); and inject a synthetic latent factor of plausible magnitude into data simulated from the fitted model to measure how much spurious radius it induces.

8.10 Uncertainty quantification

Four instruments, matched to what each is for.

1. **Sandwich variances** for all maximum-likelihood point estimates: misspecification is the norm here, so inverse-Hessian standard errors are systematically optimistic; the robust form is the default in the shipped fitters.
2. **Parametric bootstrap by simulation.** Simulate replicas from the fitted model by thinning (Section 8.2), refit, and read off the dispersion of the refits [14]. This captures finite-sample and boundary effects — κ near zero, $\rho(K)$ near its constraint — where quadratic approximations fail.
3. **Laplace or posterior summaries for (κ, θ) .** The likelihood surface in (κ, θ) develops a curved ridge as timing information degrades (Chapter 9 derives why); a Laplace approximation at the mode understates joint uncertainty along the ridge, so we prefer posterior summaries from a short MCMC or a dense profile scan.
4. **Cluster bootstrap over sectors for the choice model.** Stage-2 events within a sector share shocks (the same heat, the same latent factor), so i.i.d.-event resampling is anticonservative. Resample *sectors* with replacement, refit (8.6), and report percentile intervals; with twelve sectors, also report the jackknife-over-sectors range as a stability check.

The through-line, restated once: the decomposition is exact, so every modelling risk lives in the parametrization of a factor, and each has a named diagnostic and a named uncertainty instrument. Chapter 9 takes up the remaining failure mode: what to do when even the timestamps cannot be trusted.

Chapter 9

When the Data Are Not Right: Censored and Aggregated Observation

The event-time machinery of Chapter 8 assumes that deal timestamps mean what they say. Chapter 2 documented four ways in which ours do not: recorded `deal_date` values carry jitter of up to six days; announced and closed dates disagree, and which one populates the field varies by deal class; several vendor exports arrive as weekly files, so within-week ordering is unrecoverable; and some downstream systems expose only bucketed counts per sector, never row-level timestamps. Fitting the exact likelihood of Chapter 8 to such data quietly estimates the kernel of the *jitter*, not of the market. This chapter develops the honest alternative: a mean-behaviour approximation whose likelihood consumes exactly what the data provide — counts on a grid — and, just as importantly, a precise account of which parameters survive the aggregation and which do not.

Formal setting. Fix a sector and an observation grid $0 = o_0 < o_1 < \dots < o_W = T$ (typically weekly, $o_w - o_{w-1} = \Delta$). We observe only the bucket counts

$$C_w = N(o_w) - N(o_{w-1}), \quad w = 1, \dots, W, \quad (9.1)$$

together with point-in-time covariates x_t , while the model of interest remains the linear Hawkes process (8.1) with baseline $s(t)$ (covariate-driven) and kernel ϕ . Timestamps within buckets are unavailable or untrusted.

9.1 The mean-behaviour construction

Definition 9.1 (Mean intensity). The mean-behaviour function of a point process with conditional intensity $\lambda(t)$ is $\xi(t) = \mathbb{E}[\lambda(t)]$, the deterministic expected intensity at t (expectation over histories, covariates held fixed at their realized paths).

Proposition 9.2 (Volterra equation for the linear Hawkes mean). *For the linear Hawkes process (8.1), ξ solves the linear Volterra integral equation of the second kind*

$$\xi(t) = s(t) + \int_0^t \phi(t-u) \xi(u) \, du, \quad (9.2)$$

and in the multivariate case $\xi_\ell = s_\ell + \sum_m \phi_{\ell m} * \xi_m$ componentwise [17].

Proof sketch. Take expectations through the (linear) intensity equation: $\mathbb{E}[\lambda(t)] = s(t) + \mathbb{E} \int_0^t \phi(t-u) dN(u)$. Since $N(t) - \int_0^t \lambda(u) du$ is a martingale, $\mathbb{E}[dN(u)] = \mathbb{E}[\lambda(u)] du = \xi(u) du$, and Fubini gives (9.2). This is the step that fails for the nonlinear (log-link) models of Chapter 8: expectation does not pass through the exponential, which is why the factorization route that kept the sector layer linear is a precondition for everything in this chapter. $\square$

Definition 9.3 (Mean-behaviour Poisson process, MBPP). The MBPP companion of the Hawkes model is the inhomogeneous Poisson process with the deterministic intensity $\xi(t; \vartheta)$ obtained by solving (9.2) at parameter $\vartheta = (s(\cdot), \phi)$. It has the *same parameters* as the Hawkes model and the deterministic compensator $\Xi_w(\vartheta) = \int_{o_{w-1}}^{o_w} \xi(u; \vartheta) du$.

Because the MBPP is Poisson with a deterministic compensator, the interval-censored likelihood follows from the Watanabe characterization already used in Chapter 7: bucket counts are independent Poisson with means Ξ_w , so the negative log-likelihood is, up to constants,

$$-\ell(\vartheta) = \sum_{w=1}^W [\Xi_w(\vartheta) - C_w \log \Xi_w(\vartheta)]. \quad (9.3)$$

This is tractable, smooth in ϑ , and consumes exactly the observed (C_w) — no within-bucket ordering is invented.

What is preserved, what is lost. Preserved: the first moment ($\mathbb{E}C_w = \Xi_w$ holds exactly for the true Hawkes process, by Proposition 9.2), and therefore the *meaning* of every parameter — κ , θ , and the baseline coefficients retain their Chapter 8 interpretations; MBPP estimation is a legitimate estimator of Hawkes parameters, not a different model wearing the same names. Lost: everything beyond the first moment. Hawkes clustering makes bucket counts over-dispersed and positively autocorrelated; the MBPP likelihood treats them as independent Poisson. On the testbed the measured weekly dispersion is about 2 against the Poisson value 1, so (9.3) is a quasi-likelihood in practice and the quasi-Poisson correction of Section 9.4 is *mandatory*: uncorrected standard errors are roughly $\sqrt{2}$ too small, and likelihood-ratio model comparisons are correspondingly anticonservative.

9.2 Solving the Volterra equation

For the exponential kernel the convolution has finite memory in a strong sense: writing $h(t) = \int_0^t \kappa \theta e^{-\theta(t-u)} \xi(u) du$ so that $\xi = s + h$, differentiation gives the scalar linear ODE

$$h'(t) = \kappa \theta s(t) - \theta(1 - \kappa) h(t), \quad h(0) = 0, \quad (9.4)$$

and in the multivariate case a system of dimension M (one excitation state per stream, or per (ℓ, m) pair for full matrices). For the piecewise-constant baselines produced by weekly covariates, (9.4) has closed-form solutions on each interval — exponential relaxation toward $s/(1 - \kappa)$ at rate $\theta(1 - \kappa)$ — so ξ , Ξ_w , and their ϑ -gradients are all exact and cheap; no numerical integrator is needed. The shipped implementation (`code/` directory, `mbpp_solver`) does precisely this interval-by-interval propagation.

Proposition 9.4 (When the Volterra equation is a finite ODE). *The MBPP map $s \mapsto \xi$ of (9.2) is realizable as a finite-dimensional linear ODE with output ξ if and only if the kernel's Laplace transform $\hat{\phi}(z)$ is a strictly proper rational function — equivalently, ϕ is a finite sum of polynomial-times-exponential terms (complex-conjugate pairs allowed, yielding damped oscillations). This is the standard realization theorem of linear systems theory; the proof pointer and the state-space construction are in the appendix (Chapter 10) and mirrored in the code.*

Power-law kernels — popular in the financial-microstructure literature — are not rational and require either quadrature of (9.2) on the grid or an approximating sum of exponentials; both are implemented, but for deal flow at weekly resolution we have found no evidence preferring power-law tails over the exponential family, and the identifiability results below suggest the data could not adjudicate anyway.

9.3 Identifiability loss under censoring: the kappa–theta ridge

Censoring does not degrade all parameters equally. The branching ratio is a *level* parameter; the decay rate is a *timing* parameter; bucketing destroys timing information first.

Proposition 9.5 (The ridge). *Consider the stationary regime with slowly varying baseline and bucket width Δ . Then: (i) $\partial\Xi_w/\partial\kappa$ is of order $\Xi_w/(1-\kappa)$ — an $O(1)$ level effect, since the stationary mean is $s/(1-\kappa)$; (ii) θ enters Ξ_w only through transient (lag) terms of relative size governed by $e^{-\theta(1-\kappa)\Delta}$, so $\partial\Xi_w/\partial\theta$ is small, and oscillates in sign across buckets as the response lag shifts mass between adjacent buckets. Consequently the Fisher information of the likelihood (9.3) is nearly singular along a ridge in the (κ, θ) plane on which the predicted bucket means are constant, and the ridge worsens monotonically with Δ : as $\theta(1-\kappa)\Delta \rightarrow \infty$, θ leaves the first-moment likelihood entirely.*

Proof sketch. With constant s the stationary solution of (9.4) gives $\Xi_w = \Delta s/(1-\kappa)$ exactly — θ -free. Time variation in s reintroduces θ only through the convolution lag of (9.4), whose influence on *bucket-integrated* means decays like the relaxation factor above; differentiating the interval solution of (9.4) in θ and integrating over buckets makes both claims quantitative. Details in the appendix (Chapter 10). $\square$

The second-order structure of the counts (over-dispersion, autocorrelation) does carry θ , but the MBPP construction discards it by design; recovering θ from autocovariances is possible in principle and fragile in practice at our volumes. The operational guidance is in Table 9.1.

Bucket width	Decay θ	Branching κ	Covariate effects
1 day	estimable if $1/\theta \gtrsim$ days	yes	yes
7 days	only if $1/\theta \gtrsim$ weeks	yes	yes (weekly macro)
28 days	no	yes, wider intervals	slow-moving only

Table 9.1: What survives censoring, by bucket width, for the interval-censored likelihood (9.3) with exponential kernels. “Yes” means the profile likelihood has usable curvature at testbed volumes ($\sim 50k$ deals, 12 sectors).

Guidance. Report κ (or $\rho(K)$) and covariate effects as the primary estimands from censored data; they are what the data identify. Fix θ at a domain-informed value (deal-flow response times of one to four weeks) or give it a tight prior, and report sensitivity of κ to that choice — on the testbed $\hat{\kappa}$ moves little across reasonable θ , which is precisely the ridge seen from the other side. Recall also Warning 8.10: the confounding of excitation by latent factors is *worse* under censoring, since remedy (iii) of Chapter 8 — fine timing — is unavailable.

Remark 9.6 (Bayesian treatment of the ridge). A Laplace approximation at the MLE is exactly wrong here: the log-likelihood is flat along a curved (banana-shaped) ridge, so a quadratic expansion at the mode fabricates precision in the ridge direction. A short MCMC over (κ, θ) with the honest prior on θ produces banana-shaped posteriors that display the partial identification faithfully; marginal posterior intervals for κ remain narrow, which is the correct summary of what censored data know.

9.4 Estimation recipes and uncertainty quantification

1. **Point estimation.** Minimize (9.3) with exact gradients from the interval-wise ODE solution; enforce $\kappa \in [0, 1)$ (multivariate: $\rho(K) < 1$) by reparametrization. Multistart is cheap and recommended because of the ridge.
2. **Quasi-Poisson sandwich.** Estimate the dispersion $\hat{\varphi}$ from Pearson residuals ($\hat{\varphi} \approx 2$ weekly on the testbed) and report sandwich standard errors; this corrects the variance for over-dispersion but not for autocorrelation, so pair it with item 3 when intervals matter.
3. **Parametric bootstrap.** Simulate full Hawkes paths from the fitted parameters by thinning (Chapter 8), aggregate them to buckets through the *same* observation grid, refit the MBPP, and use the refit dispersion [14]. Because replicas are genuinely clustered, this captures both over-dispersion and autocorrelation — it is the reference UQ for censored fits.
4. **Posterior summaries.** For (κ, θ) jointly, prefer the MCMC of Remark 9.6 over any quadratic method.

9.5 Worked example: calibrating a censored sector

The recipe list above turns into numbers only when walked through once in full. Take the largest testbed sector (enterprise software): $W = 417$ weekly buckets over eight years, counts C_w between 0 and 31, about 4,100 deals in total, and the weekly macro covariates x_w of Chapter 2. The model is the univariate Hawkes process (8.1) with exponential kernel $\phi(u) = \kappa\theta e^{-\theta u}$ and log-linear piecewise-constant baseline $s(t) = \exp(x_w^\top \beta)$ for t in bucket w , so $\Delta = 7$ days.

Step 1: bucket compensators in closed form. Because s is constant on buckets, the ODE (9.4) solves bucket by bucket. Write $\gamma = \theta(1 - \kappa)$ for the relaxation rate, $h_w = h(o_{w-1})$ for the excitation state entering bucket w , and $h_w^* = \kappa s_w / (1 - \kappa)$ for the level toward which the state relaxes while $s = s_w$. The state-space update across a bucket is

$$h_{w+1} = h_w^* + (h_w - h_w^*) e^{-\gamma\Delta}, \quad (9.5)$$

and integrating $\xi = s + h$ across the bucket gives the compensator

$$\Xi_w = \frac{s_w\Delta}{1 - \kappa} + (h_w - h_w^*) \frac{1 - e^{-\gamma\Delta}}{\gamma}. \quad (9.6)$$

Initialization is $h_1 = 0$ if the window opens at the sector's birth, or $h_1 = h_1^*$ for a stationary start (used here, with the first year treated as burn-in). Equation (9.6) repays reading at its two extremes. As $\gamma\Delta \rightarrow \infty$ the second term vanishes and $\Xi_w \rightarrow s_w\Delta/(1 - \kappa)$: the compensator contains no θ at all — Proposition 9.5 in explicit form. As $\gamma\Delta \rightarrow 0$, $\Xi_w \rightarrow \Delta(s_w + h_w)$: the bucket resolves the excitation state and timing information survives. Weekly deal-flow data sit near the first extreme. Differentiating (9.5)–(9.6) in (β, κ, θ) yields recursions of the same linear form for the derivative states, so exact likelihood gradients come from the same single forward pass.

Step 2: assembling the likelihood. Substituting (9.6) into the interval-censored likelihood (9.3) gives an objective whose evaluation, with gradient, is one sweep of W closed-form updates — microseconds per call, so the multistart of Section 9.4 costs nothing. We reparametrize $\kappa = \text{logit}^{-1}(\tilde{\kappa})$ and $\theta = e^{\tilde{\theta}}$ to keep the constraints implicit and the Hessian usable.

Step 3: profiling, and the ridge one actually sees. Define the profile $\ell_p(\kappa, \theta) = \max_{\beta} \ell(\beta, \kappa, \theta)$ and plot its contours over $(\kappa, 1/\theta)$. Two features appear on the testbed sector. In the κ direction the profile is cleanly quadratic: moving κ by ± 0.05 from its optimum raises the profile deviance by about six units. In the θ direction it is a plateau: as $1/\theta$ sweeps from four days to six weeks the profile deviance changes by less than one unit — far inside noise. The plateau is not exactly parallel to the θ axis: refitting with $1/\theta$ fixed at 7, 14, and 28 days gives $\hat{\kappa} = 0.17, 0.18$, and 0.20 — the gently curved banana of Remark 9.6 seen edge-on. The mechanism is Proposition 9.5: θ enters the bucket means only by shifting covariate-response mass between adjacent buckets, and weekly macro covariates are themselves too smooth to make that shift visible.

Step 4: quasi-Poisson correction. The Pearson dispersion at the fit,

$$\hat{\phi} = \frac{1}{W - p} \sum_{w=1}^W \frac{(C_w - \hat{\Xi}_w)^2}{\hat{\Xi}_w} = 2.1 \quad (9.7)$$

with p fitted parameters, so all model-based standard errors are inflated by $\sqrt{2.1} \approx 1.45$. This magnitude is not a red flag: the fitted clustering itself predicts over-dispersion of order $(1 - \kappa)^{-2} \approx 1.5$ for wide buckets, and the parametric bootstrap of Section 9.4 reproduces dispersions near 1.9 from the fitted model — most of the excess variance is the model’s own clustering, not lack of fit. The naive standard error for κ is 0.021; the quasi-Poisson standard error is 0.030.

Step 5: reporting. The reported line for this sector reads: $\hat{\kappa} = 0.18$, quasi-Poisson 95% interval $[0.12, 0.24]$, parametric-bootstrap interval $[0.11, 0.25]$ (wider because replicas carry autocorrelation; it is the headline interval). The decay is *not* estimated: $1/\theta$ is fixed at 14 days on domain grounds (deal-flow response times of one to four weeks), and the sensitivity line $\hat{\kappa} \in [0.17, 0.20]$ for $1/\theta \in [7, 28]$ days is printed next to the estimate, verbatim. Baseline coefficients β are reported with quasi-Poisson standard errors exactly as in Chapter 7. Table 9.2 restates the reporting rule as a function of bucket width; it is the operational companion of Table 9.1.

Bucket width	Covariates	κ	θ	Recommended reporting
1 day	sharp	sharp	partial	(κ, θ) jointly, with profile contours or posterior
7 days	sharp (weekly macro)	usable	none	κ with bootstrap CI; θ fixed, sensitivity line attached
28 days	slow-moving only	weak	none	interval or posterior for κ , labelled conditional on θ

Table 9.2: Qualitative identifiability and reporting recommendations by bucket width, from the worked example of Section 9.5. “Partial” means the profile has curvature only when $1/\theta$ is comparable to the bucket width; “none” means fix θ and disclose the choice.

9.6 Model checking under censoring

The time-rescaling diagnostics of Chapter 8 require timestamps; binned counts need binned diagnostics. Four checks cover what bucket data can falsify.

Pearson residuals. Plot $r_w = (C_w - \hat{\Xi}_w)/(\hat{\varphi} \hat{\Xi}_w)^{1/2}$ against the fitted $\hat{\Xi}_w$ and against calendar time. A funnel against fitted values indicates mean–variance misfit beyond a constant dispersion; waves against time indicate baseline misspecification (seasonality or a missed macro covariate, Chapter 2); isolated spikes are usually vendor data faults, not model failure. The autocorrelation of r_w is *expected* to be mildly positive under the fitted Hawkes model — judge it against the bootstrap band below, never against white-noise limits.

PIT histograms. For discrete counts the probability integral transform is not uniform even under the true model; use the randomized version $u_w = F_w(C_w - 1) + V_w [F_w(C_w) - F_w(C_w - 1)]$ with V_w i.i.d. uniform and F_w the Poisson($\hat{\Xi}_w$) CDF [6]. Two caveats: average the histogram over many randomization draws (a single draw adds visible noise at $W \approx 400$), and expect a U-shape from over-dispersion even when the mean structure is correct — the PIT compares candidate *mean* structures; it cannot certify the Poisson second moment, which we already know is wrong.

Dispersion test. Report $\hat{\varphi}$ of (9.7), but do not refer it to its Poisson chi-squared null: the fitted Hawkes model itself implies $\varphi > 1$, so “significant over-dispersion” is not evidence against the model. The correct reference distribution is the bootstrap one.

Parametric-bootstrap check. Simulate full Hawkes paths at $\hat{\vartheta}$ by thinning (Chapter 8), bin them through the *same* observation grid, and compare observed count statistics with the replica distributions [14]: the dispersion, count autocorrelations at lags one to four, the fraction of zero buckets, and the largest weekly count. On the testbed sector the observed dispersion 2.1 falls inside the replica interval [1.6, 2.3] and the observed lag-one autocorrelation 0.16 inside [0.08, 0.21] — no evidence of misfit beyond the first moment. Choose the statistics before looking at the replicas.

Warning 9.7 (What censored data cannot falsify). None of these checks sees inside a bucket. Kernels that redistribute excitation mass *within* the bucket width produce nearly identical laws for (C_w) , so a censored fit that passes every diagnostic above is still consistent with badly wrong within-week dynamics — fast cascades over days and smooth drift over the week are indistinguishable at $\Delta = 7$ days. In particular, the fixed θ of Section 9.5 is an *assumption*, and no bucket-level diagnostic can validate it. Never let a censored fit support a timing claim (“sectors respond within days”); such claims require trusted timestamps and the machinery of Chapter 8.

9.7 Accelerating repeated solves: physics-informed operator learning

Calibration needs a few thousand Volterra solves; scenario stress testing and walk-forward simulation need millions — every macro scenario, every bootstrap replica, every counterfactual baseline is a fresh $(s, A) \mapsto \xi$ solve, where A collects the excitation parameters. A neural operator trained on solver output, with the Volterra residual of (9.2) as a physics-informed penalty, learns this map well enough to replace the ODE solver in such loops. The shipped implementation reaches sub-2% relative L^2 error for 3–8 sector systems under the full training protocol (a quick 600-epoch demo run reaches 8.2%), at about 3ms per solve — orders of magnitude faster than solving per scenario, and batchable on accelerators.

Warning 9.8 (An operator is only trustworthy inside its training range). The operator has no notion of extrapolation: it was trained on subcritical instances ($\rho(K) < 1$) within specific baseline and parameter ranges, and degrades silently outside them. Run the shipped accuracy

report against the exact solver on *your* parameter ranges before use, and keep exact solvers as the default for calibration — optimizer gradients amplify surrogate error, and calibration is a few thousand solves anyway. The operator is an accelerator for many-solve downstream loops, not a pillar of the methodology.

9.8 Choosing an observation model

Table 9.3 summarizes the choice among the three sector-layer estimators now available: the event-time Hawkes likelihood of Chapter 8, the MBPP-censored likelihood of this chapter, and the discrete count GLM of Chapter 7 [6]. The rule of thumb that survives all our experiments: trust event time only when you trust the timestamps to well within the kernel time-scale; otherwise the MBPP gives the same parameters with honest inputs, and when θ is not needed at all, the discrete GLM is the simplest thing that works.

	Event-time Hawkes	MBPP (censored)	Discrete GLM
Timestamp quality	exact, jitter $\ll 1/\theta$	any (uses counts only)	any (uses counts only)
Decay θ	estimable	fix or tight prior (Prop. 9.5)	not a parameter
Deal volume	needs many events per stream	works at low counts	works at low counts
Dispersion	implied by clustering	quasi-Poisson correction mandatory	quasi-likelihood native
Compute cost	$O(n)$ per stream, exact	interval ODE, exact gradients	GLM fit, trivial
Recommended UQ	sandwich + thinning bootstrap	parametric bootstrap; MCMC for (κ, θ)	quasi-Poisson sandwich

Table 9.3: Decision table for the sector ground process under the three observation models. Full measured comparisons are in Chapter 10.

One closing remark ties this chapter to the manual’s through-lines. Censoring is not merely a nuisance that inflates variance: it *changes what is identified* — θ exits, κ stays — and honest reporting means saying so, with posterior bananas rather than fabricated ellipses. The final decision tables, and the measured evidence behind every number quoted here, are collected in Chapter 10.

Chapter 10

Choosing a Method: Decision Framework and Testbed Evidence

Chapters 3 through 9 each argued for a method inside its own regime of observability. This chapter is the exit interview. Given a dataset, a question, and a tolerance for assumptions: which pipeline do we run, what does it estimate, how do we attach uncertainty, and what is most likely to go wrong? We compress the manual into two decision tables (Section 10.1), then present the measured evidence from the synthetic testbed behind the recommendations (Section 10.2). Section 10.3 states two standing benchmarks that any revision of the pipelines must pass or explicitly fail; Section 10.4 explains why the synthetic worlds make the causal pipeline of Chapter 5 *testable* rather than merely arguable. Sections 10.5–10.7 close with the uncertainty-quantification summary, the production checklist, and the honest list of what this manual does not solve.

10.1 Two decision tables

There are two natural entry points. Either we start from the *data situation* (what the vendor gave us, which geography, how reliable the timestamps are) and ask what is estimable at all, or we start from the *question* and ask which pipeline answers it. Table 10.1 takes the first route; Table 10.2 the second. Both cite the chapter developing each recommendation; neither is a substitute for reading it. The most common failure we observe in practice is running a question-driven pipeline on a data situation that cannot support its estimand — most often event-time excitation models on bucketed exports (row three of Table 10.1), or causal claims read off predictive fits (rows five and six).

10.2 Testbed evidence

All quantitative claims below come from the synthetic testbed that ships with the code (Appendix C), seed 0 throughout. Two worlds are simulated, both mirroring the schemas of Chapter 2. *Regime A* is well specified for our tools: a weekly two-stage DGP over 520 weeks, 12 sectors, and 8,000 firms (7,031 tracked), in which sector counts follow a count GLM with lagged excitation and firms are selected by a risk-set softmax with a 26-week cooldown indicator (true coefficient approximately -2.7); it produces 50,036 observed deals. *Regime B* is deliberately misspecified: a daily firm-level block Hawkes process with 30-day-half-life self-inhibition, 7-day-half-life sector contagion, and a latent quality mark, producing 49,788 deals fitted with the same weekly tools. Both worlds pass through an observation-noise layer calibrated to Chapter 2: 28% undisclosed deal sizes, a further 15% only estimated (18% relative error), 55% missing valuations, 3% sector mislabels, 1.2% duplicates, date jitter of up to 6

Scenario	Recommended model	Estimand	UQ method	First-order risk
EU fundamentals panel	feature-engineered XG-Boost on point-in-time panel (Ch. 4)	$\mathbb{E}[\text{success} \mid X]$; EBITDA-growth quantiles	split conformal / CQR	snapshot-field leakage
US marks, reliable day timestamps	marked Hawkes, ground $\times$ mark two-stage (Ch. 8)	excitation kernel; branching ratio	Laplace interval; parametric bootstrap	latent common factor inflates excitation
US weekly or bucketed exports	covariate count GLM; MBPP interval-censored likelihood (Ch. 7, 9)	covariate semi-elasticities; effective radius κ	quasi-Poisson sandwich; cluster bootstrap	aggregation manufactures apparent excitation
Short history or tiny sectors	pooled Poisson GLM with hierarchical shrinkage (Ch. 7)	sector rate levels and trends	parametric bootstrap	overdispersion; a few large weeks dominate
Causal: financing effect on outcomes	backdoor adjustment; AIPW / DML (Ch. 5)	ATT of a round on survival / growth	cluster bootstrap CI; E-value	unobserved quality selects into treatment
Causal: contagion between firms or sectors	event-time Hawkes with factor proxies and sensitivity analysis (Ch. 8)	excitation net of common shocks	profile CI; sensitivity band	radius absorbs latent risk appetite
Macro-effect question	count GLM with publish-lag exposure design (Ch. 6, 7)	rate semi-elasticity to macro series	HAC / sandwich	publication-lag leakage; few macro cycles

Table 10.1: Master decision table: from data situation to method. Every row assumes the point-in-time discipline of Chapter 2 has already been enforced.

Question	Pipeline
Success prediction	feature-engineered XGBoost [9] with conformal intervals [15] on the point-in-time panel (Ch. 4)
Who-gets-funded ranking	two-stage sector count process plus funded-pool conditional logit [8] with an explicit newcomer option (Ch. 8)
Contagion measurement	event-time Hawkes [1, 17] with macro-factor proxies and sensitivity analysis (Ch. 8)
Macro sensitivity	covariate count GLM with exposure and publish-lag designs (Ch. 6, 7)
Censored or bucketed exports	MBPP interval-censored likelihood; report the effective radius κ , never the raw kernel parameters θ (Ch. 9)

Table 10.2: Question-driven decision table: from estimand to pipeline.

days, and 12% of firms untracked. Regime A shows what the estimators recover when their assumptions hold but the data are dirty; regime B shows what the same numbers mean when the world is event-time and the export is not.

10.2.1 Stage 1: sector-level count model recovery

Quantity	Regime A	Regime B
Covariate β correlation with truth	0.675	0.37
Intercept bias	−0.09	—
Excitation–entry correlation	0.27	—
Fitted effective spectral radius	0.355	0.57
True effective spectral radius	0.172	(not weekly-defined)
Held-out Pearson dispersion	—	2.72

Table 10.3: Stage-1 recovery (seed 0). Regime A is well specified; regime B is a daily block Hawkes fitted with weekly tools.

Table 10.3 carries three lessons. First, observation noise alone attenuates: in regime A, where the fitted class contains the truth, the covariate coefficients correlate 0.675 with the true β — roughly 25% attenuation, consistent with errors-in-variables from estimated sizes and jittered dates. Signs and rankings survive; magnitudes are lower bounds. Second, the fitted effective radius of 0.355 against a true 0.172 is *not* noise: it is the latent serially correlated risk-appetite factor being absorbed into the excitation term, exactly the confounding mechanism of Section 10.4. Third, regime B shows what misspecification does: coefficient correlation collapses to 0.37, the fitted radius of 0.57 is mostly a daily-to-weekly aggregation artifact, and the held-out Pearson dispersion of 2.72 flags the failure — hence the dispersion check on the production checklist. What generalizes: the direction of every bias (attenuation, radius inflation, dispersion under aggregation). What is synthetic-specific: the exact factors, which depend on the noise calibration above.

10.2.2 Stage 2: funded-pool conditional logit

Metric	Regime A	Regime B
Held-out NLL per event	3.346	3.796
Random-choice NLL per event	4.011	4.172
Top-5 accuracy	0.446	0.316
Log-gap coefficient	+1.146	+0.437
Newcomer ASC	6.856	4.691
Newcomer share of events	≈ 0.27	≈ 0.27
Dropped immediate-repeat events	35	2

Table 10.4: Choice-model performance: 64-candidate sampled softmax on truthful risk sets, held-out weeks, seed 0.

The model in Table 10.4 is the funded-pool conditional logit with a 64-candidate sampled softmax (consistency argument in Appendix A) on truthful risk sets. In regime A the recovered feature weights are: raised-to-date 0.336 (truth 0.30), employees 0.253 (truth 0.30), age 0.054 (truth −0.15), stage 0.132 (truth 0.15). Three of four are recovered to within attenuation; the age sign is missed because the 26-week cooldown indicator and the fitted log-gap term (+1.146) absorb the age-adjacent recency structure — the mechanism is captured but attributed to timing features rather than to age. The large newcomer alternative-specific constant is not

a pathology: it prices the size of the unobserved entrant pool, and by the integrated-entrant identity of Appendix A it is interpretable only jointly with the assumed pool size. In regime B every number degrades but the ordering of models is preserved, which is the property we actually rely on when ranking candidates.

10.2.3 Risk-set contamination

Risk-set construction $\mathcal{R}_t$	Held-out NLL	Weight behaviour
Naive observed-active mask	3.886	inverted: recency +1.94 vs. true 0
Oracle membership mask	3.558	all weights recovered

Table 10.5: Risk-set contamination, regime A. The only difference between rows is who is deemed at risk at each event time.

Warning 10.1. Risk-set validity is first-order. In Table 10.5 the naive mask — treating a firm as at risk only when the vendor happens to show it active — does not merely cost likelihood; it *inverts* the recency weight to +1.94 against a true value of 0, because firms enter the observed mask by transacting. No feature engineering repairs an estimate whose conditioning set is wrong. Audit $\mathcal{R}_t$ before touching the utility specification.

10.2.4 A stability episode, and the operator surrogate

Feedback specification	Eff. radius	Outcome	Consequence
Log-link, raw-count feedback	0.35	metastable	50k-event world tipped to 169k events
Log-link, raw-count feedback	0.15	stable	simulation matches fitted moments

Table 10.6: Stability episode. Linear-Hawkes intuition (explosion only at radius ≥ 1) does not transfer to log-link feedback on raw counts.

Table 10.6 records the incident behind checklist item 4: a log-link model feeding back raw counts at effective radius 0.35 — comfortably subcritical by linear intuition — is metastable, and one excursion tipped a 50,000-event world to 169,000 events. Stability theory for nonlinear self-excitation [5, 6] makes clear that the linear threshold is not the relevant boundary once the link is convex in the feedback; bounded transforms such as $\log(1 + n)$ restore contractivity. Separately, the PINO operator surrogate for the MBPP Volterra equation (derivation in Appendix A) reaches sub-2% relative L^2 error across 3–8 sector configurations under the full training protocol, at roughly 3 ms per solve; a 600-epoch demonstration run reaches only 8.2%, so the cheap configuration is a demo, not a substitute.

10.3 Standing benchmarks

We maintain two permanent falsifiability devices. They are not experiments run once; they are challenges every revision of the pipelines must confront, with targets frozen from Section 10.2.

Benchmark 1 (XGBoost ranking baseline). Fit a gradient-boosted ranker [9] on *identical* choice sets — same 64 candidates, same truthful risk sets, same temporal splits — and compare top-5 accuracy against the conditional logit’s 0.446 (regime A) and 0.316 (regime B). Decision rule: if the boosted ranker wins materially, the linear-in-features utility is missing features or nonlinearity, and the remedy is feature work, not model exotica; if it does not, we keep the logit for its interpretable weights and the ASC decomposition. Either outcome is informative; a run that skips the identical-choice-set discipline is informative about nothing.

Benchmark 2 (non-homogeneous Poisson null). For any claimed excitation effect, fit the nested ladder: covariates-only inhomogeneous Poisson (simulated by thinning [4]), then covariates plus excitation, then the full event-time timing model, and decompose the held-out improvement across the rungs. If the excitation rung contributes approximately nothing at this data scale, drop it: an excitation term that does not pay its way in held-out likelihood is a liability in simulation (Table 10.6) and an attractive nuisance in interpretation (Table 10.3). The residual-analysis toolkit of [3] supplies the diagnostics for each rung.

10.4 Causal validation on synthetic ground truth

For a manager deciding whether to trust the causal layer, the testbed changes the conversation: because the DGP is fully known, the pipeline of Chapter 5 is *testable*, not merely defensible. The protocol: (i) generate a binary treatment (say, receiving a growth round) with known selection on observables *and* on the latent quality mark; (ii) estimate the effect four ways — naive difference in means, backdoor-adjusted regression [10], AIPW, and double/debiased ML [11, 13] — and compare each to the known truth; (iii) attach cluster bootstrap confidence intervals [14]; (iv) compute E-values [16] and check them against the known strength of the latent confounding. The code asserts the expected qualitative outcomes as hypotheses: the naive estimate is biased in the direction of selection; backdoor adjustment removes the observable component; AIPW matches backdoor with better robustness to nuisance misspecification; and the residual gap to truth is attributable to latent quality, with magnitude consistent with the computed E-value. Where an assertion fails, the pipeline — not the world — is wrong.

The count-process analogue is already in evidence: the fitted effective radius of 0.355 against a true 0.172 (Table 10.3) is the point-process version of a confounded treatment effect. “Past events excite future events” is a causal claim in which past events are the treatment; the latent serially correlated risk-appetite factor is the unmeasured common cause, and its omission violates exchangeability in precisely the sense of [12]. The excitation-vs-latent-factor experiment therefore doubles as the standing reminder that no in-sample fit statistic distinguishes contagion from common shocks; only design (factor proxies, sensitivity bands, and, on the simulator, do-style interventions) does.

10.5 Uncertainty quantification: summary

Method	Technique	Guarantee	Key assumptions
Count GLM (Ch. 7)	quasi-Poisson sandwich	asymptotic SE validity for mean parameters	correct mean function; dispersion as nuisance
Two-stage / choice (Ch. 8)	cluster bootstrap by week	asymptotic CI coverage	weak dependence across clusters; clustering at the right level
XGBoost prediction (Ch. 4)	split conformal / CQR [15]	finite-sample marginal coverage	exchangeability; weakened under temporal splits
Hawkes MLE (Ch. 8)	Laplace / posterior interval	local asymptotic normality	correct specification; stationarity
MBPP (Ch. 9)	parametric bootstrap	coverage under the fitted DGP	model correctness; report κ , not θ

Table 10.7: UQ summary: what each interval means and what it silently assumes.

Table 10.7 is deliberately blunt about guarantees. Only the conformal row carries a finite-sample statement, and even that one is marginal (not conditional) and degrades when temporal splits break exchangeability. Every other interval in this manual is asymptotic and

specification-dependent; the dispersion of 2.72 in Table 10.3 means naive Poisson standard errors in regime B are understated by about $\sqrt{2.72} \approx 1.65$ before correction.

10.6 Production checklist

1. **Point-in-time audit.** Reconstruct every feature from data knowable at the prediction date; join macro series on `publish_date`, never `ref_date` (Ch. 2, 6).
2. **Deduplication.** Apply the 3-day-window convention before any counting; the testbed carries 1.2% duplicates and date jitter up to 6 days, and both alias into spurious short-lag excitation.
3. **Risk-set truthfulness audit.** Validate the membership mask $\mathcal{R}_t$ against a labelled subsample of known entrants and exits; contamination inverts weights (Table 10.5).
4. **Effective-radius guard.** Refuse to simulate any fitted model with effective radius ≥ 0.95 ; for log-link raw-count feedback refuse well below that (metastability observed at 0.35, Table 10.6), or use bounded transforms.
5. **Dispersion correction.** Report the held-out Pearson dispersion and scale standard errors accordingly; treat dispersion far above 1 as a specification alarm, not a nuisance.
6. **Universe coverage and newcomer calibration.** Track the untracked-firm share (12% in the testbed) and recalibrate the newcomer ASC whenever the pool definition changes; the ASC is meaningless detached from the pool (Appendix A).
7. **Temporal splits only.** No random splits anywhere in the stack — not for model selection, not for conformal calibration, not for the benchmarks of Section 10.3.
8. **Seeds and reproducibility.** Every reported number carries a seed (this chapter: seed 0) and a regeneration command; Appendix C maps each table here to its entry point.

10.7 Limitations and roadmap

Four limitations stand. First, entity resolution is single-vendor: without a second source, `company_id` merges and splits cannot be audited, and every firm-level count inherits that fragility. Second, latent-factor confounding is mitigated, not solved: the 0.355-versus-0.172 radius gap persists under our best factor proxies, and we know of no observational design that closes it; sensitivity reporting is the honest ceiling. Third, cross-border generalization is untested: the two synthetic worlds mirror the EU-fundamentals and US-marks regimes respectively, and nothing here licenses transporting a model fitted in one to the other. Fourth, the outcome space is too coarse: “next event” conflates a follow-on round with an acquisition and with quiet failure. The next modelling step is competing risks — cause-specific hazards for M&A versus next round in the spirit of [7], marked into the point-process machinery of [2] — for which the survival scaffolding in the code base (Appendix C) is already laid.

Chapter 11

The Code Plan: What We Test, How, and What We Have Tested

Chapter 10 chose the methods; this chapter is the engineering contract that keeps them honest. It answers three questions in order. What kinds of tests does the code base run, and what does each kind actually certify (Section 11.1)? What has already been tested on the synthetic worlds, and what did we learn — including from the tests the data generator itself failed (Section 11.2)? What remains to be run before the calibration campaign is complete, stated as pass/fail specifications rather than intentions (Section 11.3)? Section 11.4 closes with the continuous-integration gates and the definition of done. Where a number already appears in a table of Chapter 10 we cite it rather than repeat it; this chapter adds the numbers that belong to the engineering log rather than to the decision framework.

11.1 Testing philosophy and harness

A calibration code base fails in two distinct ways: the code computes the wrong number, or the code computes the right number for a procedure that cannot recover the truth. Conventional software tests catch only the first. We therefore run four layers of tests, each certifying something the previous layer cannot.

Layer 1: unit tests on hand-computable micro-examples. Every nontrivial data transformation has a companion example small enough to verify by hand. Three anchor cases: (i) a three-firm choice set with known funding histories, where the pools, the excluded most-recently-funded firm, the newcomer row, and the winner labels produced by `build_choice_sets` are asserted against a worked-out table; (ii) a two-bucket compensator for an exponential-kernel intensity, checked against the closed-form integral to machine precision; (iii) deduplication on a constructed deal table containing exact duplicates, 2-day near-duplicates, and legitimate consecutive-week repeats, asserting which rows survive. These tests certify arithmetic and indexing, nothing more — but when a recovery test fails, they are what lets us exonerate the plumbing in minutes.

Layer 2: equivalence tests between fast and reference fitters. The package ships slow, transparent loop fitters and vectorized fast fitters (`fit_sector_glm_fast`, `fit_choice_fast`). The fast versions exploit structure: the sector GLM is separable per receiving sector, so twelve small concave problems replace one large one; the ranker is a padded-tensor batch over events and candidates. Speed is worthless if the optimum moves, so on subsamples both implementations must agree at the optimum — same loss to 10^{-6} , parameters to matching

tolerance — before a fast fitter may be used in any experiment. The check is rerun whenever either implementation changes.

Layer 3: statistical recovery tests. These are the scientific tests. We fit on synthetic data whose ground truth we wrote down, and a run passes when the truth lies within 2 standard errors of the estimate across 5 seeds. A recovery failure with passing layers 1–2 is a finding about the *method* — identifiability, confounding, misspecification — and several such findings became sections of this manual. Recovery tests are the only layer that can certify the phrase “the pipeline works.”

Layer 4: invariant and leakage audits. Two standing assertions run over the whole feature stack. The point-in-time assertion recomputes every feature using only data truncated at time t and requires the result to be bit-identical to the production value: any divergence means some path read the future. The oracle-isolation assertion requires that no fitting path opens `ground_truth.npz` or `private_deal_map.csv`; oracle files are for evaluation only, and a fit that touches them is void regardless of its metrics. Both audits are structural: they do not check that a number is right, they check that it was legal to compute.

Layer	What it certifies	Cadence
Unit (micro-examples)	arithmetic and indexing of each transformation	every commit
Equivalence (fast vs. loop)	the vectorized optimum is the reference optimum	every commit
Recovery (synthetic truth)	the procedure can find truth within 2 SE, 5 seeds	nightly, small configs
Invariant / leakage	features are point-in-time; oracle files untouched	blocking, every commit

Table 11.1: The four test layers and what a green result actually means.

Three reproducibility rules bind everything above. First, every number reported in this manual is regenerable from one seeded entry point — for the worlds themselves,

```
PYTHONPATH=. python -m synthetic.generate --regime A --seed 0
```

and analogously one script per experiment table with a `--seed` argument. Second, temporal splits only, everywhere: train weeks [0, 416) against test weeks [416, 520) on the weekly track, an 80% day quantile on the choice track, and no random split anywhere in the stack, including model selection. Third, runtimes are logged alongside results, because a fitter that silently becomes ten times slower is a regression even when its numbers do not move.

11.2 What we have tested so far

Everything in this section is measured, seed 0, on the two synthetic worlds of Chapter 1: regime A, the well-specified weekly two-stage world (50,036 deals), and regime B, the deliberately misspecified daily block Hawkes world (49,788 deals), both spanning 520 weeks, 12 sectors, and 8,000 firms, and both passed through the full observation-noise stack. The headline recovery tables live in Chapter 10; here we record the log entries around them — what broke, what was measured on the way, and what each episode taught the production design.

11.2.1 Tests the world itself failed

The generator was not born stable. We present its construction-time failures as what they were: recovery tests run against the simulator, which the simulator failed until fixed. Each failure is a preview of a production incident, because every mechanism involved — log-link feedback, cumulative covariates, deduplication — exists in the real pipeline too.

First, metastability. With log-link raw-count feedback at an effective spectral radius of 0.35 — comfortably subcritical by linear-Hawkes intuition — the world was metastable: a single ordinary burst tipped the roughly 50k-event trajectory into a supercritical excursion that finished at 169,436 events. The shipped generator runs at radius 0.15 with a self-consistent intercept calibration, and the episode is analyzed as a stability result in Chapter 10. Second, rich-get-richer collapse. With unbounded cumulative features (raised-to-date, employees) feeding positive selection weights and no exit mechanism, deal flow concentrated absurdly: one firm collected 422 deals. The fix was structural, not a tuning knob — size-damped employee growth capped at 2,000 and stage-graduation exits that retire persistent winners — after which the maximum per-firm count fell to 32 and the 99th percentile to 17. Third, deduplication overreach. A naive 7-day same-company window, meant to absorb the 1.2% injected duplicates under 6-day date jitter, destroyed roughly 38k legitimate consecutive-week deals — most of the repeat-funding signal the model exists to capture. The shipped window is 3 days, validated on the layer-1 constructed table.

Failure	Symptom	Fix	Post-fix check
Log-link metastability	radius 0.35 world tipped from ~50k to 169,436 events	radius 0.15; self-consistent intercept calibration	event counts stable across seeds
Rich-get-richer collapse	one firm with 422 deals	size-damped employee growth (cap 2,000); stage-graduation exits	max 32; p99 = 17
Dedup overreach	7-day window destroyed ~38k legitimate repeat deals	3-day window	constructed-table unit test

Table 11.2: Generator failures during construction, treated as recovery tests the world initially failed.

Remark 11.1. These are not embarrassments to hide but the cheapest incidents we will ever have. Each cost hours on synthetic data; the identical mechanisms in production would cost quarters of misallocated attention before anyone noticed. The production lessons are collected in Table 11.4.

11.2.2 Stage-1 recovery: the weekly sector GLM

Training on weeks $[0, 416)$, the weekly sector GLM recovers regime A — the world it is specified for — up to observation noise: covariate β correlation with truth 0.675, roughly 25% attenuation from the errors-in-variables induced by estimated sizes and jittered dates; intercept bias -0.09 ; excitation-entry correlation 0.27. The one systematic failure is causal, not numerical: the fitted effective spectral radius is 0.355 against a true 0.172, because the latent serially correlated risk-appetite factor is absorbed as spurious excitation. On regime B the same tools face a daily block Hawkes world and degrade as they should: β correlation 0.37, fitted radius 0.57 (largely a daily-to-weekly aggregation artifact), and a held-out Pearson dispersion of 2.72 that correctly flags the misspecification. The full table is in Chapter 10; the log entry here is the engineering one: the fast fitter reproduces the loop fitter’s optimum (layer 2) and completes in 0.2 seconds where the loop fitter takes minutes, which is what makes the 5-seed, multi-variant forward plan of Section 11.3 affordable at all.

11.2.3 Choice-model recovery: the funded-pool conditional logit

The stage-2 model is the funded-pool conditional logit on truthful risk sets with a 64-candidate sampled softmax [8], split at the 80% day quantile. On regime A it reaches a held-out NLL per event of 3.346 against 4.011 for random choice and a top-5 accuracy of 0.446; on regime B, 3.796 against 4.172 and top-5 of 0.316. The newcomer option wins roughly 27% of events in both worlds. The exclusion rule dropped 35 immediate-repeat events in A and 2 in B. Construction of the choice tensors takes 6.4 seconds and the fit about 12 seconds. The recovered regime-A weights are worth a table, because the pattern of what recovers and what transmutes is a finding in itself.

Feature	Fitted (regime A)	Truth
Raised-to-date	0.336	0.30
Employees	0.253	0.30
Age	0.054	-0.15
Stage	0.132	0.15
Log-gap (time since last funding)	+1.146	(26-week cooldown ≈ -2.7)
Newcomer ASC	6.856	(prices the entrant pool)

Table 11.3: Choice-model weight recovery, regime A, seed 0. Regime B analogues: log-gap +0.437, newcomer ASC 4.691.

Raised, employees, and stage recover to within attenuation. The age sign is missed because the true mechanism there is recency, and the fitted positive log-gap weight is the model’s translation of the true cooldown: a 26-week suppression indicator of about -2.7 reappears as “longer gap, higher score,” which is the same economics in a different parametrization. In regime B the softer log-gap weight of +0.437 matches the smooth 30-day-half-life inhibition of that world. Whether the log form is the *right* carrier of this structure — as opposed to binned dummies or an explicit exponential decay — is exactly experiment E2 of the forward plan.

11.2.4 Risk-set contamination

One experiment was promoted from finding to assumption. With a naive observed-active mask — a firm counted as at risk only when the vendor shows it active — the ranker does not merely lose likelihood (held-out NLL 3.886 versus 3.558 for the oracle mask); it *inverts* the economics, fitting a recency weight of +1.94 where the truth is 0, because firms enter the observed mask by transacting. The oracle mask recovers all weights. The full table and the warning it earned are in Chapter 10. The consequence for the code plan: truthful risk sets are now a stated design assumption, and the contaminated-mask fit is kept as a standing audit — if a production risk set is ever inferred rather than given, this is the first experiment to rerun.

11.2.5 Operator smoke test

The physics-informed neural operator machinery for the compensator equations runs end to end: data generation, training, and inference are wired and reproducible. A deliberately cheap 600-epoch demonstration reaches 8.2% relative L^2 error — machinery confirmed, accuracy not yet claimed. Under the full training protocol the operator reaches sub-2% for 3–8 sector configurations at roughly 3 ms per solve, which is the regime that matters for the many-repeated-solves role it is assigned in experiment E4.

Finding (measured)	Production lesson
Log-link raw-count feedback metastable at radius 0.35 (50k $\rightarrow$ 169,436 events)	guard simulation well below the linear threshold for log-link feedback; prefer bounded transforms of counts
One firm collected 422 deals under unbounded cumulative features with no exits	cap or damp cumulative covariates; model exits explicitly in any forward-simulation loop
7-day dedup destroyed $\sim$ 38k legitimate repeat deals	dedup windows are modelling decisions; validate against constructed cases before counting anything
Fitted radius 0.355 vs. true 0.172 under a latent risk-appetite factor	never read fitted excitation causally without factor proxies and a sensitivity band
Naive risk-set mask inverted the recency weight (+1.94 vs. true 0)	treat risk-set truthfulness as an audited input, not a fittable nuisance

Table 11.4: The five log entries of this section, restated as production rules.

11.2.6 From findings to production rules

11.3 The forward test plan

What follows is the committed test plan: each entry states its purpose, its procedure in two or three lines, and a pass criterion frozen *before* the run, so that results can fail. Unless stated otherwise, every experiment runs on both regimes at 5 seeds with error bars, under the splits and audit rules of Section 11.1. All entries are currently *planned*; Table 11.5 summarizes.

E1 — When/where layer: weekly GLM versus event-time Hawkes. *Purpose:* quantify what day-resolution timestamps buy over weekly buckets, and how badly latent confounding contaminates fitted excitation. *Procedure:* fit the weekly GLM and a multivariate event-time sector Hawkes (exponential kernel, log-linear macro baseline) on the same streams; rerun the event-time fit with dates bucketed at 1d, 7d, and 28d, reporting bias and SE of the excitation scale κ and decay θ at each width; separately, regenerate regime A with `latent_strength` $\in \{0, 0.15, 0.3\}$ at fixed true radius and plot fitted effective radius against latent strength, with and without mitigations (lagged aggregate count covariate, off-diagonal penalties). *Pass:* event-time fit beats the weekly GLM on held-out NLL on regime B; θ recovered within 2 SE at day resolution on B while the κ - θ ridge visibly reopens by 28d; the confounding curve is monotone and yields a stated bucket-width recommendation for real data.

E2 — Which-firm layer: choice-model sub-experiments. *Purpose:* settle the four open design choices inside the conditional logit before it is frozen for production. *Procedure and pass criteria, per sub-experiment:*

1. *Gap functional form.* Fit log-gap versus binned gap dummies (<1m, 1–3m, 3–6m, 6–12m, >12m) versus exponential decay $\exp(-\text{days}/\tau)$ with $\tau \in \{30, 90, 180\}$ days; select by held-out NLL and plot each implied hazard-versus-gap curve against truth. Pass: the binned fit recovers the true 26-week step in regime A and the smooth exponential decay in regime B.
2. *Newcomer designs N1–N4.* Compare the bare ASC against the context-covariate ASC (sector heat and macro features), the representative-entrant construction (plug-in and integrated Gaussian variants), the nested two-step, and the immigrant/offspring split that aligns first financings with the Hawkes background stream. Pass: the recommended design calibrates quarterly newcomer share within 2 percentage points, with incumbent weights materially unchanged.

3. *Sampled-softmax stability.* Refit at `max_candidates` $\in \{32, 64, 128\}$ and with the full denominator on a 10% subsample. Pass: weights stable across candidate counts, consistent with the subsampled-denominator consistency result.
4. *Exclusion-rule sensitivity.* Vary the exclude-most-recent rule (sector-scoped, global, none); count dropped repeats and measure weight shifts. Pass: the effect is pinned with a number (expected negligible at 2–35 events), and the rule is stated in the production spec either way.

E3 — Single-layer block Hawkes versus the two-stage factorization. *Purpose:* the central architecture decision — given exact dates and truthful risk sets, what does the two-stage factorization still buy, and what does it cost in joint-intensity coherence? *Procedure:* fit the full firm-level event-time block Hawkes (log-link baseline, self-inhibition, size-normalized sector field) on day-resolution timestamps; evaluate it and the two-stage pipeline on identical held-out events for joint NLL, next-deal sector prediction, and funded-firm ranking conditional on the event. *Pass:* on regime B the single-layer fit recovers the self-inhibition ordering with rank correlation at least 0.9; the head-to-head produces a written recommendation with numbers on all three metrics, whichever way it falls.

E4 — Operator benchmark. *Purpose:* establish where the neural-operator surrogate pays for itself now that exact event-time MLE removes its calibration role. *Procedure:* train under the full protocol; report accuracy versus sector count M and the wall-clock crossover against the exact solvers under repeated forward solves; benchmark differentiable-solver inverse calibration against the MLE for bias, variance, and wall-clock on identical data. *Pass:* sub-2% relative L^2 reproduced at full protocol; the crossover point (in solves and in M) is documented; the inverse-calibration table states plainly whether MLE wins at this scale.

E5 — Walk-forward backtest with covariate ablation. *Purpose:* the end-to-end rehearsal of the production loop under strictly forward information. *Procedure:* freeze the best when/where and which-firm models; walk forward weekly over the final 20% of days, emitting sector intensities, ranked candidate lists, and newcomer probabilities; score top-1/5/10 and MRR against oracle and random, PIT count calibration with the quasi-Poisson dispersion correction, and lead time (weeks a funded firm spends in the top decile before its deal); rerun with macro covariates removed and on the `covariate_scale` 1.5 world. *Pass:* the pipeline beats random throughout and degrades gracefully toward oracle; the covariate ablation turns “covariates help” into a signed number in both directions.

B1 — XGBoost ranking benchmark. *Purpose:* a model-free discriminative ceiling for stage 2, per the standing benchmark of Chapter 10. *Procedure:* train boosted rankers on the identical choice sets and splits in two variants — feature parity (exactly the six logit features, isolating functional form) and kitchen sink (all point-in-time-legal raw fields, with snapshot columns explicitly forbidden); score with per-event renormalization; modest hyperparameter search with a time-ordered validation tail. *Pass:* the comparison against top-5 of 0.446 (A) and 0.316 (B) is reported with the frozen decision rule: a material win means the logit is missing features or nonlinearity, and the remedy is feature work; a narrow result keeps the interpretable logit.

B2 — Non-homogeneous Poisson null. *Purpose:* show, not assume, that self-excitation earns its complexity at this data scale. *Procedure:* fit the nested ladder — covariates-only inhomogeneous Poisson (simulable by thinning [4]), covariates plus excitation, full event-time timing — plus a seasonal-naive baseline; decompose held-out improvement across the rungs

with residual diagnostics per rung [3]. *Pass*: the decomposition table exists with error bars over 5 seeds; if the excitation rung contributes approximately nothing, the recommendation to drop it is written into the decision tables of Chapter 10.

Causal-pipeline validation. *Purpose*: test the causal stack of Chapter 5 against known truth, per the protocol sketched in Chapter 10. *Procedure*: generate a treatment with known selection on observables and on the latent quality mark; estimate naive, backdoor-adjusted, and AIPW effects; attach cluster-bootstrap confidence intervals and E-values. *Pass*: the naive estimate is biased in the selection direction, adjustment removes the observable component, AIPW matches backdoor with better nuisance robustness, and the residual gap to truth is consistent with the computed E-value — each asserted in code, so a failure is a red test, not a discussion.

Test	Pass criterion (frozen)	Status
E1 bucketing	θ within 2 SE at day resolution on B; ridge visible by 28d	planned
E1 confounding curve	monotone fitted-radius curve over latent strength; mitigations quantified	planned
E2.1 gap form	binned fit recovers 26-week step (A) and smooth decay (B)	planned
E2.2 newcomer N1–N4	quarterly newcomer share within 2pp; incumbent weights stable	planned
E2.3 sampled softmax	weights stable across 32/64/128/full-on-subsample	planned
E2.4 exclusion rule	sensitivity pinned with a number	planned
E3 head-to-head	rank-corr ≥ 0.9 on B; three-metric comparison written up	planned
E4 operator	sub-2% rel- L^2 ; crossover and inverse-calibration tables	planned
E5 backtest	beats random throughout; ablation deltas reported	planned
B1 XGBoost	decision rule applied to top-5 0.446 (A) / 0.316 (B)	planned
B2 Poisson null	improvement decomposition with error bars	planned
Causal validation	all four coded assertions pass against truth	planned

Table 11.5: Forward test plan at a glance. All rows: both regimes where applicable, 5 seeds, temporal splits, audits of Section 11.1 in force.

11.4 Continuous integration and definition of done

The four layers of Section 11.1 map onto three CI gates. On every commit, the unit and equivalence suites run and must pass; they are fast because the micro-examples are tiny and the equivalence checks run on subsamples. Nightly, the recovery suite runs on reduced configurations — fewer weeks, fewer seeds — as an early-warning system; a nightly recovery regression blocks the next release, not the next commit. The leakage audit is a blocking check at both cadences: a commit that makes any feature non-point-in-time, or that lets a fitting path touch `ground_truth.npz` or `private_deal_map.csv`, does not merge, whatever its metrics say.

The artifact policy is equally plain. Results tables are versioned CSVs, written by the experiment entry points and diffed like code, so a silently shifted number is a visible change in review. Figures are regenerated from those CSVs by scripts and are never hand-edited; a figure that cannot be regenerated is deleted rather than trusted.

The campaign is done when three conditions hold. Every row of Table 11.5 has moved from *planned* to *pass* or *fail* with a number attached — not a narrative, a number. The benchmarks B1 and B2 are reported in the decision tables of Chapter 10, so that the method choices there cite measured evidence rather than expectation. And every failed pass criterion is either fixed

and rerun, or documented as a limitation alongside those of Chapter 10 — because a plan whose failures disappear quietly was never a test plan at all.

Appendix A

Detailed Proofs

This appendix gives complete proofs of every numbered mathematical claim in Chapters 3–9, together with precise statements and proofs of results the main text uses in display form only: the selection identities of Chapter 6, the instability of log-link count feedback behind Warning 7.9, and the explicit κ – θ ridge computation behind Proposition 9.5. Each claim is restated compactly — keeping its original number, which the restatement references — and then proved in full: every interchange of limit, integral, and expectation is justified, and every appeal to the literature is explicit. Where the fully general theorem is genuinely beyond the scope of this manual (the general marked-space Watanabe theory, stability of nonlinear Hawkes processes, the do-calculus in full), we prove the special case the manual actually uses and cite the source for the general statement. Helper results proved here carry appendix numbers and labels prefixed `app`.

A.1 Standing conventions and two lemmas

Throughout, $(\Omega, \mathcal{F}, \mathbb{P})$ carries a filtration $(\mathcal{H}_t)_{t \geq 0}$ satisfying the usual conditions, and N is a *simple* point process adapted to it: a right-continuous counting path with unit jumps at strictly increasing event times $t_1 < t_2 < \dots$ having no finite accumulation point. We assume N admits a nonnegative predictable intensity λ in the sense of Definition 7.1, with

$$\mathbb{E}[\Lambda(T)] = \mathbb{E}\left[\int_0^T \lambda(u) \, du\right] < \infty \quad \text{for every finite } T, \quad (\text{A.1})$$

and write $M = N - \Lambda$ for the compensated process.

Lemma A.1 (Compensator calculus). *Under (A.1): (i) M is a martingale on every $[0, T]$; (ii) for every bounded predictable real- or complex-valued process H , the integral $t \mapsto \int_0^t H(s) \, dM(s)$ is a martingale on $[0, T]$; in particular it has mean zero.*

Lemma A.1 is the defining property of the compensator together with the elementary theory of Stieltjes integration against point-process martingales; we use it without proof and refer to [5] (the integration theory recalled there) and [2, Ch. 7]. Complex integrands are covered by applying the real statement to real and imaginary parts. The special case we use most often we prove in full.

Lemma A.2 (Unbiasedness against deterministic integrands). *Let $f : [0, T] \rightarrow [0, \infty]$ be Borel measurable. Then $\mathbb{E} \int_0^T f(s) \, dN(s) = \mathbb{E} \int_0^T f(s) \lambda(s) \, ds$, both sides possibly infinite. For bounded signed measurable f the same identity holds with both sides finite.*

Proof. Define two measures on the Borel subsets of $[0, T]$ by $\mu_1(B) = \mathbb{E} \int_0^T \mathbf{1}\{s \in B\} \, dN(s)$ and $\mu_2(B) = \mathbb{E} \int_0^T \mathbf{1}\{s \in B\} \lambda(s) \, ds$; both are finite by (A.1). For $B = (a, b]$, Lemma A.1(i)

gives $\mathbb{E}[N(b) - N(a)] = \mathbb{E}[\Lambda(b) - \Lambda(a)]$, so $\mu_1 = \mu_2$ on the π -system of half-open intervals, and both vanish on $\{0\}$. Two finite measures agreeing on a generating π -system agree on the generated σ -algebra (Dynkin), so $\mu_1 = \mu_2$. The identity extends to simple nonnegative f by linearity; to measurable $f \geq 0$ by monotone convergence along simple $f_n \uparrow f$, each side being an integral against μ_i by Tonelli; and to bounded signed f by splitting $f = f_+ - f_-$. $\square$

A.2 Proofs for Chapter 3

Claim (Proposition 3.3). *The arc growth rate (3.2) is defined for all sign patterns, bounded in $[-2, 2]$, antisymmetric under swapping $F_y \leftrightarrow F_{y+h}$, and for $F_y, F_{y+h} > 0$ satisfies $g = \log(F_{y+h}/F_y) + O(\log^3(F_{y+h}/F_y))$ near zero growth.*

Proof. Write $a = F_y$, $b = F_{y+h}$, $g = (b - a)/\frac{1}{2}(|a| + |b|)$.

Definedness. The denominator vanishes iff $|a| + |b| = 0$ iff $a = b = 0$, the case covered by the convention $g = 0$.

Bounds, with equality characterization. By the triangle inequality $|b - a| \leq |a| + |b|$, so $|g| \leq 2$. Squaring, $|b - a|^2 \leq (|a| + |b|)^2$ is equivalent to $-2ab \leq 2|a||b|$, with equality iff $ab = -|a||b|$, i.e. iff $ab \leq 0$. Thus $|g| = 2$ exactly for sign-crossing (or from-zero) transitions, the reading of Example 3.4.

Antisymmetry. Swapping a and b negates the numerator and fixes the denominator.

Expansion. For $a, b > 0$ set $u = \log(b/a)$, so $b = ae^u$ and

$$g = \frac{a(e^u - 1)}{\frac{1}{2}a(e^u + 1)} = 2 \frac{e^u - 1}{e^u + 1} = 2 \tanh(u/2).$$

Since $\tanh$ is analytic with bounded derivatives of every order on $\mathbb{R}$, Taylor's theorem with Lagrange remainder gives $\tanh x = x - x^3/3 + O(x^5)$ uniformly on compacts, hence $g = u - u^3/12 + O(u^5)$ and in particular $g - u = O(u^3)$: arc growth agrees with log growth to second order at zero growth. $\square$

A.3 Proofs for Chapter 4

Claim (Proposition 4.2). *Let Y have distribution function F with $\mathbb{E}|Y| < \infty$ and let $\varphi(a) = \mathbb{E}[\rho_\tau(Y - a)]$ with ρ_τ the pinball loss (4.3). Then a^* minimizes φ if and only if $F(a^{*-}) \leq \tau \leq F(a^*)$; if F is continuous and strictly increasing, the minimizer is unique and equals $F^{-1}(\tau)$.*

Proof. Since $0 \leq \rho_\tau(u) \leq \max(\tau, 1 - \tau)|u|$, φ is finite everywhere. Fix $y \in \mathbb{R}$ and let $\psi_y(a) = \rho_\tau(y - a)$: on $\{a < y\}$ it equals $\tau(y - a)$, on $\{a > y\}$ it equals $(1 - \tau)(a - y)$, so ψ_y is Lipschitz with constant $\max(\tau, 1 - \tau)$ and derivative $\mathbf{1}\{y < s\} - \tau$ at every $s \neq y$. By the fundamental theorem of calculus for Lipschitz functions, for $a < b$, $\psi_y(b) - \psi_y(a) = \int_a^b (\mathbf{1}\{y < s\} - \tau) ds$. Taking expectations and applying Fubini — the integrand is bounded by 1 on the product of a probability space with $[a, b]$ —

$$\varphi(b) - \varphi(a) = \int_a^b (\mathbb{P}(Y < s) - \tau) ds = \int_a^b (F(s) - \tau) ds, \quad (\text{A.2})$$

using $\mathbb{P}(Y < s) = F(s^-) = F(s)$ off the (countable, Lebesgue-null) set of discontinuities.

Let a^* satisfy $F(a^{*-}) \leq \tau \leq F(a^*)$. For $b > a^*$, monotonicity gives $F(s) \geq F(a^*) \geq \tau$ on (a^*, b) , so (A.2) yields $\varphi(b) \geq \varphi(a^*)$; for $b < a^*$, $F(s) \leq F(a^{*-}) \leq \tau$ on (b, a^*) , so $\varphi(a^*) - \varphi(b) \leq 0$. Hence a^* is a global minimizer. Conversely, if $F(a) < \tau$, right-continuity gives $\delta > 0$ with $F < \tau$ on $[a, a + \delta)$, so $\varphi(a + \delta) < \varphi(a)$ and a is not a minimizer; if $F(a^-) > \tau$,

there is $\delta > 0$ with $F > \tau$ on $(a - \delta, a)$, so $\varphi(a - \delta) < \varphi(a)$. When F is continuous and strictly increasing, the condition $F(a^-) \leq \tau \leq F(a)$ reads $F(a) = \tau$, which has the unique solution $a = F^{-1}(\tau)$. $\square$

Lemma A.3 (Quantile of exchangeable scores). *Let $V_1, \dots, V_{n+1}$ be exchangeable real random variables, $\alpha \in (0, 1)$, and $k = \lceil (1 - \alpha)(n + 1) \rceil \leq n$. Let Q be the k -th smallest of $V_1, \dots, V_n$. Then $\mathbb{P}(V_{n+1} \leq Q) \geq 1 - \alpha$, and if the V_j are almost surely distinct, also $\mathbb{P}(V_{n+1} \leq Q) \leq 1 - \alpha + 1/(n + 1)$.*

Proof. For $j \leq n + 1$ let $R_j = \#\{i \neq j : V_i < V_j\}$, computed in the pooled sample of size $n + 1$. First, $\{V_{n+1} > Q\} \subseteq \{R_{n+1} \geq k\}$: if V_{n+1} exceeds the k -th smallest of the first n values, then at least k of those values are $\leq Q < V_{n+1}$. Second, deterministically $\#\{j : R_j \geq k\} \leq n + 1 - k$: if $R_j \geq k$ then V_j strictly exceeds at least k pooled values, hence exceeds the k -th smallest pooled value W ; and at least k indices have $V_i \leq W$, so at most $n + 1 - k$ indices can satisfy $V_j > W$. Third, by exchangeability the events $\{R_j \geq k\}$ have a common probability (swapping coordinates j and $n + 1$ maps one to the other and preserves the joint law), so

$$\mathbb{P}(R_{n+1} \geq k) = \frac{1}{n + 1} \sum_{j=1}^{n+1} \mathbb{P}(R_j \geq k) = \frac{\mathbb{E} \#\{j : R_j \geq k\}}{n + 1} \leq \frac{n + 1 - k}{n + 1} \leq \alpha,$$

the last step because $k \geq (1 - \alpha)(n + 1)$. This proves the lower bound. For the upper bound with almost surely distinct values: $\{V_{n+1} \leq Q\}$ then equals, up to a null set, $\{V_{n+1} < Q\} \subseteq \{R_{n+1} \leq k - 1\}$ (at most $k - 1$ of the first n values lie below V_{n+1} , since all values below V_{n+1} lie below Q and exactly $k - 1$ of the first n do); with distinct values $\#\{j : R_j \geq k\} = n + 1 - k$ exactly, so $\mathbb{P}(R_{n+1} \leq k - 1) = 1 - (n + 1 - k)/(n + 1) = k/(n + 1) \leq 1 - \alpha + 1/(n + 1)$. $\square$

Claim (Theorem 4.3). *In split CQR with calibration set I_2 of size n_2 , if the calibration pairs $\{(X_j, Y_j)\}_{j \in I_2}$ and the test pair (X, Y) are exchangeable, then $\mathbb{P}\{Y \in C(X)\} \geq 1 - \alpha$; if the conformity scores are almost surely distinct, also $\mathbb{P}\{Y \in C(X)\} \leq 1 - \alpha + 1/(n_2 + 1)$ [15].*

Proof. Condition on the proper training set I_1 ; the fitted quantile functions $\hat{q}_{\alpha/2}, \hat{q}_{1-\alpha/2}$ are then fixed measurable functions. Define the score map $s(x, y) = \max\{\hat{q}_{\alpha/2}(x) - y, y - \hat{q}_{1-\alpha/2}(x)\}$ and set $E_j = s(X_j, Y_j)$ for $j \in I_2$ and $E_{n_2+1} = s(X, Y)$. A fixed measurable function applied coordinatewise to an exchangeable vector preserves exchangeability, so $(E_j)_{j \in I_2}$ together with E_{n_2+1} are exchangeable given I_1 . The algorithm's threshold $Q_{1-\alpha}$ is the k -th smallest calibration score with $k = \lceil (n_2 + 1)(1 - \alpha) \rceil$ (if $k > n_2$, set $Q_{1-\alpha} = +\infty$ and the coverage bound is trivial). The coverage event is exactly a score comparison:

$$Y \in C(X) \iff \hat{q}_{\alpha/2}(X) - Y \leq Q_{1-\alpha} \text{ and } Y - \hat{q}_{1-\alpha/2}(X) \leq Q_{1-\alpha} \iff E_{n_2+1} \leq Q_{1-\alpha}.$$

Lemma A.3, applied conditionally on I_1 , gives $\mathbb{P}\{Y \in C(X) \mid I_1\} \geq 1 - \alpha$ and, under almost-sure distinctness, the matching upper bound. Integrating over I_1 preserves both inequalities. The finite-sample guarantee is *marginal* and rests entirely on exchangeability, which temporal drift breaks — Warning 4.4 is the operational consequence, not a footnote. $\square$

A.4 Proofs for Chapter 5

We work with discrete variables and a strictly positive joint law; the continuous case is identical with densities. Let G be a DAG over $V = (V_1, \dots, V_p)$ and let P factorize as $P(v) = \prod_j P(v_j \mid \text{pa}_j)$. Following [10], the interventional law is *defined* by the truncated factorization

$$P_a(v) = \prod_{j: V_j \neq A} P(v_j \mid \text{pa}_j) \Big|_{A=a} \quad \text{on } \{v : v_A = a\}, \quad (\text{A.3})$$

and $\mathbb{E}[Y \mid \text{do}(A = a)]$ is the mean of Y under P_a .

Lemma A.4 (Interventions do not move non-descendants). *Let W be the set of non-descendants of A (excluding A). Then $P_a(w) = P(w)$. In particular $P_a(z) = P(z)$ for any $Z \subseteq W$.*

Proof. No parent of a W -variable can be A or a strict descendant D -variable of A : a child of A is a descendant, and a variable with a parent in D is itself a descendant. Likewise every parent of A is in W (a parent that is a descendant would close a directed cycle). Hence the order (W in topological order, then A , then the strict descendants D in topological order) is a topological order of G , and

$$P(w, a, d) = \left[\prod_{j \in W} P(w_j \mid \text{pa}_j) \right] P(a \mid \text{pa}_A) \left[\prod_{j \in D} P(d_j \mid \text{pa}_j) \right], \quad P_a(w, d) = \left[\prod_{j \in W} \left[\prod_{j \in D} \right] \right]_{A=a}.$$

Summing $P_a(w, d)$ over the D -coordinates in reverse topological order telescopes (each conditional sums to one), leaving $P_a(w) = \prod_{j \in W} P(w_j \mid \text{pa}_j)$. The same telescoping in the observational law (sum over D , then over a) gives $P(w) = \prod_{j \in W} P(w_j \mid \text{pa}_j)$. Summing out $W \setminus Z$ gives the claim for Z , which contains no descendant of A by backdoor condition (i). $\square$

Lemma A.5 (Backdoor blocking survives edge deletion). *If Z satisfies the backdoor criterion (Definition 5.4) for (A, Y) in G , then Z d -separates A from Y in the graph $G_{\underline{A}}$ obtained from G by deleting all edges out of A .*

Proof. Let π be any A – Y path in $G_{\underline{A}}$. Its first edge points into A (all out-edges are deleted), so π is also a path of G and is a backdoor path there; by criterion (ii) it is blocked by Z in G : some node W on π is either (a) a non-collider on π with $W \in Z$, or (b) a collider on π with neither W nor any G -descendant of W in Z . Collider status depends only on the orientations of the two π -edges at W , which are the same in both graphs. In case (a), π is blocked at W in $G_{\underline{A}}$ as well. In case (b), the $G_{\underline{A}}$ -descendants of W form a subset of its G -descendants, so again neither W nor any $G_{\underline{A}}$ -descendant of W lies in Z , and π is blocked. Hence every A – Y path of $G_{\underline{A}}$ is blocked by Z . $\square$

Claim (Theorem 5.5). *If Z satisfies the backdoor criterion relative to (A, Y) , then $\mathbb{E}[Y \mid \text{do}(A = a)] = \mathbb{E}_Z[\mathbb{E}[Y \mid A = a, Z]]$, and consequently $\tau_{\text{ATE}} = \mathbb{E}_Z[\mathbb{E}[Y \mid A = 1, Z] - \mathbb{E}[Y \mid A = 0, Z]]$.*

Proof. Decompose $P_a(y) = \sum_z P_a(y \mid z) P_a(z)$. Lemma A.4 gives $P_a(z) = P(z)$. For the conditional we invoke the invariance half of Rule 2 of the do-calculus: if Z d -separates A from Y in $G_{\underline{A}}$, then $P_a(y \mid z) = P(y \mid a, z)$. This is proved in [10] from the truncated factorization (A.3) via the soundness of d -separation for DAG-factorized laws; we delegate exactly this step, and Lemma A.5 verifies its graphical hypothesis from the backdoor definition. Combining, $P_a(y) = \sum_z P(y \mid a, z) P(z)$, i.e. $\mathbb{E}[Y \mid \text{do}(A = a)] = \sum_z P(z) \mathbb{E}[Y \mid A = a, Z = z]$; subtracting the $a = 0$ and $a = 1$ versions yields the ATE formula.

Two special cases are worth recording because they are fully self-contained. (1) If $Z = \text{pa}_A$, no graphical lemma is needed: on $\{A = a\}$, $P(v) = P_a(v) P(a \mid z)$ directly from (A.3), so $P_a(y, z) = P(y, z, a) / P(a \mid z) = P(y \mid a, z) P(z)$, which is the formula. (2) Under Assumption 5.3 with potential-outcome semantics ($\mathbb{E}[Y \mid \text{do}(A = a)] := \mathbb{E}[Y(a)]$), the formula is three lines: $\mathbb{E}[Y(a)] = \mathbb{E}_Z \mathbb{E}[Y(a) \mid Z] = \mathbb{E}_Z \mathbb{E}[Y(a) \mid A = a, Z] = \mathbb{E}_Z \mathbb{E}[Y \mid A = a, Z]$, using ignorability and then consistency (Assumption 5.2). Case (2) is the form under which Chapter 5 actually operates. $\square$

Claim (Proposition 5.7). *Let ψ be the AIPW score (5.3) built from working nuisances (μ_1, μ_0, e) with $\delta \leq e(Z) \leq 1 - \delta$ for some $\delta > 0$ and $\mathbb{E}|Y| + \mathbb{E}|\mu_a(Z)| < \infty$. Under Assumptions 5.2–5.3, $\mathbb{E}[\psi] = \tau_{\text{ATE}}$ if either $(\mu_1, \mu_0) = (\mu_1^*, \mu_0^*)$ is the true outcome pair or $e = e^*$ is the true propensity.*

Proof. Write $\psi = \psi_1 - \psi_0$ with $\psi_1 = \mu_1(Z) + A(Y - \mu_1(Z))/e(Z)$ and $\psi_0 = \mu_0(Z) + (1 - A)(Y - \mu_0(Z))/(1 - e(Z))$; all terms are integrable because the denominators are bounded away from zero. We show $\mathbb{E}[\psi_1] = \mathbb{E}[Y(1)]$ under either hypothesis; the ψ_0 half is symmetric, and subtracting gives the claim.

True outcome model. Recall $\mu_1^*(Z) = \mathbb{E}[Y | A = 1, Z] = \mathbb{E}[Y(1) | A = 1, Z] = \mathbb{E}[Y(1) | Z]$, by consistency and then ignorability. Conditioning on Z and using that $e(Z)$ is Z -measurable,

$$\mathbb{E}\left[\frac{A(Y - \mu_1^*(Z))}{e(Z)} \mid Z\right] = \frac{\mathbb{P}(A=1 \mid Z)}{e(Z)} \mathbb{E}[Y - \mu_1^*(Z) \mid A = 1, Z] = \frac{e^*(Z)}{e(Z)} \cdot 0 = 0,$$

regardless of the working e . Hence $\mathbb{E}[\psi_1] = \mathbb{E}[\mu_1^*(Z)] = \mathbb{E}[\mathbb{E}[Y(1) | Z]] = \mathbb{E}[Y(1)]$.

True propensity. With $e = e^*$ and arbitrary μ_1 ,

$$\mathbb{E}[\psi_1 \mid Z] = \mu_1(Z) + \frac{e^*(Z) \mathbb{E}[Y \mid A = 1, Z] - e^*(Z) \mu_1(Z)}{e^*(Z)} = \mathbb{E}[Y \mid A = 1, Z] = \mathbb{E}[Y(1) \mid Z],$$

the working μ_1 cancelling exactly; take expectations over Z . $\square$

Proposition A.6 (Influence-function variance of AIPW). *Let both nuisances be correct, $\mathbb{E}[Y^2] < \infty$, and let $O_1, \dots, O_n$ be i.i.d. Write $\tau(Z) = \mu_1^*(Z) - \mu_0^*(Z)$ and $\sigma_a^2(Z) = \text{Var}(Y \mid A = a, Z)$. Then $\hat{\tau}_{\text{AIPW}} = n^{-1} \sum_i \psi(O_i)$ satisfies $\sqrt{n}(\hat{\tau}_{\text{AIPW}} - \tau_{\text{ATE}}) \rightarrow \mathcal{N}(0, V)$ with*

$$V = \text{Var}(\psi) = \text{Var}(\tau(Z)) + \mathbb{E}\left[\frac{\sigma_1^2(Z)}{e^*(Z)}\right] + \mathbb{E}\left[\frac{\sigma_0^2(Z)}{1 - e^*(Z)}\right].$$

Proof. Asymptotic normality is the Lindeberg–Levy central limit theorem for the i.i.d. mean of the square-integrable ψ , whose mean is τ_{ATE} by Proposition 5.7. For the variance, write $\psi = \tau(Z) + U_1 - U_0$ with $U_1 = A(Y - \mu_1^*(Z))/e^*(Z)$ and $U_0 = (1 - A)(Y - \mu_0^*(Z))/(1 - e^*(Z))$. Both U_a have conditional mean zero given Z (shown in the previous proof), so they are uncorrelated with the Z -measurable $\tau(Z)$, and $U_1 U_0 = 0$ pointwise because $A(1 - A) = 0$. By conditioning on (A, Z) and using $\mu_1^*(Z) = \mathbb{E}[Y \mid A = 1, Z]$,

$$\mathbb{E}[U_1^2 \mid Z] = \frac{\mathbb{E}[A(Y - \mu_1^*(Z))^2 \mid Z]}{e^*(Z)^2} = \frac{e^*(Z) \sigma_1^2(Z)}{e^*(Z)^2} = \frac{\sigma_1^2(Z)}{e^*(Z)},$$

and symmetrically for U_0 . Adding the three orthogonal contributions gives V . This V is the semiparametric efficiency bound for the ATE, and the cross-fitted estimator of Section 5.4 attains it under the product-rate condition of Remark 5.8; see [13]. The firm-clustered variance (5.4) is the sample analogue of $\text{Var}(\sum_y \psi_{i,y})$ with firms as independent blocks, and the cluster bootstrap of Section 5.5 estimates the same quantity by resampling [14]. $\square$

A.5 Proofs for Chapter 6

Chapter 6 uses three display-level identities — the selection-bias covariance formula (6.5), the worst-case bounds (6.6), and the unbiasedness of the inverse-probability-of-observation weights below (6.7). We record them with proofs.

Proposition A.7 (Selection identities). *Let g be integrable, $D \in \{0, 1\}$ with $p = \mathbb{P}(D=1) \in (0, 1)$. Then: (i) $\mathbb{E}[g \mid D=1] - \mathbb{E}[g] = \text{Cov}(g, \mathbf{1}\{D=1\})/p$; (ii) if $g \in [g_{\text{lo}}, g_{\text{hi}}]$ almost surely, then $\mathbb{E}[g]$ lies in the interval (6.6); (iii) under Assumption 6.3, with $\pi = \mathbb{P}(D=1 \mid \mathcal{H}_{T_{i,k}}) > 0$ almost surely, $\mathbb{E}[Dg/\pi] = \mathbb{E}[g]$.*

Proof. (i) $\text{Cov}(g, \mathbf{1}\{D=1\}) = \mathbb{E}[g\mathbf{1}\{D=1\}] - p\mathbb{E}[g] = p\mathbb{E}[g \mid D=1] - p\mathbb{E}[g]$; divide by p . (ii) By the law of total expectation, $\mathbb{E}[g] = p\mathbb{E}[g \mid D=1] + (1-p)\mathbb{E}[g \mid D=0]$, and $\mathbb{E}[g \mid D=0] \in [g_{\text{lo}}, g_{\text{hi}}]$ by the almost-sure bounds. (iii) By the tower property and the conditional independence $g \perp\!\!\!\perp D \mid \mathcal{H}_{T_{i,k}}$,

$$\mathbb{E}\left[\frac{Dg}{\pi}\right] = \mathbb{E}\left[\frac{1}{\pi}\mathbb{E}[Dg \mid \mathcal{H}_{T_{i,k}}]\right] = \mathbb{E}\left[\frac{1}{\pi}\mathbb{E}[D \mid \mathcal{H}_{T_{i,k}}]\mathbb{E}[g \mid \mathcal{H}_{T_{i,k}}]\right] = \mathbb{E}[\mathbb{E}[g \mid \mathcal{H}_{T_{i,k}}]] = \mathbb{E}[g],$$

all steps valid because π is $\mathcal{H}_{T_{i,k}}$ -measurable and positive. $\square$

Remark A.8. The closed form (6.7) for π is the survival function of the first event after $T_{i,k}$: with an internal (self-exciting) history, the intensity along the no-new-event continuation is an $\mathcal{H}_{T_{i,k}}$ -measurable function of elapsed time, and the hazard representation of the inter-event distribution [2, Ch. 7.2] gives $\mathbb{P}(T_{i,k+1} > \tau_{\text{end}} \mid \mathcal{H}_{T_{i,k}}) = \exp(-\int \lambda_i)$. If external covariates evolve stochastically over the gap, (6.7) holds with the exponential replaced by its conditional expectation.

A.6 Proofs for Chapter 7

Claim (Theorem 7.2). *Let N be a simple point process adapted to $\mathcal{H}_t$ with continuous compensator Λ . Then N is an inhomogeneous Poisson process — independent increments, $N(b) - N(a) \sim \text{Poisson}(\Lambda(b) - \Lambda(a))$ — if and only if Λ is deterministic.*

Proof. *Deterministic implies Poisson (the direction the manual uses).* Assume Λ is a continuous deterministic nondecreasing function, finite on compacts, with $N - \Lambda$ a martingale (Lemma A.1(i); here (A.1) holds automatically since Λ is finite). Fix $u \in \mathbb{R}$ and $a \geq 0$, and define for $t \geq a$

$$Z_t = \exp\left\{iu(N(t) - N(a)) - (e^{iu} - 1)(\Lambda(t) - \Lambda(a))\right\} = e^{iu(N(t) - N(a))} g(t), \quad (\text{A.4})$$

with g continuous, deterministic, of bounded variation, and bounded away from 0 and ∞ on compacts. Both factors are of bounded variation, so Stieltjes integration by parts (with g continuous) gives $dZ_t = e^{iu(N(t^-) - N(a))} dg(t) + g(t) d(e^{iu(N - N(a))})_t$. The first differential is $-(e^{iu} - 1)Z_{t-} d\Lambda(t)$ by the chain rule for continuous BV functions (and Z_{t-} may replace Z_t in a $d\Lambda$ -integral because Λ , being continuous, charges no countable set, in particular not the jump times). The second is a pure-jump differential: at an event, e^{iuN} is multiplied by e^{iu} , so $d(e^{iu(N - N(a))})_t = (e^{iu} - 1)e^{iu(N(t^-) - N(a))} dN(t)$. Adding,

$$Z_t = Z_a + \int_{(a,t]} Z_{s-} (e^{iu} - 1) d(N - \Lambda)(s).$$

The integrand $H_s = Z_{s-}(e^{iu} - 1)$ is predictable (left limit of an adapted cadlag process) and bounded on $[a, b]$ by $|e^{iu} - 1|e^{|e^{iu} - 1|\Lambda(b)}$, a deterministic constant. By Lemma A.1(ii) the integral is a martingale, so $\mathbb{E}[Z_t \mid \mathcal{H}_a] = Z_a = 1$, i.e.

$$\mathbb{E}[e^{iu(N(t) - N(a))} \mid \mathcal{H}_a] = \exp\{(e^{iu} - 1)(\Lambda(t) - \Lambda(a))\},$$

the characteristic function of $\text{Poisson}(\Lambda(t) - \Lambda(a))$, and it is *deterministic*: by uniqueness of characteristic functions the increment is Poisson distributed, and since the conditional characteristic function is constant, the increment is independent of $\mathcal{H}_a$. For disjoint intervals $(a_1, b_1], \dots, (a_m, b_m]$ (ordered), iterating the tower property from the last interval inward factorizes the joint characteristic function of the increments into the product of Poisson characteristic functions, proving joint independence. Hence N is inhomogeneous Poisson with mean function Λ .

Poisson implies deterministic. If N has independent Poisson increments with continuous mean function m , then $N(t) - m(t)$ is a martingale ($\mathbb{E}[N(t) - N(s) \mid \mathcal{H}_s] = m(t) - m(s)$ by independence), and m is a continuous deterministic — hence predictable — increasing process. The compensator is the unique predictable increasing process making N minus it a martingale (Doob–Meyer uniqueness, which we take from [5]), so $\Lambda = m$ almost surely: deterministic. For the general marked-space version of both directions see [2, Ch. 7]. $\square$

Claim (Theorem 7.3). *Let N have event times $t_1 < t_2 < \dots$ and conditional intensity $\lambda(t) > 0$ with $\Lambda(t) \rightarrow \infty$ almost surely. Then $\tau_k = \Lambda(t_k)$ are the event times of a unit-rate Poisson process; equivalently $\delta_k = \tau_k - \tau_{k-1}$ are i.i.d. $\text{Exp}(1)$.*

Proof. Step 1: the time change. Λ is continuous (an integral) and strictly increasing: for $a < b$, $\Lambda(b) - \Lambda(a) = \int_a^b \lambda > 0$ because λ is strictly positive. Since $\Lambda(\infty) = \infty$ a.s., for each $s \geq 0$ there is a unique finite $\tau(s)$ with $\Lambda(\tau(s)) = s$, and $s \mapsto \tau(s)$ is continuous and increasing. Each $\tau(s)$ is a stopping time: $\{\tau(s) \leq t\} = \{\Lambda(t) \geq s\} \in \mathcal{H}_t$. Define $\mathcal{G}_s = \mathcal{H}_{\tau(s)}$ (a filtration, since τ is monotone) and $\tilde{N}(s) = N(\tau(s))$. Because N is right-continuous and adapted, it is progressively measurable, so $\tilde{N}(s)$ is $\mathcal{G}_s$ -measurable (the standard measurability of a progressive process at a stopping time). $\tilde{N}$ is a simple counting path whose jump times are exactly $\{\Lambda(t_k)\} = \{\tau_k\}$, since the strictly increasing Λ maps distinct event times to distinct values.

Step 2: integrability. Fix s and let $T = \tau(s)$. For each n , $T \wedge n$ is a bounded stopping time, and Doob’s optional sampling applied to the martingale $M = N - \Lambda$ on $[0, n]$ gives $\mathbb{E}[N(T \wedge n)] = \mathbb{E}[\Lambda(T \wedge n)] \leq s$. Monotone convergence ($T < \infty$ a.s., N and Λ nondecreasing) yields $\mathbb{E}[\tilde{N}(s)] = \mathbb{E}[\Lambda(T)] = s < \infty$.

Step 3: $\tilde{N}(s) - s$ is a $\mathcal{G}$ -martingale. Fix $s' < s$, put $T' = \tau(s')$, $T = \tau(s)$, and take $A \in \mathcal{G}_{s'} = \mathcal{H}_{T'}$. For each n , $T' \wedge n \leq T \wedge n$ are bounded stopping times, so optional sampling gives $\mathbb{E}[M(T \wedge n) \mid \mathcal{H}_{T' \wedge n}] = M(T' \wedge n)$. The event $A_n = A \cap \{T' \leq n\}$ belongs to $\mathcal{H}_{T' \wedge n}$ (for $t < n$, $A_n \cap \{T' \wedge n \leq t\} = A \cap \{T' \leq t\} \in \mathcal{H}_t$; for $t \geq n$ it lies in $\mathcal{H}_n \subseteq \mathcal{H}_t$), whence $\mathbb{E}[\mathbf{1}\{A_n\}M(T \wedge n)] = \mathbb{E}[\mathbf{1}\{A_n\}M(T' \wedge n)]$. As $n \rightarrow \infty$: $\mathbf{1}\{A_n\} \uparrow \mathbf{1}\{A\}$ since $T' < \infty$ a.s.; $M(T \wedge n) \rightarrow M(T)$ and $M(T' \wedge n) \rightarrow M(T')$ a.s.; and $|M(T \wedge n)| \leq N(T) + s$ with $N(T) \in L^1$ by Step 2 (similarly for T'), so dominated convergence applies on both sides: $\mathbb{E}[\mathbf{1}\{A\}M(T)] = \mathbb{E}[\mathbf{1}\{A\}M(T')]$ for all $A \in \mathcal{G}_{s'}$. Since $M(T') = \tilde{N}(s') - s'$ is $\mathcal{G}_{s'}$ -measurable, this says $\mathbb{E}[\tilde{N}(s) - s \mid \mathcal{G}_{s'}] = \tilde{N}(s') - s'$.

Step 4: conclude. $\tilde{N}$ is a simple counting process adapted to $(\mathcal{G}_s)$ whose compensator is the continuous deterministic function s . By the direction of Theorem 7.2 proved above (whose proof used only the martingale property of $N - \Lambda$ with Λ continuous deterministic), $\tilde{N}$ is a unit-rate Poisson process. Its jump times are the τ_k , and the equivalence of a unit-rate Poisson process with i.i.d. standard-exponential interarrivals is the classical description of the Poisson process [2, Ch. 2]; the residual-analysis use is due to [3]. $\square$

Claim (Corollary 7.4). *Under correct specification, every fitted model class in Part III yields rescaled inter-arrivals $\hat{\delta}_k$ that are i.i.d. $\text{Exp}(1)$, and $u_k = 1 - e^{-\hat{\delta}_k}$ that are i.i.d. $\text{Uniform}(0, 1)$.*

Proof. Every model class specifies $\lambda(t; \theta)$ explicitly with $\lambda > 0$; under correct specification the data are generated by $\lambda(\cdot; \theta_0)$, and Theorem 7.3 applied at θ_0 makes the rescaled inter-arrivals i.i.d. $\text{Exp}(1)$. The probability integral transform finishes: $\mathbb{P}(1 - e^{-\delta} \leq u) = \mathbb{P}(\delta \leq -\log(1 - u)) = u$ for $u \in (0, 1)$, and measurable transformations of independent variables are independent. (In practice $\hat{\theta}$ replaces θ_0 ; the tests are then conservative screens, exact only asymptotically.) $\square$

Claim (Proposition 7.5). With $x_w = (1, X_w^\top)^\top$, the weekly log-likelihood (7.6) is concave in $\gamma = (a, \beta)$, strictly concave iff $\{x_w\}$ spans $\mathbb{R}^{p+1}$; the MLE, when it exists, is then unique.

Proof. Write $\eta_w = x_w^\top \gamma$, so $\ell(\gamma) = \sum_w [n_w \eta_w - \Delta_w e^{\eta_w}]$. Then $\nabla \ell = \sum_w (n_w - \Delta_w e^{\eta_w}) x_w$ and $\nabla^2 \ell = -\sum_w \Delta_w e^{\eta_w} x_w x_w^\top$. For any v , $v^\top \nabla^2 \ell v = -\sum_w \Delta_w e^{\eta_w} (x_w^\top v)^2 \leq 0$, so ℓ is concave; the form is zero iff $x_w^\top v = 0$ for all w . If the x_w span, no such $v \neq 0$ exists and $\nabla^2 \ell \prec 0$ everywhere: ℓ is strictly concave, and two distinct maximizers would contradict strict concavity at their midpoint, so the maximizer is unique. Conversely, if some $v \neq 0$ is orthogonal to every x_w , then $\ell(\gamma + tv) = \ell(\gamma)$ for all t : ℓ is constant along a line, hence not strictly concave and its maximizer (if any) not unique. $\square$

Claim (Proposition 7.6). The Fisher information of (7.6) is $I(a, \beta) = \sum_w \Delta_w \lambda_w x_w x_w^\top$ with $\lambda_w = e^{a+\beta^\top X_w}$, and after profiling out a the information for β is the intensity-weighted temporal covariance (7.8). A covariate constant over the window is not identified separately from a ; a near-linear trend leaves as effective information the weighted variance of its detrended residual.

Proof. The Hessian computed in the previous proof does not involve the counts n_w , so observed and expected information coincide: $I(\gamma) = -\nabla^2 \ell = \sum_w \omega_w x_w x_w^\top$ with $\omega_w = \Delta_w \lambda_w$. Partition by the intercept: $I_{aa} = \sum_w \omega_w$, $I_{a\beta} = \sum_w \omega_w X_w^\top$, $I_{\beta\beta} = \sum_w \omega_w X_w X_w^\top$. The information for β with a unknown is the Schur complement $I_\beta = I_{\beta\beta} - I_{\beta a} I_{aa}^{-1} I_{a\beta}$ (the inverse of $[I^{-1}]_{\beta\beta}$, by the block-inverse formula). With $\bar{X}_\omega = \sum_w \omega_w X_w / \sum_w \omega_w$,

$$\sum_w \omega_w (X_w - \bar{X}_\omega)(X_w - \bar{X}_\omega)^\top = \sum_w \omega_w X_w X_w^\top - \left(\sum_w \omega_w \right) \bar{X}_\omega \bar{X}_\omega^\top = I_{\beta\beta} - I_{\beta a} I_{aa}^{-1} I_{a\beta},$$

which is (7.8). (i) If coordinate j of X_w is constant, then $X_w^{(j)} - \bar{X}_\omega^{(j)} = 0$ for all w , so I_β has a zero row and column: singular. The non-identification is exact, not asymptotic: replacing $(a, \beta_j) \rightarrow (a - tc, \beta_j + t)$ leaves every linear predictor η_w , hence the entire likelihood, unchanged. (ii) More generally, let U collect nuisance columns (intercept and an explicit trend t_w) and endow $\mathbb{R}^W$ with the inner product $\langle f, g \rangle_\omega = \sum_w \omega_w f_w g_w$. The Schur complement of the U -block in the full Gram matrix is the Gram matrix of the residuals of the X -columns after $\langle \cdot, \cdot \rangle_\omega$ -orthogonal projection onto $\text{span}(U)$ — because $I_{\beta U} I_{UU}^{-1} I_{U\beta}$ is precisely the Gram matrix of those projections. Hence the profile information for β is $\sum_w \omega_w r_w r_w^\top$ with r_w the ω -weighted regression residual of X_w on $(1, t_w)$: a covariate that is nearly a linear trend has small residuals and correspondingly little information, as claimed. $\square$

Proposition A.9 (Instability of log-link raw-count feedback (Mechanism B)). Let $a \in \mathbb{R}$, $b > 0$, and let $Y_t \mid \mathcal{H}_t \sim \text{Poisson}(\lambda_t)$ with $\lambda_{t+1} = \exp(a + bY_t)$ and any initial $\lambda_1 > 0$. Define $G(\lambda) = e^a \exp\{\lambda(e^b - 1)\}$. Then: (i) $\mathbb{E}[\lambda_{t+1} \mid \mathcal{H}_t] = G(\lambda_t)$ exactly, with G increasing, strictly convex, and $G'(\lambda^*) = \lambda^*(e^b - 1)$ at any fixed point λ^* ; (ii) G has a fixed point iff $e^a(e^b - 1) \leq e^{-1}$, in which case there are at most two, $\lambda_-^* \leq \lambda_+^*$, with $\lambda_-^*(e^b - 1) \leq 1$ (the local stability condition of the mean recursion) and $\lambda_+^*(e^b - 1) \geq 1$ (repelling); (iii) in all cases $\mathbb{E}[\lambda_t] \rightarrow \infty$ as $t \rightarrow \infty$: no fixed point is globally attracting for the mean dynamics; (iv) by contrast, under linear feedback $\lambda_{t+1} = c + \rho Y_t$ with $c > 0$ and $0 \leq \rho < 1$, $\mathbb{E}[\lambda_t] \rightarrow c/(1 - \rho)$ geometrically from every initial condition.

Proof. (i) The Poisson moment generating function: for $Y \sim \text{Poisson}(\lambda)$ and any real s , $\mathbb{E}[e^{sY}] = \sum_{k \geq 0} e^{sk} e^{-\lambda} \lambda^k / k! = e^{\lambda(e^s - 1)}$ (the series is the exponential series at λe^s). Hence $\mathbb{E}[\lambda_{t+1} \mid \mathcal{H}_t] = e^a \mathbb{E}[e^{bY_t} \mid \mathcal{H}_t] = G(\lambda_t)$. Direct differentiation gives $G' = (e^b - 1)G > 0$ and $G'' = (e^b - 1)^2 G > 0$; at a fixed point $G(\lambda^*) = \lambda^*$, so $G'(\lambda^*) = \lambda^*(e^b - 1)$.

(ii) Write $c_0 = e^a$, $d = e^b - 1 > 0$. Fixed points solve $c_0 e^{d\lambda} = \lambda$. The strictly convex function $g(\lambda) = G(\lambda) - \lambda$ has $g(0) = c_0 > 0$ and $g(\lambda) \rightarrow \infty$; it attains its minimum where $G' = 1$, i.e. at

$\tilde{\lambda}$ with $c_0 d e^{d\tilde{\lambda}} = 1$, giving $\tilde{\lambda} = d^{-1} \log(1/(c_0 d))$ when positive and $g(\tilde{\lambda}) = 1/d - \tilde{\lambda}$. Thus g has a zero iff $\tilde{\lambda} \geq 1/d$, i.e. iff $c_0 d \leq e^{-1}$, which is the stated condition. A strictly convex function has at most two zeros; at the smaller, g crosses from positive to non-positive, so $g'(\lambda_-^*) \leq 0$, i.e. $G'(\lambda_-^*) \leq 1$; at the larger the crossing is upward, $G'(\lambda_+^*) \geq 1$.

(iii) Note $\lambda_t \geq e^a$ for every $t \geq 2$. Choose an integer k with $\lambda' := e^{a+bk}$ strictly above every fixed point (possible: fixed points are finite, or absent). Then $G(\lambda) > \lambda$ for all $\lambda \geq \lambda'$, and the deterministic iterates $G^{on}(\lambda')$ increase; were they bounded, their limit would be a fixed point $\geq \lambda'$, a contradiction — so $G^{on}(\lambda') \uparrow \infty$. Poisson tails are stochastically increasing in the mean (superpose an independent $\text{Poisson}(\mu' - \mu)$), so $p_0 := \mathbb{P}(\text{Poisson}(e^a) \geq k) > 0$ lower-bounds $\mathbb{P}(Y_2 \geq k \mid \mathcal{H}_2)$. By conditional Jensen with the convex increasing G and induction on n , $\mathbb{E}[\lambda_{3+n} \mid \mathcal{H}_3] \geq G^{on}(\lambda_3)$ (allowing $+\infty$ throughout). On the event $B = \{Y_2 \geq k\}$ we have $\lambda_3 = e^{a+bY_2} \geq \lambda'$, hence

$$\mathbb{E}[\lambda_{3+n}] \geq \mathbb{E}[\mathbf{1}\{B\} \mathbb{E}[\lambda_{3+n} \mid \mathcal{H}_3]] \geq \mathbb{P}(B) G^{on}(\lambda') \geq p_0 G^{on}(\lambda') \longrightarrow \infty.$$

(iv) Linearity passes through expectations exactly: $\mathbb{E}[\lambda_{t+1} \mid \mathcal{H}_t] = c + \rho \lambda_t$, so $\mathbb{E}[\lambda_t] - c/(1-\rho) = \rho^{t-1}(\mathbb{E}[\lambda_1] - c/(1-\rho)) \rightarrow 0$. The contrast with (iii) is precisely the Jensen gap: G is convex, so rare large counts dominate the mean, while the linear map has no gap. Note also $e^b - 1 > b$, so the true mean-map derivative $\lambda^*(e^b - 1)$ strictly exceeds the naive effective radius $\rho_{\text{eff}} = b\lambda^*$ of (7.12): a configuration comfortably subcritical by ρ_{eff} can be metastable, which is the measured episode of Warning 7.9 and Table 10.6. Genuine ergodicity theory for log-linear count autoregressions (with $\log(1+Y)$ feedback) is in [6]; for nonlinear Hawkes stability see [5]. $\square$

A.7 Proofs for Chapter 8

Lemma A.10 (Branching moments and local finiteness). *Consider a cluster rooted at a point at time v : generation 0 is the root; each point of generation n independently produces $\text{Poisson}(\kappa)$ generation- $(n+1)$ points at lags drawn i.i.d. from the density ϕ/κ on $(0, \infty)$. Let Z_n be the size of generation n and S_n a sum of n i.i.d. lags. Then (i) $\mathbb{E}Z_n = \kappa^n$, and for $\kappa < 1$ the mean total cluster size is $\sum_n \kappa^n = 1/(1-\kappa)$; (ii) the expected number of generation- n points within $[v, v+T]$ is $\kappa^n \mathbb{P}(S_n \leq T)$; (iii) for every T and every κ , $\sum_n \kappa^n \mathbb{P}(S_n \leq T) < \infty$, so clusters are almost surely finite on compact windows.*

Proof. (i) Wald's identity for a Poisson number of nonnegative i.i.d. terms: if K is independent of the i.i.d. sequence (X_i) , conditioning on K and applying Tonelli gives $\mathbb{E} \sum_{i \leq K} X_i = \mathbb{E}K \cdot \mathbb{E}X_1$. Applying it generation by generation, $\mathbb{E}Z_{n+1} = \mathbb{E}[\mathbb{E}[Z_{n+1} \mid \text{gen } n]] = \kappa \mathbb{E}Z_n$, so $\mathbb{E}Z_n = \kappa^n$, and the total follows by Tonelli over n . (ii) Let m_n be the mean measure of generation- n displacements from the root. Then $m_0 = \delta_0$ and, by the same conditioning plus Tonelli, $m_{n+1}(B) = \iint \mathbf{1}\{u \in B\} \phi(u-p) du m_n(dp)$: the density of m_n is κ^n times the n -fold convolution of ϕ/κ , i.e. the law of S_n scaled by κ^n ; evaluate on $[0, T]$. (iii) Lags are a.s. strictly positive, so by dominated convergence $\mathbb{E}e^{-\eta S_1} \downarrow 0$ as $\eta \uparrow \infty$; choose η with $r := \kappa \mathbb{E}e^{-\eta S_1} < 1$. Markov's inequality gives $\mathbb{P}(S_n \leq T) \leq e^{\eta T} (\mathbb{E}e^{-\eta S_1})^n$, hence $\sum_n \kappa^n \mathbb{P}(S_n \leq T) \leq e^{\eta T} / (1-r) < \infty$. Finite expectation implies almost-sure finiteness. $\square$

Claim (Theorem 8.2). *For $\phi \geq 0$ with $\kappa = \int \phi < 1$, the Hawkes process (8.1) is equal in law to the Poisson cluster process (immigrants at rate $s(\cdot)$, i.i.d. $\text{Poisson}(\kappa)$ offspring at lag density ϕ/κ , superposed over generations). Under constant s it admits a stationary version iff $\kappa < 1$, with mean intensity $s/(1-\kappa)$ and mean cluster size $1/(1-\kappa)$; the multivariate statement holds with $\rho(K) < 1$, mean vector $(I-K)^{-1}s$.*

Proof. Construction. On a suitable product space let immigrants form a Poisson process with locally integrable intensity $s(\cdot)$ on $[0, \infty)$, and attach to each immigrant an independent cluster

as in Lemma A.10. Let N_c be the superposition of all points. By Lemma A.10(ii)–(iii) and Campbell's formula over immigrants, $\mathbb{E}N_c[0, T] \leq (\int_0^T s) \sum_n \kappa^n \mathbb{P}(S_n \leq T) < \infty$, so N_c is almost surely locally finite; it is simple because all lags and immigrant positions have nonatomic laws, so any two prescribed points coincide with probability zero and there are countably many pairs.

Intensity with respect to the genealogical filtration. Let $\mathcal{F}_t$ be generated by all points born up to t together with their genealogy. Decompose N_c into the immigrant stream and the direct-offspring streams N^p of each born point p . By construction, given p , N^p is a Poisson process on (p, ∞) with intensity $\phi(\cdot - p)$, independent of all other building blocks; by independence of Poisson increments, $N^p(t) - \int_p^{t \vee p} \phi(u - p) du$ is an $\mathcal{F}$ -martingale, as is the compensated immigrant stream. Partial sums over the countably many streams are martingales, and they converge in L^1 for each t (monotonically in the counting and compensating parts separately, with finite limit expectation $\mathbb{E}N_c(t) < \infty$), so the limit $N_c(t) - \int_0^t \lambda_c(u) du$, with $\lambda_c(u) = s(u) + \sum_{p < u} \phi(u - p)$, is an $\mathcal{F}$ -martingale (L^1 -limits pass through conditional expectations).

Reduction to the internal history. λ_c is exactly the Hawkes formula (8.1) evaluated on the points of N_c : it depends on point positions only, not on the genealogy, so it is predictable for the internal filtration $\mathcal{H}_t = \sigma(N_c(s), s \leq t)$, and $\mathcal{H}_t \subseteq \mathcal{F}_t$. The tower property then transfers the martingale property: for $s < t$, $\mathbb{E}[N_c(t) - \int_0^t \lambda_c | \mathcal{H}_s] = \mathbb{E}[\mathbb{E}[\cdot | \mathcal{F}_s] | \mathcal{H}_s] = \mathbb{E}[N_c(s) - \int_0^s \lambda_c | \mathcal{H}_s] = N_c(s) - \int_0^s \lambda_c$, the last equality because the variable is $\mathcal{H}_s$ -measurable. Hence N_c has internal-history intensity (8.1).

Law identification. A simple point process whose internal-history intensity coincides almost surely with a fixed predictable functional of the past configuration has its law on every $[0, T]$ uniquely determined — the intensity determines the successive conditional hazard functions of inter-event times, hence all finite-dimensional joint densities. We take this uniqueness from [2, Ch. 7.2–7.3]. Therefore N_c equals in law the Hawkes process of Definition 8.1; this is the cluster representation, originally implicit in [1].

Stationarity. Let s be constant. For $\kappa < 1$, run the same construction with immigrants homogeneous Poisson on all of $\mathbb{R}$. The construction mechanism is invariant under time shifts, so the law of N_c is stationary, and by Wald plus Campbell the mean rate is $s \cdot \mathbb{E}[\text{cluster size}] = s/(1 - \kappa) < \infty$ (Lemma A.10(i)). Conversely, suppose a stationary version with finite mean intensity $\bar{\xi} = \mathbb{E}N(0, 1]$ exists for some κ . Then $\mathbb{E}\lambda(t)$ is finite and constant in t (stationarity; and $\mathbb{E}\int_0^1 \lambda = \mathbb{E}N(0, 1]$ by Lemma A.2 with $f \equiv 1$), so $\mathbb{E}\lambda(t) = \bar{\xi}$ for a.e. t . Taking expectations in (8.1) and using Campbell's formula for the stationary process ($\mathbb{E}\sum_{t_k < t} \phi(t - t_k) = \bar{\xi} \int_0^\infty \phi$, [2]) gives $\bar{\xi} = s + \kappa \bar{\xi}$, i.e. $(1 - \kappa)\bar{\xi} = s$. For $s > 0$ this has no finite nonnegative solution when $\kappa \geq 1$; when $\kappa < 1$ it returns $\bar{\xi} = s/(1 - \kappa)$, consistent with the construction. The multivariate case repeats every step with matrix kernels: generation means are K^n , $\sum_n K^n = (I - K)^{-1}$ converges iff $\rho(K) < 1$ (Gelfand's formula), the mean vector is $(I - K)^{-1}s$, and the first-moment identity reads $(I - K)\bar{\xi} = s$. For stability theory beyond the linear case see [5]. $\square$

Claim (Proposition 8.4). *Let $\lambda(t) = s(t) + \sum_{t_k < t} \phi(t - t_k)$ with s bounded. If $\phi < 0$ on a set of positive Lebesgue measure, there exist finite histories on which the formula is strictly negative; hence (8.1) defines a valid intensity for all histories only if $\phi \geq 0$ almost everywhere.*

Proof. Suppose ϕ is not nonnegative a.e. Then there are $\epsilon > 0$ and a Borel set $A \subseteq (0, \infty)$ of positive Lebesgue measure with $\phi < -\epsilon$ on A . Let $s_{\max} = \sup_t s(t) < \infty$ and fix an integer $n > s_{\max}/\epsilon$. A set of positive measure is infinite, so we may choose n distinct lags $a_1, \dots, a_n \in A$; fix any t larger than all of them and place events at $x_i = t - a_i$, a simple finite configuration. Evaluating the formula at t : $\lambda(t) \leq s_{\max} + \sum_{i=1}^n \phi(a_i) < s_{\max} - n\epsilon < 0$. A conditional intensity is by definition nonnegative, so no point process can have (8.1) as its intensity functional on *all* finite histories unless $\phi \geq 0$ a.e. $\square$

Remark A.11 (Why the all-histories qualifier is essential). Requiring only *almost-sure* non-negativity along realized histories does not force $\phi \geq 0$: with constant baseline s and $\phi = -s \mathbf{1}\{0 < u \leq 1\}$, the induced dynamics are a renewal process with a unit dead time — after each event the formula sits at exactly zero for one time unit, so no two events are ever closer than 1 and the offending configurations of the proof are never visited. Such exactly tuned signed kernels are fragile (any perturbation of s or ϕ breaks the cancellation), evade none of the substantive losses — the cluster representation of Theorem 8.2 requires $\phi \geq 0$ to define offspring densities, and truncating at zero, $\lambda = (s + \sum \phi)_+$, destroys the linearity on which the Volterra first-moment equation of Chapter 9 rests — and none of the estimation theory of this manual applies to them. The factorization route (b) of Section 8.3 avoids the problem wholesale.

Claim (Theorem 8.5). *Let N_s be the ground process of sector- s deals with intensity $\lambda_s^g(t)$ and let each event carry a firm mark. Then $\lambda_i(t) = \lambda_{s(i)}^g(t) p(i | s(i), t, \mathcal{H}_t)$ exactly, the log-likelihood of the marked stream splits into a ground term plus a mark term, and the two factors can be estimated separately without loss.*

Proof. We prove the finite-mark-space case, which is the case used: the marks are the finitely many firms of a sector. Represent the marked stream as the multivariate point process $(N_i)_{i \in \mathcal{M}}$, $|\mathcal{M}| < \infty$, on the common internal history $(\mathcal{H}_t)$, each N_i simple with predictable intensity λ_i satisfying (A.1), and no two components jumping together (simplicity of the marked stream).

Ground intensity. $N_g = \sum_i N_i$ is a simple counting process and $N_g - \int_0^\cdot \sum_i \lambda_i$ is a finite sum of martingales, hence a martingale; the integrand $\lambda_g := \sum_i \lambda_i$ is predictable and nonnegative, so λ_g is the intensity of N_g .

The mark kernel. Define $p(i | t) = \lambda_i(t)/\lambda_g(t)$ on $\{\lambda_g(t) > 0\}$ and let $p(\cdot | t)$ be an arbitrary fixed probability vector on $\{\lambda_g(t) = 0\}$. Then $p(\cdot | t)$ is a probability vector for every t and $\lambda_i = \lambda_g p(i | \cdot)$ everywhere: on $\{\lambda_g = 0\}$ both sides vanish because $0 \leq \lambda_i \leq \lambda_g$. This is (8.5), an identity, with the firm-level intensity λ_i and $\lambda_s^g = \sum_{j:s(j)=s} \lambda_j$. Its interpretation as the conditional mark law is the Campbell-type identity: for every bounded predictable $H(t, i)$, Lemma A.1(ii) applied componentwise gives

$$\mathbb{E} \sum_k H(t_k, m_k) = \sum_i \mathbb{E} \int_0^T H(t, i) \lambda_i(t) dt = \mathbb{E} \int_0^T \lambda_g(t) \sum_i H(t, i) p(i | t) dt :$$

conditionally on an event at t and on $\mathcal{H}_{t-}$, the mark is distributed as $p(\cdot | t)$ [2, Ch. 6.4].

Likelihood factorization. For a realization $(t_1, m_1), \dots, (t_n, m_n)$ on $[0, T]$, the likelihood of a multivariate point process with internal history is

$$L_T = \left[\prod_{k=1}^n \lambda_{m_k}(t_k) \right] \exp \left(- \int_0^T \lambda_g(u) du \right),$$

the product of successive conditional densities of (next time, next mark) given the past; we take this representation from [2, Ch. 7.2–7.3], where it follows from the hazard representation of the internal-history intensity. Substituting $\lambda_{m_k} = \lambda_g p(m_k | \cdot)$,

$$\log L_T = \underbrace{\sum_k \log \lambda_g(t_k) - \int_0^T \lambda_g du}_{\text{ground term}} + \underbrace{\sum_k \log p(m_k | t_k)}_{\text{mark term}}.$$

If λ_g carries parameters θ_1 and p carries parameters θ_2 with (θ_1, θ_2) variation independent, the two terms share no parameters and joint maximization separates into the two stage-wise maximizations: the split is lossless at population level, exactly as Remark 8.6 asserts. For general (non-finite) mark spaces the same theorem holds with p a probability kernel; see [2, Ch. 6–7]. $\square$

Claim (Proposition 8.7). *Fix m and replace, for each event, the risk set in (8.6) by the winner plus a uniform random subsample of m non-winners drawn independently of utilities. The score of the resulting conditional likelihood has mean zero at the true w^* , and its maximizer is consistent.*

Proof. Assume events are i.i.d. draws of (risk set $\mathcal{R}$ with $|\mathcal{R}| = r$, features $\{z_j\}_{j \in \mathcal{R}}$, winner c), with $\mathbb{P}(c = i \mid \mathcal{R}, z) = e^{u_i} / \sum_{j \in \mathcal{R}} e^{u_j}$, $u_j = w^{*\top} z_j$; integrability $\mathbb{E}[\max_{j \in \mathcal{R}} \|z_j\|] < \infty$; and the identification condition that for every $v \neq 0$, the within-sample-set utilities $v^\top z_j$, $j \in D$, are non-constant with positive probability. (In the sequential setting of Chapter 8 the same argument runs conditionally on the event filtration, the score becomes a martingale difference array, and consistency holds under ergodicity; we give the i.i.d. proof.)

Step 1: exact conditional likelihood (uniform conditioning). Let $D = \{c\} \cup S$ where S is a uniform m -subset of $\mathcal{R} \setminus \{c\}$ drawn independently of everything else given $(\mathcal{R}, c)$. For a fixed $(m+1)$ -subset $d \subseteq \mathcal{R}$ and $i \in d$,

$$\mathbb{P}(c = i, D = d \mid \mathcal{R}, z) = \frac{e^{u_i}}{\sum_{j \in \mathcal{R}} e^{u_j}} \cdot \binom{r-1}{m}^{-1},$$

and the second factor is the same for every $i \in d$ — McFadden’s uniform-conditioning property [8]. Dividing by the sum over $i' \in d$, $\mathbb{P}(c = i \mid D = d, \mathcal{R}, z) = e^{u_i} / \sum_{j \in d} e^{u_j}$: the sampled softmax is an *exact* conditional likelihood, not an approximation.

Step 2: zero-mean score. Let $\ell(w; c, D) = w^\top z_c - \log \sum_{j \in D} e^{w^\top z_j}$, with gradient $z_c - \sum_j p_w(j \mid D) z_j$, bounded by $2 \max_j \|z_j\|$, hence integrable. For each fixed d , $\sum_{i \in d} p_{w^*}(i \mid d) \nabla_w \log p_w(i \mid d) |_{w=w^*} = \nabla_w \sum_i p_w(i \mid d) |_{w^*} = 0$ (a finite sum, differentiated termwise), so the score has conditional mean zero given $(D, \mathcal{R}, z)$ at w^* , and therefore unconditional mean zero.

Step 3: concavity and identification. $\ell(\cdot; c, d)$ is concave: its Hessian is $-\text{Cov}_{p_w(\cdot \mid d)}(z) \preceq 0$ (log-sum-exp of affine functions is convex). Let $L(w) = \mathbb{E}[\ell(w)]$. By Step 1 the model is correctly specified conditionally on D , so $L(w^*) - L(w) = \mathbb{E}[\text{KL}(p_{w^*}(\cdot \mid D) \parallel p_w(\cdot \mid D))] \geq 0$ (Gibbs’ inequality conditionally, then take expectations), with equality iff $p_w(\cdot \mid D) = p_{w^*}(\cdot \mid D)$ a.s., i.e. iff $(w - w^*)^\top (z_i - z_j) = 0$ for all $i, j \in D$ a.s. — excluded for $w \neq w^*$ by the identification condition. So w^* is the unique maximizer of L .

Step 4: consistency. Let ℓ_n be the sample average criterion. For each fixed w , $\ell_n(w) \rightarrow L(w)$ a.s. (strong law, integrability from Step 2). Pointwise almost-sure convergence of concave functions on a countable dense subset of $\mathbb{R}^d$ implies uniform convergence on compact sets (a standard fact of convex analysis). Fix $\epsilon > 0$ and let $\delta = L(w^*) - \max_{\|w - w^*\| = \epsilon} L(w) > 0$ (positive by uniqueness and continuity). Eventually, uniformly on the sphere, $\ell_n(w) \leq L(w) + \delta/3 \leq L(w^*) - 2\delta/3 \leq \ell_n(w^*) - \delta/3$; a concave function that is larger at the center than everywhere on a sphere attains its maximum inside the ball, so every maximizer $\hat{w}_n$ satisfies $\|\hat{w}_n - w^*\| \leq \epsilon$ eventually, almost surely. $\square$

Lemma A.12 (Gaussian moment generating function). *If $z \sim \mathcal{N}(\mu, \Sigma)$ in $\mathbb{R}^q$ and $w \in \mathbb{R}^q$, then $\mathbb{E}[\exp(w^\top z)] = \exp(w^\top \mu + \frac{1}{2} w^\top \Sigma w)$.*

Proof. Factor $\Sigma = LL^\top$ (possible for any positive semidefinite matrix) and write $z \stackrel{d}{=} \mu + L\varepsilon$ with $\varepsilon \sim \mathcal{N}(0, I_q)$. Then $w^\top z = w^\top \mu + a^\top \varepsilon$ with $a = L^\top w$, and by independence of the coordinates $\mathbb{E} e^{a^\top \varepsilon} = \prod_i \mathbb{E} e^{a_i \varepsilon_i}$. The univariate computation completes the square: $\mathbb{E} e^{a_i \varepsilon_i} = (2\pi)^{-1/2} \int e^{a_i x - x^2/2} dx = e^{a_i^2/2} (2\pi)^{-1/2} \int e^{-(x-a_i)^2/2} dx = e^{a_i^2/2}$. Multiply and use $\|a\|^2 = w^\top \Sigma w$. $\square$

Claim (Proposition 8.8). *With N_{pool} latent entrants whose features are i.i.d. $\mathcal{N}(\mu, \Sigma)$ independent of $\mathcal{H}_t$, replacing the entrant block of the softmax denominator by its expectation yields the single pseudo-candidate score (8.7).*

Proof. By linearity and Lemma A.12, $\mathbb{E} \sum_{j=1}^{N_{\text{pool}}} e^{w^\top z_j} = N_{\text{pool}} e^{w^\top \mu + w^\top \Sigma w / 2}$, and matching $e^{q_{\text{new}}}$ to this expectation gives $q_{\text{new}} = \log N_{\text{pool}} + w^\top \mu + \frac{1}{2} w^\top \Sigma w$, which is (8.7). The accuracy of the mean-field replacement is quantified as follows. Condition on $\mathcal{H}_t$ and let $A > 0$ be the (now fixed) funded-pool denominator $\sum_{j \in \mathcal{R}_{s,t}} e^{q_{j,t}}$ and $T = \sum_j e^{w^\top z_j}$ the entrant sum, independent of $\mathcal{H}_t$ by assumption. Any funded candidate's choice probability is $e^{q_i} \mathbb{E}[(A+T)^{-1}]$. The map $f(x) = (A+x)^{-1}$ on $[0, \infty)$ has $0 < f''(x) = 2(A+x)^{-3} \leq 2A^{-3}$; a second-order Taylor expansion of f about $\mathbb{E}T$ with Lagrange remainder, followed by taking expectations (the linear term has mean zero), gives

$$|\mathbb{E}[f(T)] - f(\mathbb{E}T)| \leq \frac{\text{Var}(T)}{A^3}, \quad \text{Var}(T) = N_{\text{pool}} e^{2w^\top \mu + w^\top \Sigma w} (e^{w^\top \Sigma w} - 1),$$

the variance being the standard lognormal computation $\text{Var}(e^X) = e^{2m+v}(e^v - 1)$ for $X \sim \mathcal{N}(m, v)$, from Lemma A.12 applied at w and $2w$. When A and $\mathbb{E}T$ both grow like the pool size, $f \asymp N^{-1}$ while the error is $O(N^{-2})$: the relative error of the mean-field step is $O(1/N_{\text{pool}})$, negligible at the pool sizes of Chapter 8. $\square$

Proposition A.13 (Pool-size non-identifiability, made precise). *In any parametrization in which the entrant block enters the choice probabilities only through the scalar score $q_{\text{new}} = b_0 + \log N_{\text{pool}} + w^\top \mu + \frac{1}{2} w^\top \Sigma w$ with a constant pool, the likelihood of every dataset is invariant under $(b_0, \log N_{\text{pool}}) \mapsto (b_0 + c, \log N_{\text{pool}} - c)$ for every real c : the constant b_0 and the pool size (equivalently, the pool size and the entrant mean utility) are not separately identified. If instead $N_{\text{pool},t}$ varies over events and enters as a known offset, $q_{\text{new},t} = b_0 + \log N_{\text{pool},t} + \dots$, then b_0 is identified.*

Proof. Every choice probability — of a funded candidate or of the newcomer alternative — depends on the entrant parameters only through $e^{q_{\text{new}}}$ in the softmax; the stated transformation leaves q_{new} , hence every probability, hence the likelihood of any dataset, unchanged. This is exact non-identification, not a flat-likelihood approximation. With known offsets, two parameter values produce identical probabilities only if $b_0 + \log N_{\text{pool},t} = b'_0 + \log N_{\text{pool},t}$ for all t , forcing $b_0 = b'_0$. This is why the production checklist of Chapter 10 requires recalibrating the newcomer constant whenever the pool definition changes: the fitted constant is a statement about $b_0 + \log N_{\text{pool}}$, never about entrant quality alone. $\square$

A.8 Proofs for Chapter 9

Claim (Proposition 9.2). *For the linear Hawkes process (8.1) with locally bounded baseline $s \geq 0$ and integrable kernel $\phi \geq 0$, the mean intensity $\xi(t) = \mathbb{E}[\lambda(t)]$ is locally integrable and solves the Volterra equation (9.2); it is its unique locally integrable solution. The multivariate statement holds componentwise.*

Proof. Realize the process by the cluster construction in the proof of Theorem 8.2 (subcritical case) or note that (A.1) holds by Lemma A.10(iii) for every κ ; either way $\mathbb{E}N(T) < \infty$ for all T , so Lemma A.2 applies. Writing the history term as an integral against dN and applying Lemma A.2 with the nonnegative measurable $f(u) = \phi(t-u)\mathbf{1}\{u < t\}$,

$$\xi(t) = s(t) + \mathbb{E} \int_{(0,t)} \phi(t-u) dN(u) = s(t) + \mathbb{E} \int_0^t \phi(t-u) \lambda(u) du = s(t) + \int_0^t \phi(t-u) \xi(u) du,$$

the last step by Tonelli (everything nonnegative). Local integrability: $\int_0^T \xi = \mathbb{E}\Lambda(T) = \mathbb{E}N(T) < \infty$, again by Lemma A.2 with $f \equiv 1$; in particular ξ is finite a.e. and the equation holds with finite right-hand side a.e. Uniqueness: if ξ_1, ξ_2 are locally integrable solutions, $d = |\xi_1 - \xi_2|$ satisfies $d \leq \phi * d$ a.e., hence by iteration $d \leq \phi^{*n} * d$ for all n ; integrating over $[0, T]$ and using $\int_0^T (\phi^{*n} * d) \leq (\int_0^T \phi^{*n})(\int_0^T d)$ with $\int_0^T \phi^{*n} = \kappa^n \mathbb{P}(S_n \leq T) \rightarrow 0$ (Lemma A.10(iii)) forces $\int_0^T d = 0$ for every T . The linearity of (8.1) is what lets the expectation pass through — for the log-link models of Chapter 8 the first step fails, exactly as stated there. For the multivariate form see also [17]. $\square$

Claim (Proposition 9.4). *The MBPP map $s \mapsto \xi$ of (9.2) is realizable as a finite-dimensional time-invariant linear ODE with output ξ iff the kernel's Laplace transform $\hat{\phi}(z)$ is a strictly proper rational function, equivalently iff ϕ is a finite sum of polynomial-times-exponential terms (complex-conjugate pairs allowed).*

Proof. Call the map *realizable* if there exist finite n and (A, b, c) , $A \in \mathbb{R}^{n \times n}$, $b, c \in \mathbb{R}^n$, such that for every locally bounded input s , the solution of $x' = Ax + b\xi$, $x(0) = 0$, with $\xi = s + c^\top x$, reproduces the Volterra solution. Substituting the output equation, this is the ordinary ODE $x' = (A + bc^\top)x + bs$ with output $\xi = c^\top x + s$: a finite linear system driven by s .

Realizability is a property of the kernel. By variation of constants, the feedback loop computes $c^\top x(t) = \int_0^t g(t-u)\xi(u) du$ with $g(t) = c^\top e^{At}b$ (differentiate to verify; e^{At} is the matrix exponential). The system solves (9.2) for all inputs iff $\int_0^t (g - \phi)(t-u)\xi(u) du = 0$ for all t and all solutions ξ ; since any continuous ξ arises from the input $s = \xi - \phi * \xi$, the du Bois-Reymond argument (test against approximate identities) forces $\phi = g$ a.e. So realizability means precisely $\phi(t) = c^\top e^{At}b$ a.e. for some finite triple.

Realizable kernels have strictly proper rational transforms. For $\operatorname{Re} z > \|A\|$ the integral $\int_0^\infty e^{-zt} e^{At} dt$ converges absolutely and equals $(zI - A)^{-1}$ (multiply by $zI - A$ and integrate by parts, or integrate the exponential series termwise). Hence $\hat{\phi}(z) = c^\top (zI - A)^{-1}b = c^\top \operatorname{adj}(zI - A)b / \det(zI - A)$: a ratio of a polynomial of degree at most $n-1$ to one of degree n — rational and strictly proper.

Conversely. Let $\hat{\phi} = q/p$ be strictly proper rational. Partial fractions over $\mathbb{C}$: $\hat{\phi}(z) = \sum_j \sum_{r=1}^{r_j} \alpha_{jr} (z - z_j)^{-r}$. Termwise, $(z - z_j)^{-r}$ is the Laplace transform of $t^{r-1} e^{z_j t} / (r-1)!$ (direct integration for $\operatorname{Re} z > \operatorname{Re} z_j$), so the polynomial-times-exponential function $g(t) = \sum_{j,r} \alpha_{jr} t^{r-1} e^{z_j t} / (r-1)!$ has $\hat{g} = \hat{\phi}$; by injectivity of the Laplace transform on locally integrable functions of exponential order (a classical uniqueness theorem), $\phi = g$ a.e. Realization: for a single term, the $r \times r$ Jordan block J at z_j satisfies $e^{Jt} = e^{z_j t} (t^{k-l} / (k-l)!)_{l \leq k}$ (upper triangular), so $e_1^\top e^{Jt} e_r = t^{r-1} e^{z_j t} / (r-1)!$; a direct sum of such blocks (conjugate pairs combined into real $2r \times 2r$ blocks) with the matching weights in c gives $\phi(t) = c^\top e^{At}b$. Finally, the exponential kernel recovers the scalar ODE (9.4): $\phi = \kappa\theta e^{-\theta t}$ is realized by $A = -\theta$, $b = 1$, $c = \kappa\theta$, and the closed loop $x' = (-\theta + \kappa\theta)x + s\kappa\theta / (\kappa\theta)$ — written on $h = cx$ — is $h' = \kappa\theta s - \theta(1 - \kappa)h$, exactly (9.4). Power-law kernels have branch-cut (non-rational) transforms and admit no finite realization, which is why they require quadrature or exponential-sum approximation, as stated in Section 9.2. $\square$

We now make the κ - θ ridge quantitative. Recall the closed-form bucket objects of Section 9.5: with equal buckets of width Δ , piecewise-constant baseline $s_w \in [s_{\min}, s_{\max}]$, $0 < s_{\min}$, relaxation rate $\gamma = \theta(1 - \kappa)$ and $\rho = e^{-\gamma\Delta}$, the state $h_w = h(o_{w-1})$ obeys (9.5) with target levels $h_w^* = \kappa s_w / (1 - \kappa)$, and the bucket compensator is (9.6). (Both displays follow from (9.4) by solving the linear ODE on a bucket where s is constant and integrating $\xi = s + h$; (9.4) itself follows from (9.2) by writing $h(t) = \kappa\theta e^{-\theta t} \int_0^t e^{\theta u} \xi(u) du$ and differentiating.)

Proposition A.14 (The limiting singularity of the censored information matrix). *Fix $\kappa \in (0, 1)$ and the baseline range $0 < s_{\min} \leq s_w \leq s_{\max}$, take the stationary initialization $h_1 = h_1^*$, and let $C_0 = \kappa(s_{\max} - s_{\min})/(1 - \kappa)$. Then for all w : (i) $|h_w - h_w^*| \leq C_0$ and $\Xi_w = \Delta s_w/(1 - \kappa) + R_w$ with $|R_w| \leq C_0/\gamma$; (ii) $|\partial \Xi_w / \partial \theta| \leq C_0(1 - \kappa)(2\Delta\rho/\gamma + \gamma^{-2})$; (iii) $\partial \Xi_w / \partial \kappa = \Delta s_w(1 - \kappa)^{-2} + r_w$ with $|r_w| \leq C_1(\Delta\rho + 1/\gamma)$, where C_1 depends only on $(\kappa, s_{\min}, s_{\max})$. Consequently, whenever $\gamma \geq \gamma_0 > 0$ and $\gamma\Delta \rightarrow \infty$,*

$$\frac{\sup_w |\partial \Xi_w / \partial \theta|}{\inf_w \partial \Xi_w / \partial \kappa} \rightarrow 0,$$

and the per-bucket Fisher information of the Poisson likelihood (9.3) in (κ, θ) , $I = \sum_w \Xi_w^{-1}(\nabla \Xi_w)(\nabla \Xi_w)^\top$, becomes singular: $I_{\theta\theta}$ and $I_{\kappa\theta}$ vanish relative to $I_{\kappa\kappa}$, and the profile information for θ , $I_{\theta\theta} - I_{\theta\kappa}^2/I_{\kappa\kappa}$, tends to zero per bucket. In the regime $\theta \rightarrow \infty$ with Δ fixed, $\partial \Xi_w / \partial \theta \rightarrow 0$ absolutely while $\partial \Xi_w / \partial \kappa \rightarrow \Delta s_w(1 - \kappa)^{-2}$, bounded away from zero: θ leaves the first-moment likelihood entirely.

Proof. Fisher information formula. For independent $C_w \sim \text{Poisson}(\Xi_w(\eta))$ the score in a parameter η is $\sum_w (C_w/\Xi_w - 1) \partial_\eta \Xi_w$; the summands are independent with mean zero and variance $(\partial_\eta \Xi_w)^2/\Xi_w$ (since $\text{Var} C_w = \Xi_w$), giving the stated I .

(i) The levels h_w^* lie in $[\kappa s_{\min}/(1 - \kappa), \kappa s_{\max}/(1 - \kappa)]$, an interval of length C_0 . By (9.5), h_{w+1} is a convex combination of h_w and h_w^* ; with $h_1 = h_1^*$ in the interval, induction keeps every h_w in it, so $|h_w - h_w^*| \leq C_0$. Then $R_w = (h_w - h_w^*)(1 - \rho)/\gamma$ obeys $|R_w| \leq C_0/\gamma$.

(ii) Only γ and ρ carry θ ; h_w^* does not. Differentiating (9.5) in θ , the derivative state $D_w = \partial h_w / \partial \theta$ satisfies $D_1 = 0$ and $D_{w+1} = \rho D_w + (h_w - h_w^*) \partial \rho / \partial \theta$ with $\partial \rho / \partial \theta = -(1 - \kappa)\Delta\rho$, so $|D_{w+1}| \leq \rho|D_w| + (1 - \kappa)\Delta\rho C_0$ and by the geometric recursion $|D_w| \leq (1 - \kappa)\Delta\rho C_0/(1 - \rho)$ for all w . Also $\partial_\theta[(1 - \rho)/\gamma] = (1 - \kappa)[\Delta\rho/\gamma - (1 - \rho)/\gamma^2]$, bounded in modulus by $(1 - \kappa)(\Delta\rho/\gamma + \gamma^{-2})$. Hence

$$\left| \frac{\partial \Xi_w}{\partial \theta} \right| \leq |D_w| \frac{1 - \rho}{\gamma} + C_0(1 - \kappa) \left(\frac{\Delta\rho}{\gamma} + \frac{1}{\gamma^2} \right) \leq C_0(1 - \kappa) \left(\frac{2\Delta\rho}{\gamma} + \frac{1}{\gamma^2} \right).$$

(iii) The leading term of Ξ_w differentiates to $\Delta s_w(1 - \kappa)^{-2}$. For the remainder, $\partial_\kappa h_w^* = s_w/(1 - \kappa)^2 \leq s_{\max}(1 - \kappa)^{-2}$, and the derivative state $F_w = \partial h_w / \partial \kappa$ obeys $F_{w+1} = (1 - \rho) \partial_\kappa h_w^* + \rho F_w + (h_w - h_w^*) \partial \rho / \partial \kappa$ with $\partial \rho / \partial \kappa = \theta \Delta\rho$, whence, using $\theta = \gamma/(1 - \kappa)$ and the geometric recursion again, $|F_w| \leq s_{\max}(1 - \kappa)^{-2} + C_0 \Delta\rho \gamma [(1 - \kappa)(1 - \rho)]^{-1}$. Combining with $\partial_\kappa[(1 - \rho)/\gamma] = \theta[\Delta\rho/\gamma - (1 - \rho)/\gamma^2]$, every term of $\partial_\kappa R_w$ is bounded by a constant multiple (depending only on $\kappa, s_{\min}, s_{\max}$) of $\Delta\rho$ or $1/\gamma$, which is (iii).

The displayed ratio limit follows by dividing (ii) by (iii): under $\gamma \geq \gamma_0$ and $\gamma\Delta \rightarrow \infty$, the numerator terms $\Delta\rho/\gamma$ and γ^{-2} are both $o(\Delta)$ ($\Delta\rho \rightarrow 0$ superexponentially, and $\gamma^{-2} \leq (\gamma_0\gamma\Delta)^{-1}\gamma\Delta \cdot \gamma^{-1} \dots$ more simply $\gamma^{-2}/\Delta = (\gamma\Delta)^{-1}\gamma^{-1} \leq (\gamma\Delta)^{-1}\gamma_0^{-1} \rightarrow 0$), while the denominator is bounded below by $\Delta s_{\min}(1 - \kappa)^{-2}$ minus a vanishing correction. Since Ξ_w stays in a fixed positive interval around $\Delta s_w/(1 - \kappa)$ by (i), the information entries inherit the same ratios: $I_{\theta\theta}/I_{\kappa\kappa} \rightarrow 0$, $|I_{\kappa\theta}|/I_{\kappa\kappa} \rightarrow 0$, and the profile information for θ is at most $I_{\theta\theta} \rightarrow 0$ per bucket. By the Cramer–Rao bound, the variance of any unbiased estimator of θ from bucket means diverges in this limit. The $\theta \rightarrow \infty$, Δ fixed statement reads off (ii) and (iii) directly. $\square$

Claim (Proposition 9.5). *In the stationary regime with slowly varying baseline and bucket width Δ : (i) $\partial \Xi_w / \partial \kappa$ is an order-one level effect of size $\Xi_w/(1 - \kappa)$; (ii) θ enters Ξ_w only through transient terms of relative size governed by $e^{-\theta(1 - \kappa)\Delta}$, so the Fisher information of (9.3) is nearly singular along a ridge in the (κ, θ) plane, worsening monotonically with Δ ; as $\theta(1 - \kappa)\Delta \rightarrow \infty$, θ leaves the first-moment likelihood entirely.*

Proof. This is Proposition A.14 restated qualitatively. For (i): $\Delta s_w(1 - \kappa)^{-2} = [\Delta s_w/(1 - \kappa)] \cdot (1 - \kappa)^{-1} = \Xi_w/(1 - \kappa)$ up to the $O(1/\gamma)$ correction of part (i) of that proposition. For (ii): the bounds in part (ii) are proportional to $\Delta\rho/\gamma$ and γ^{-2} with $\rho = e^{-\theta(1 - \kappa)\Delta}$, both

vanishing relative to the κ -score as $\gamma\Delta \rightarrow \infty$; the sign oscillation across buckets follows from the derivative state recursion for D_w , which alternates in sign as the lagged response $h_w - h_w^*$ changes sign with the baseline steps. The near-singularity of the information matrix and the escape of θ from the likelihood are the two displayed conclusions of Proposition A.14. What censored data retain of θ beyond the first moment lives in autocovariances, which the MBPP likelihood discards by construction — Remark 9.6 and the reporting rules of Table 9.2 are the operational consequences. $\square$

Appendix B

Schema reference

This appendix is the lookup companion to Chapter 2: field-level types and typical missingness for the three PitchBook relations, a macro-feed sample showing the publish-lag structure, and coverage by deal class. Missingness figures are representative magnitudes for a US-heavy export, matching the calibration of the synthetic observation layer; re-profile any given export before modelling.

B.1 The deals table

B.2 The companies and investors tables

B.3 Macro feed sample

B.4 Field coverage by deal class

Field	Type	Typical missingness / notes
<i>Identifiers</i>		
company_id, deal_id	string key	none; join keys
company_name	string	none; display only, never a key
deal_date	date	none, but jitter up to 6 days
<i>Industry and universe</i>		
primary_industry_sector	categorical	~1%; ~3% mislabelled
primary_industry_group, primary_industry_code	categorical	~2%
company_universe	categorical	none; snapshot semantics
current_financing_status	categorical	snapshot — forbidden as feature
<i>Deal terms</i>		
deal_type, deal_class, deal_status	categorical	none
deal_type2, deal_type3	categorical	40%, 70%
deal_size	float (MUSD)	28% undisclosed; 15% of rest estimated (18% rel. error)
deal_size_status	categorical	none; flags estimated sizes
pre_money_valuation, post_valuation	float (MUSD)	55%
post_valuation_status	categorical	follows valuations
pct_acquired	float	60%; mostly M&A rows
vc_round, vc_round.up_down_flat	categorical	VC only; 65% for up/down/flat
type_of_stock, sellers	string	70%+
<i>Capital structure</i>		
total_invested_capital, total_invested_equity	float	45%
debt, total_new_debt, total_debt	float	80% on VC; better on debt
raised_to_date, total_raised_to_date	float	30%; recompute from history
total_investor_ownership	float	75%
<i>Financials at deal</i>		
revenue, ebitda	float	70%, 80% (US); far lower in EU
gross_profit, net_income	float	85%
valuation_ebitda, valuation_revenue, valuation_cash_flow	float	85%+
deal_size_ebitda, deal_size_revenue, deal_size_cash_flow	float	85%+
revenue_growth_since_last_deal	float	80%
number_of_employees	int	40%
<i>Firmography carried on deal rows</i>		
year_founded	int	5%
hq_location, hq_country, hq_region, hq_sub_region	string	2%

Table B.1: Deals table: fields, types, and representative missingness.

Field	Type	Notes
company_id	string key	join key; 12% of real firms untracked
description	string	free text; 8% missing
year_founded	int	5% missing
employees	int	<i>current</i> snapshot; forbidden as historical feature
financing_status, business_status, ownership_status	categorical	snapshots
total_money_raised	float	snapshot; recompute from deals
last_financing_date, last_financing_size, last_financing_valuation	mixed	snapshots of latest deal
investor_id	string key	investors relation
investorname	string	display only
investorstatus	categorical	current snapshot
investorsince	date	dated; usable point-in-time

Table B.2: Companies table (top block) and investors relation (bottom block). Snapshot fields may validate history but must never enter features (Chapter 2).

series_code	ref_date	publish_date	value
US_GDP_QOQ_SAAR	2023-12-31	2024-01-25	3.3
US_CPI_YOY	2024-01-31	2024-02-13	3.1
US_UNRATE	2024-01-31	2024-02-02	3.7
US_10Y_YIELD	2024-02-15	2024-02-15	4.24
EA_GDP_QOQ	2023-12-31	2024-01-30	0.0
US_PE_DRYPOWDER	2023-12-31	2024-03-15	955.0

Table B.3: Macro feed sample rows. The `ref_date`-to-`publish_date` gap — zero for market series, weeks to months for releases — is the point-in-time problem of Chapter 6: all joins key on `publish_date`.

Field group	VC	M&A	Debt
<code>deal_size</code> disclosed	65%	45%	75%
Valuations (<code>pre_money</code> , <code>post</code>)	55%	20%	5%
<code>pct_acquired</code>	15%	70%	10%
<code>vc_round</code> fields	95%	—	—
Debt / capital-structure fields	15%	35%	90%
Financials at deal	20%	45%	40%

Table B.4: Approximate non-missing coverage by deal class. No field is well populated across all classes; class-conditional models or class indicators are mandatory (Chapter 2).

Appendix C

Code map and quickstart

C.1 Directory layout

The accompanying code lives under `handbook/code`. The core package implements the estimators; the `synthetic` package implements the two testbed worlds of Chapter 10 and their observation-noise layer.

<code>handbook/code/</code>	
<code>hawkes_calibration/</code>	core package
<code>eventtime/</code>	event-time Hawkes: kernels, MLE, thinning simulation
<code>mbpp/</code>	Volterra solver, interval-censored likelihood
<code>models/</code>	<code>sector_ranker.py</code> , <code>sector_hazard.py</code> , <code>sector_survival.py</code> , <code>sector_backtest.py</code>
<code>operators/</code>	<code>pino.py</code> (operator surrogate), training loops
<code>synthetic/</code>	
<code>macro.py</code>	latent macro factors and published series
<code>firms.py</code>	firm universe, entry and exit
<code>marks.py</code>	deal marks: size, valuation, stage
<code>regime_a.py</code>	weekly two-stage DGP (well specified)
<code>regime_b.py</code>	daily firm-level block-Hawkes DGP (misspecified)
<code>observe.py</code>	noise layer: missingness, jitter, duplicates
<code>loaders.py</code>	<code>load_dataset</code> , <code>build_choice_sets</code>
<code>fast_fit.py</code>	<code>fit_sector_glm_fast</code> , <code>fit_choice_fast</code>
<code>generate.py</code>	command-line entry point
<code>EXPERIMENTS.md</code>	experiment briefings, seeds, expected outputs
<code>REVIEW.md</code>	review notes and open issues

C.2 Quickstart

Generate both synthetic worlds (seed 0 reproduces every number in Chapter 10):

```
cd handbook/code
PYTHONPATH=. python -m synthetic.generate --regime A --seed 0
PYTHONPATH=. python -m synthetic.generate --regime B --seed 0
```

Load, build choice sets, and fit both stages:

```
from synthetic.loaders import load_dataset, build_choice_sets
from synthetic.fast_fit import fit_sector_glm_fast, fit_choice_fast
```

```

ds = load_dataset("A", seed=0)
glm = fit_sector_glm_fast(ds)          # Stage 1: sector count GLM
cs = build_choice_sets(ds, n_candidates=64, cooldown_weeks=26)
cl = fit_choice_fast(cs)              # Stage 2: conditional logit

```

C.3 Regenerating the numbers

Every table in Chapter 10 regenerates from these entry points: Tables 10.3 and 10.4 from the two fits above on both regimes; Table 10.5 by passing naive versus oracle membership masks to `build_choice_sets`; Table 10.6 from the simulation guard in `hawkes_calibration/models.EXPERIMENTS.md` lists each experiment with its command, seed, runtime, and the assertion it must satisfy — including the causal-validation hypotheses — and `REVIEW.md` records known deviations; if a regenerated number drifts, trust the code.

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